

Dynamic behavior of the model

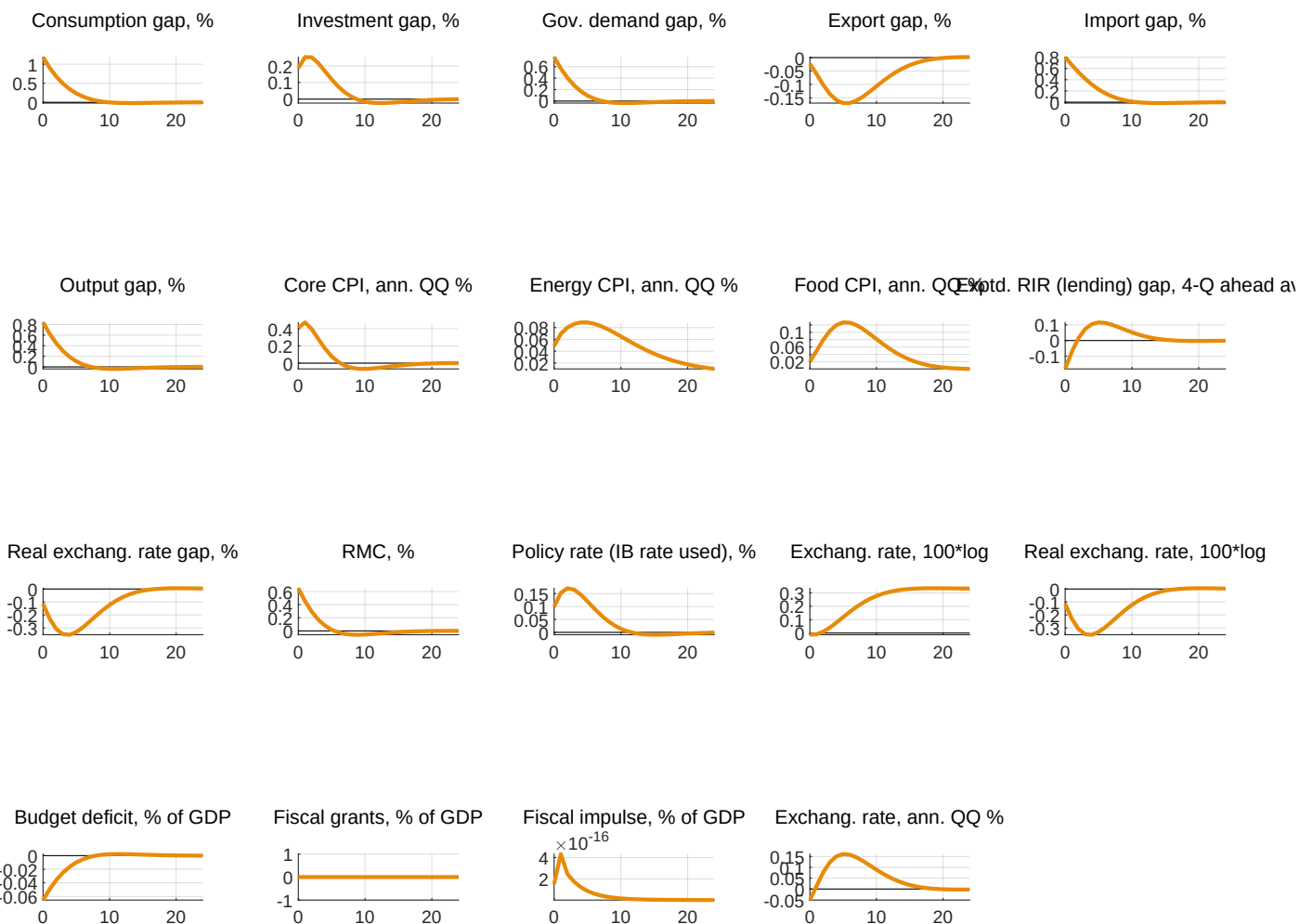
Round: 2025 Sept Forecast, time: 08-Jan-2026 16:59:36.

1 Steady states

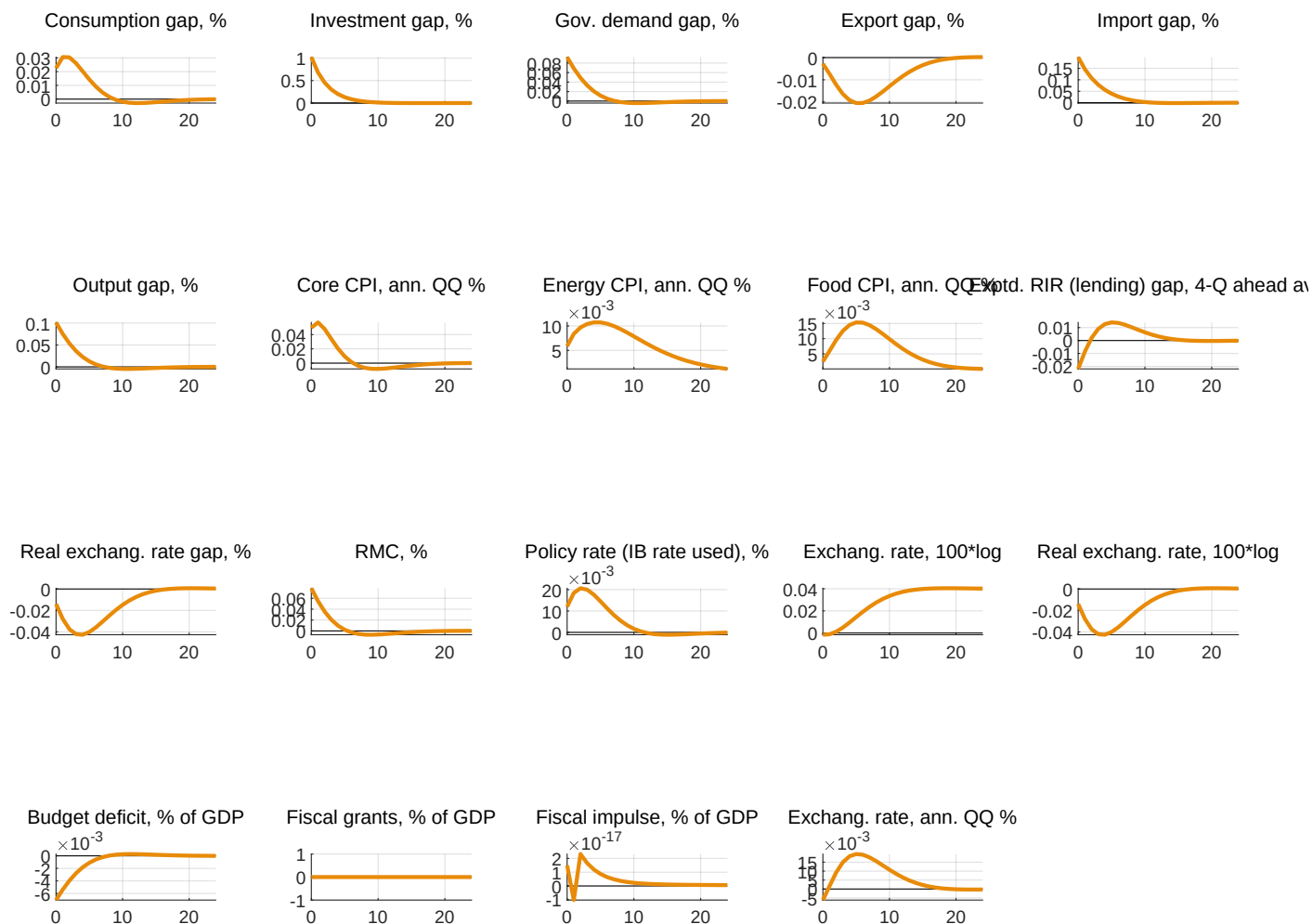
Consumption gap, % [l_cons_gap]	0.00
Investment gap, % [l_inv_gap]	0.00
Gov. demand gap, % [l_gdem_gap]	0.00
Export gap, % [l_exp_gap]	-0.00
Output gap, % [l_y_gap]	0.00
Real exchang. rate gap, % [l_z_gap]	-0.00
RIR (policy) gap, % [r_gap]	0.00
RMC, % [rmc]	-0.00
Budget deficit, % of GDP [def_y]	11.00
Struct. deficit, % of GDP [def_y_str]	11.00
Discr. deficit, % of GDP [def_y_discr]	0.00
Fiscal grants, % of GDP [grants_y]	5.00
Core rel. price gap, % [l_rp_cpi_core_gap]	-0.00
Food rel. price gap, % [l_rp_cpi_food_gap]	0.00
Energy rel. price gap, % [l_rp_cpi_ener_gap]	0.00
Lending premium gap, % [prem_d_gap]	0.00
Inflation target, YoY % [d4l_cpi_tar]	4.88
Headline CPI, ann. QQ % [dl_cpi]	4.88
Core CPI, ann. QQ % [dl_cpi_core]	4.48
Food CPI, ann. QQ % [dl_cpi_food]	6.86
Energy CPI, ann. QQ % [dl_cpi_ener]	4.88
Policy rate (IB rate used), % [i]	6.48
Policy rate tnd (IB rate used), % [i_tnd]	6.48
Real interest (policy) rate, % [r]	2.00
RIR (policy) trend, % [r_tnd]	2.00
Core rel. price tnd., ann. QQ % [dl_rp_cpi_core_tnd]	-0.40
Food rel. price tnd., ann. QQ % [dl_rp_cpi_food_tnd]	1.98
Energy rel. price tnd., ann. QQ % [dl_rp_cpi_ener_tnd]	0.00

2 Impulse response functions

Responses to Cons. gap shock, % [shock_1_cons_gap]



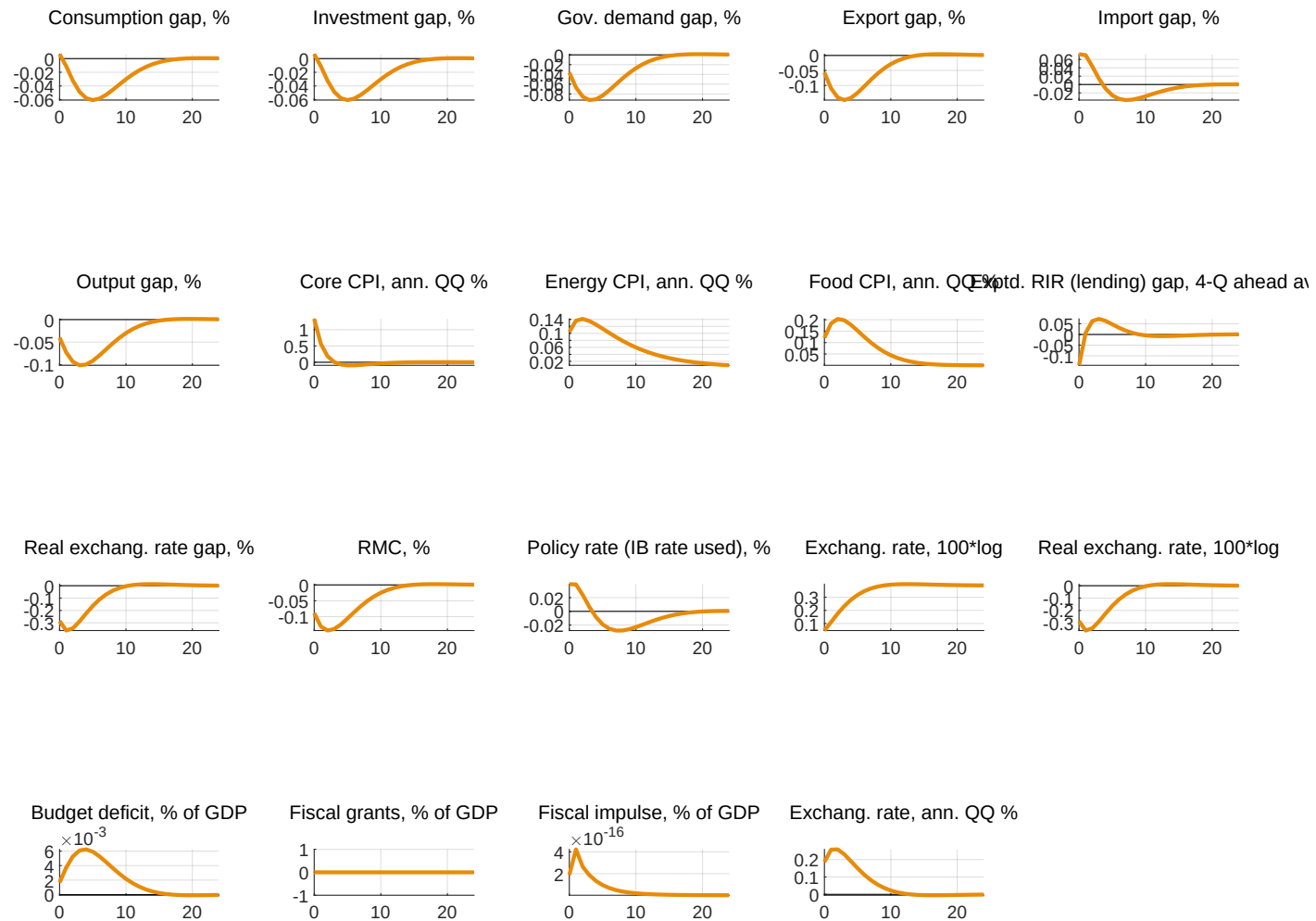
Responses to Inv. gap shock, % [shock_1_inv_gap]



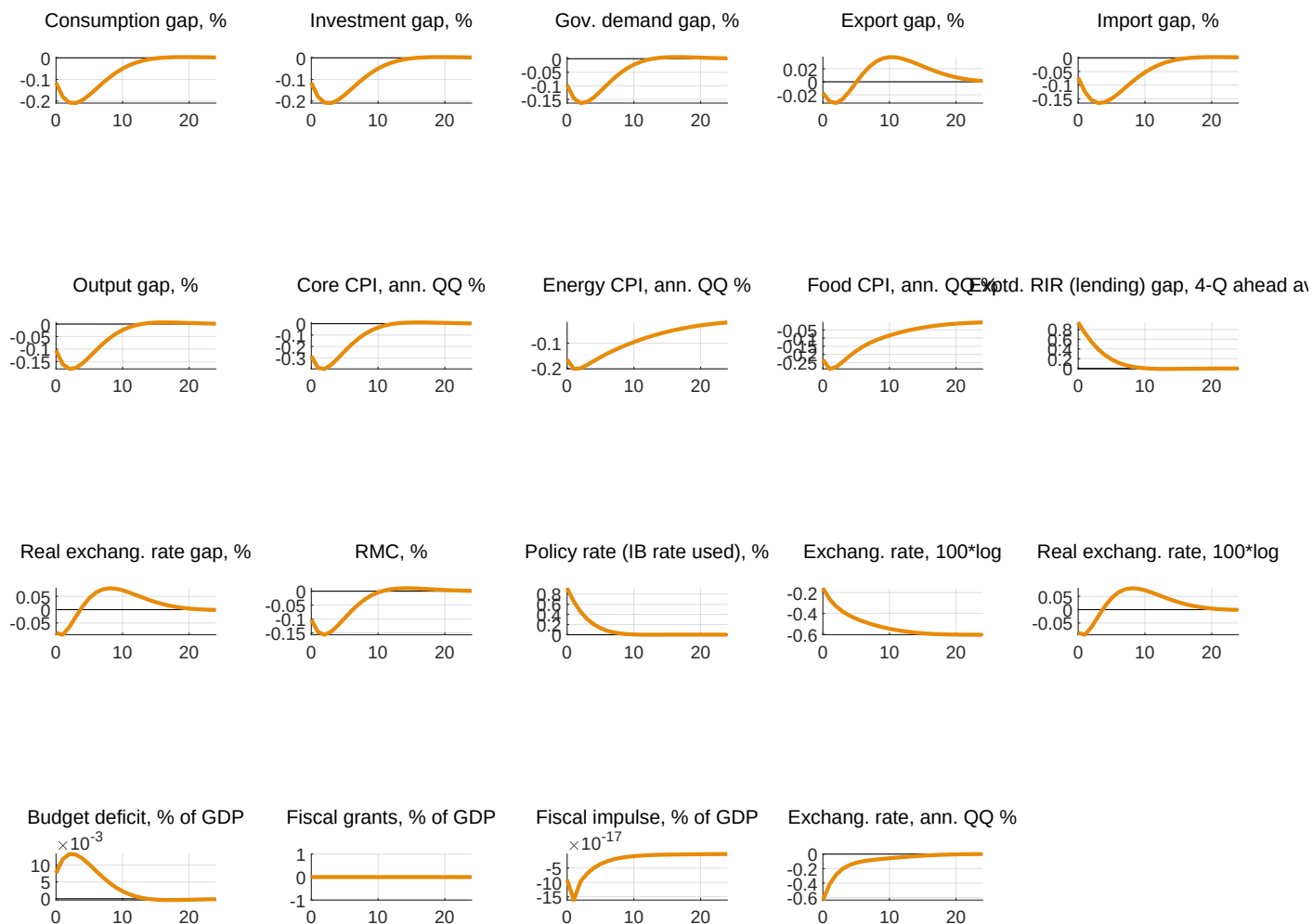
Responses to Export gap shock, % [shock_1_exp_gap]



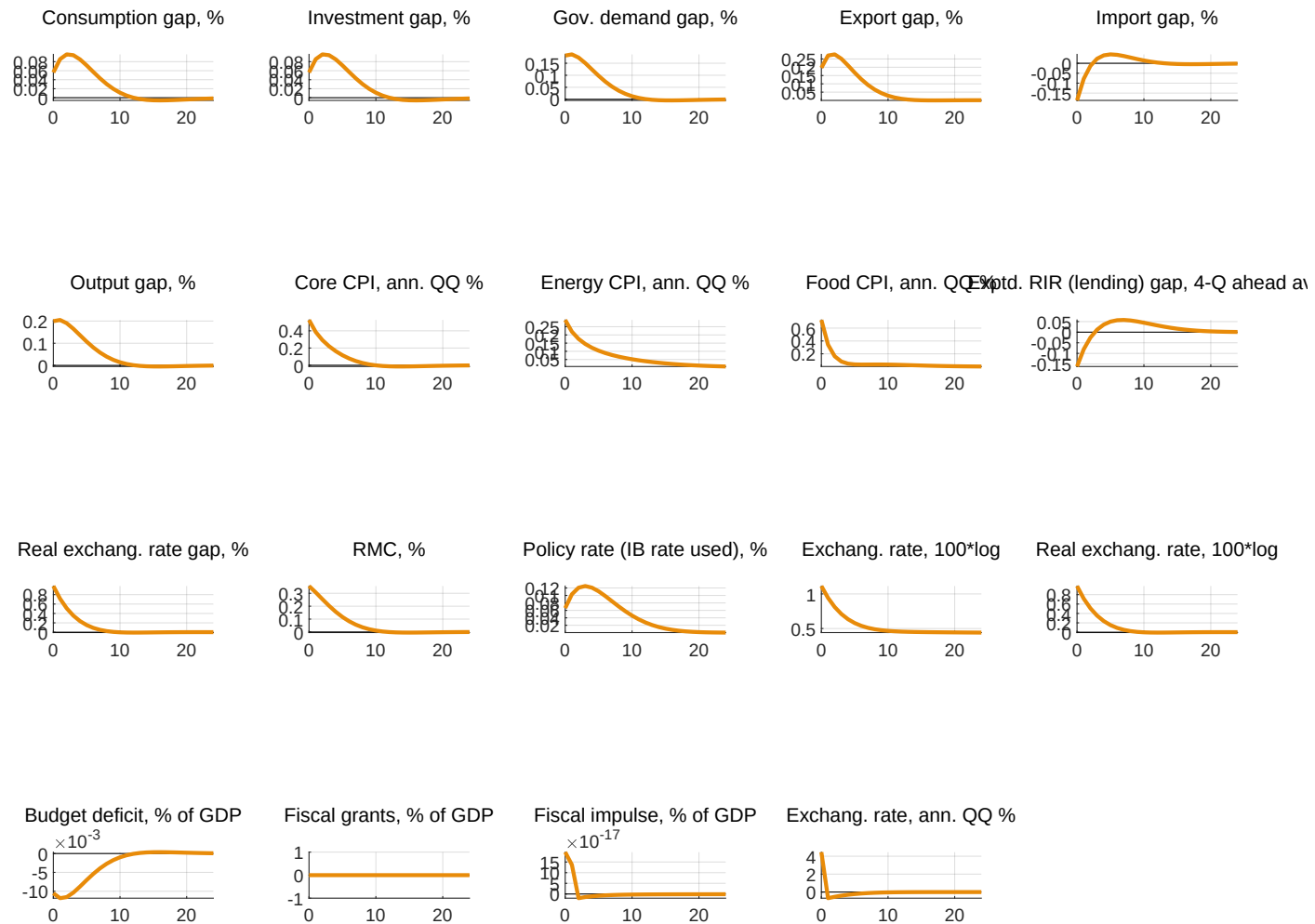
Responses to Core infl. shock, ann. QQ % [shock_dl_cpi_core]



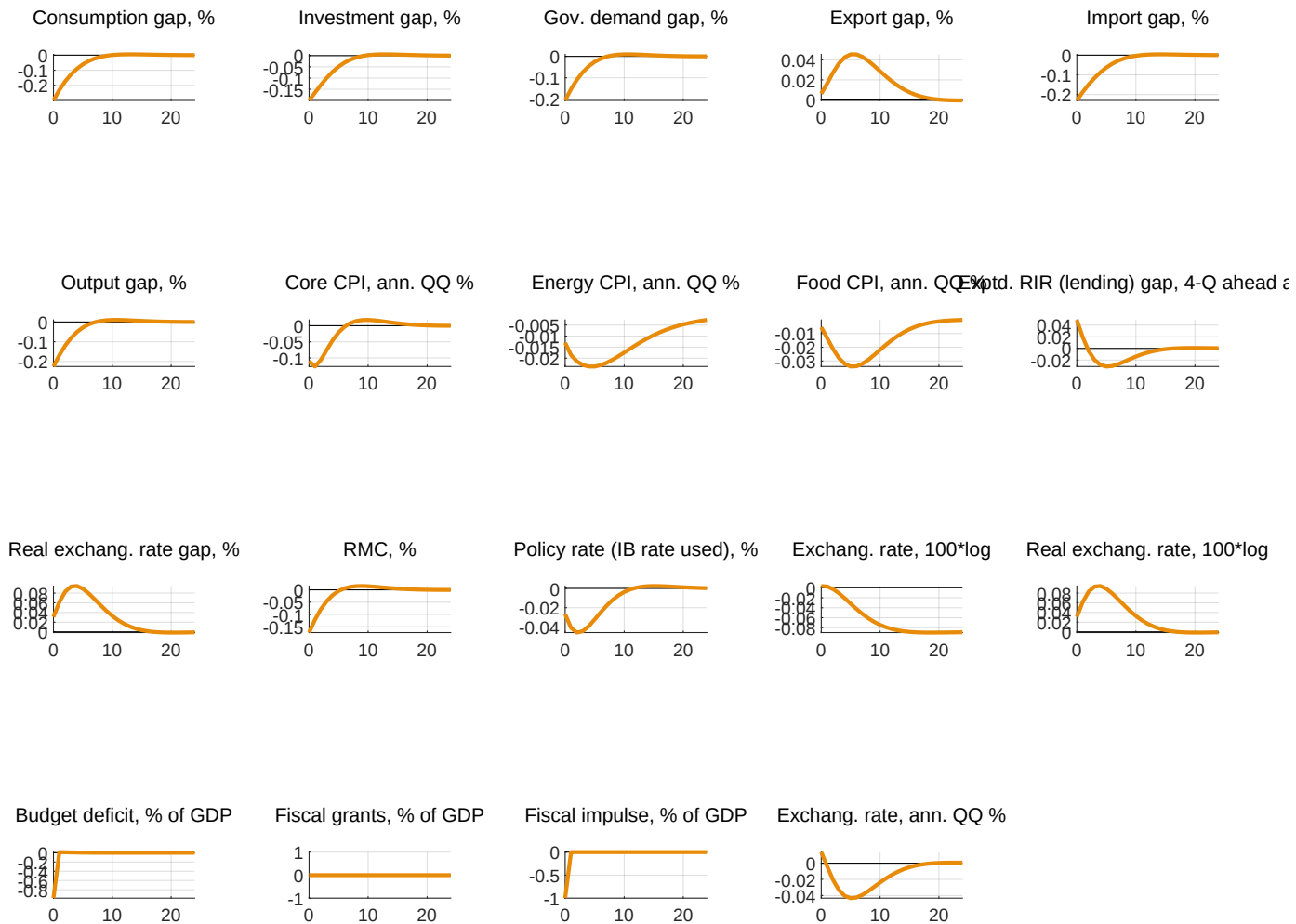
Responses to Policy rate shock, % [shock_i]



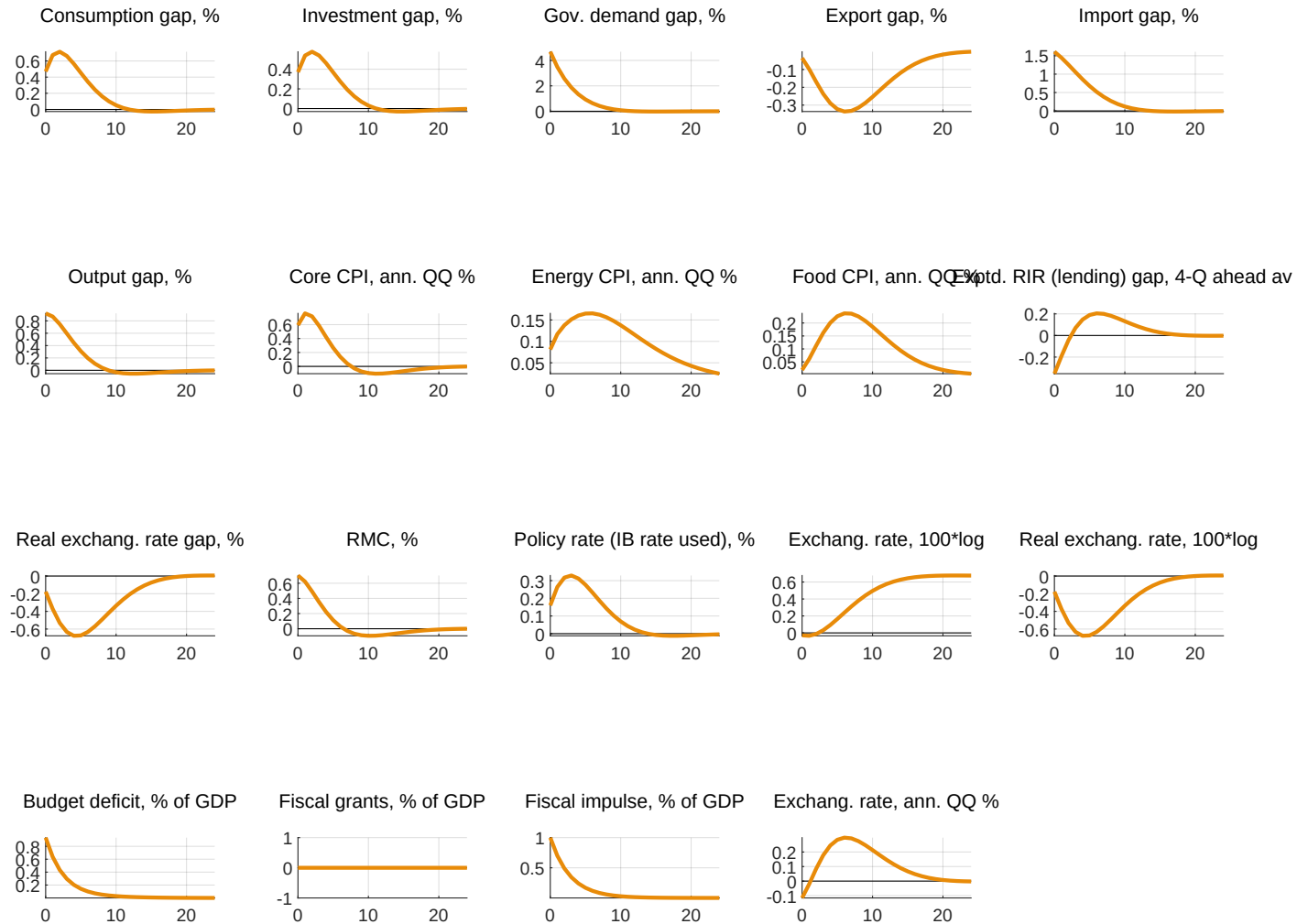
Responses to Exchng. rate shock, 100*log [shock_1_s]



Responses to Gov rev. discr. shock, % of GDP [shock_grev_y_discr]



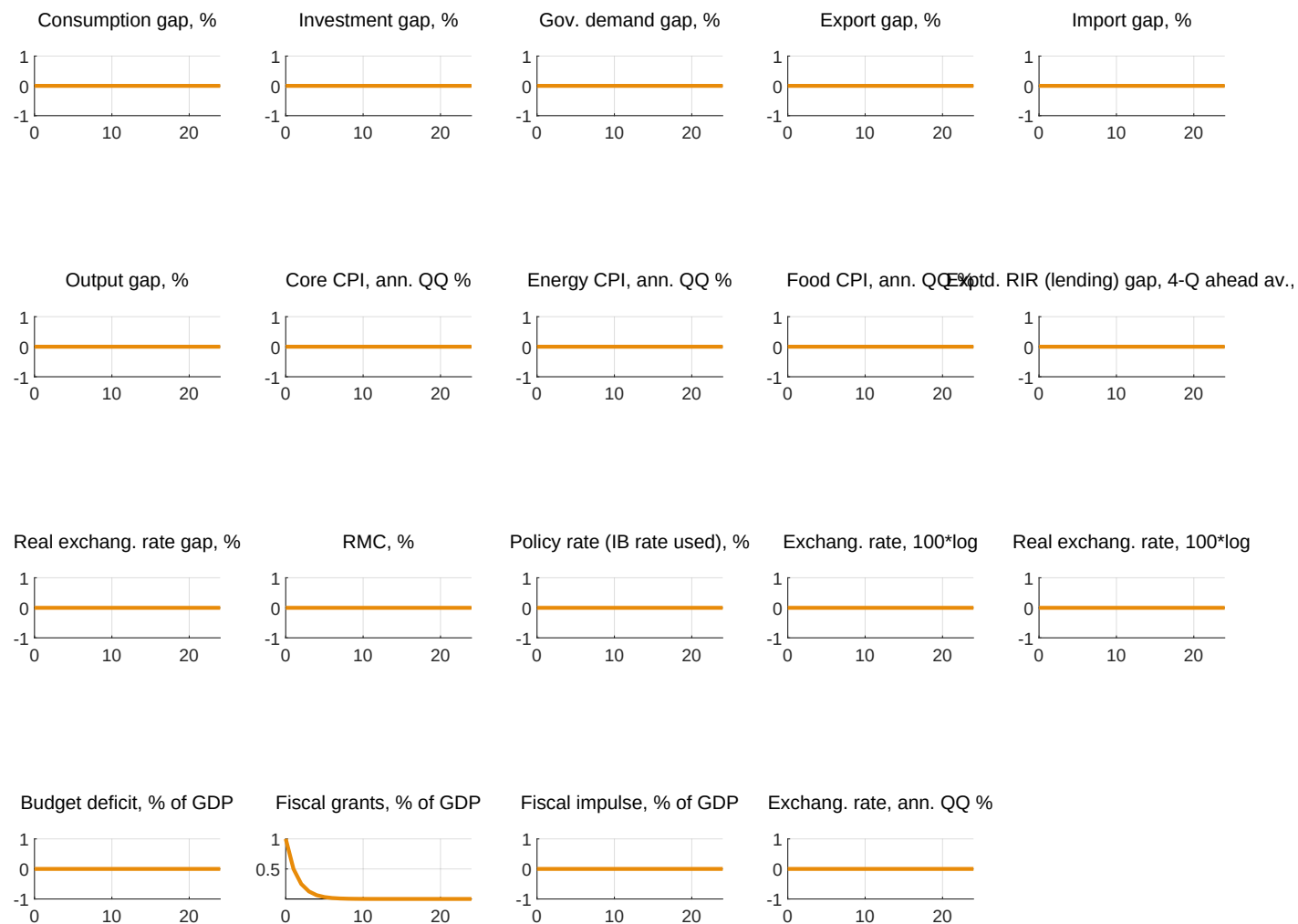
Responses to Gdem. discr. shock, % of GDP [shock_gdem_y_discr]



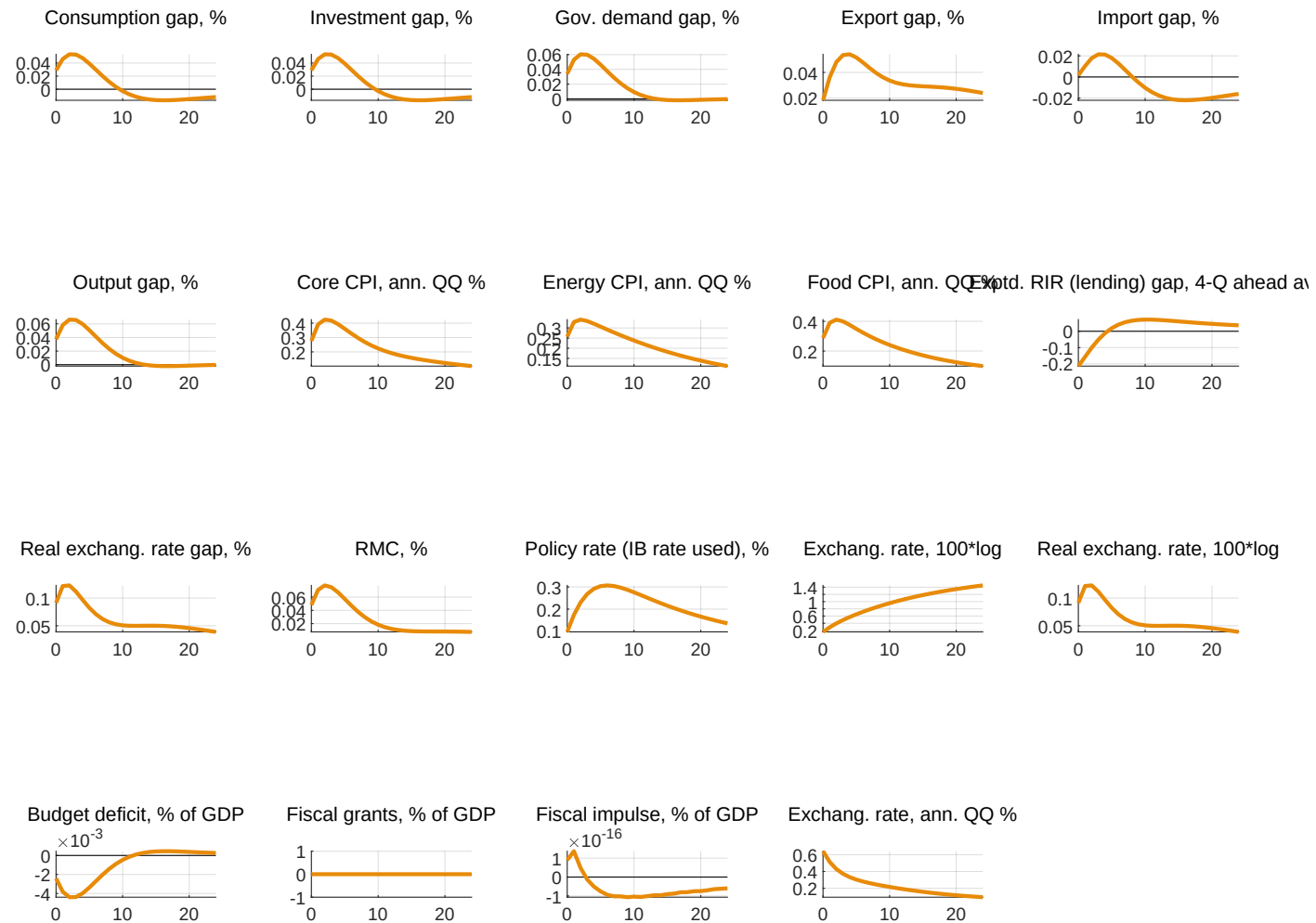
Responses to Gov exp. discr. shock, % of GDP [shock_oexp_y_discr]



Responses to Fiscal grants, % of GDP [shock_grants_y]



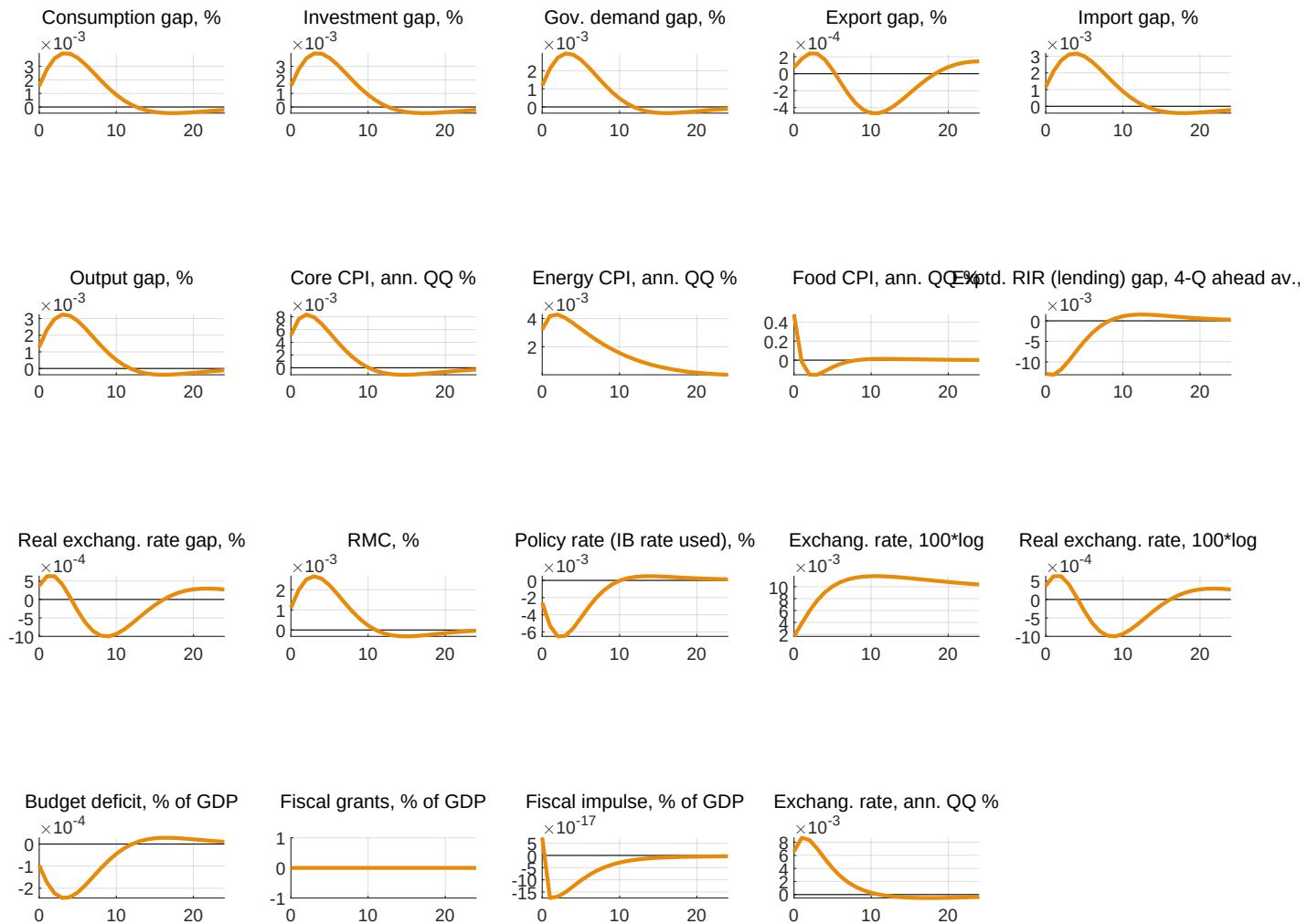
Responses to Foreign intr. rate shock, % [shock_istar]



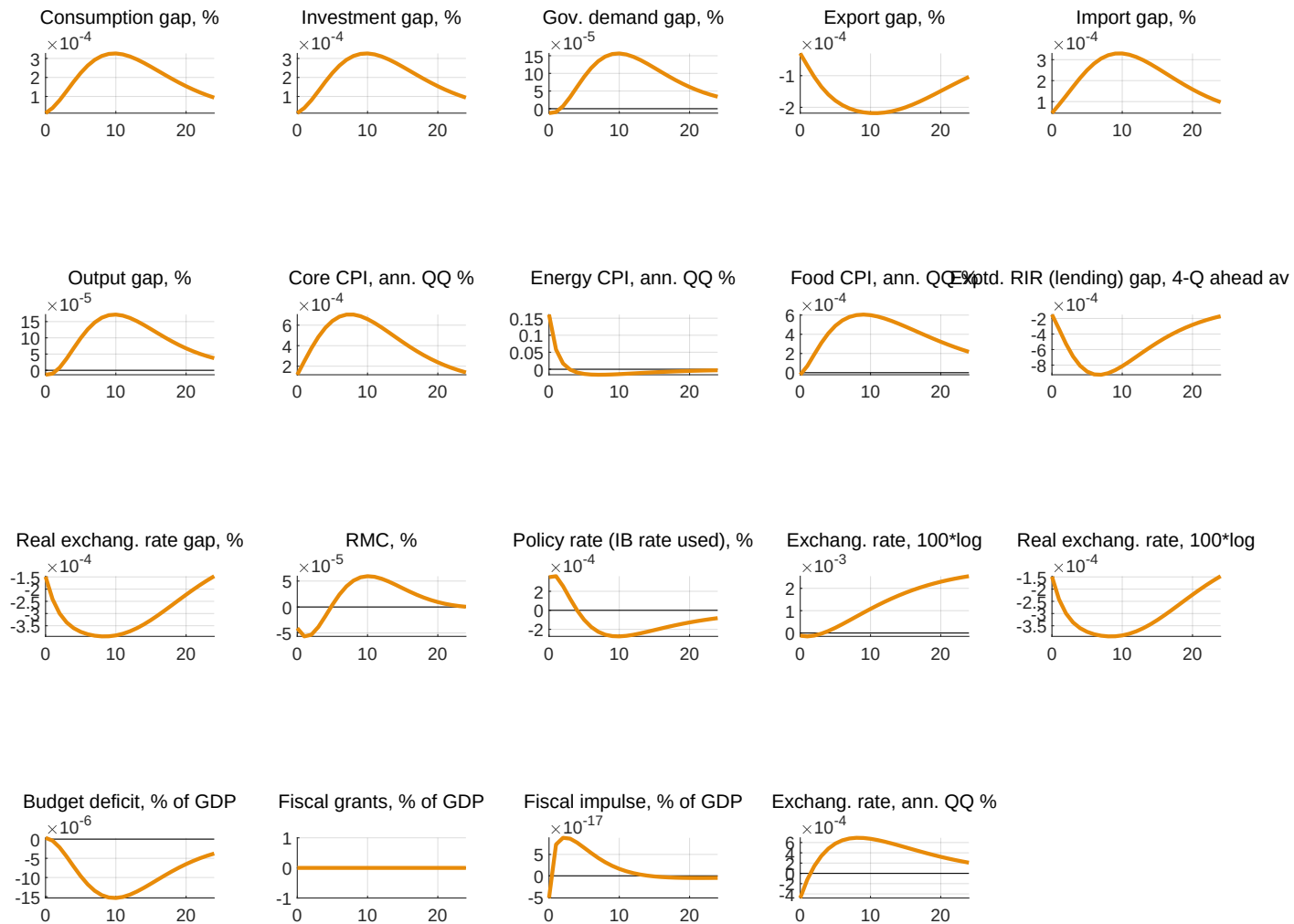
Responses to Foreign CPI shock, ann. QQ % [shock_dl_cpistar]



Responses to Foreign rel. food price gap shock, % [shock_l_rp_foodstar_gap]

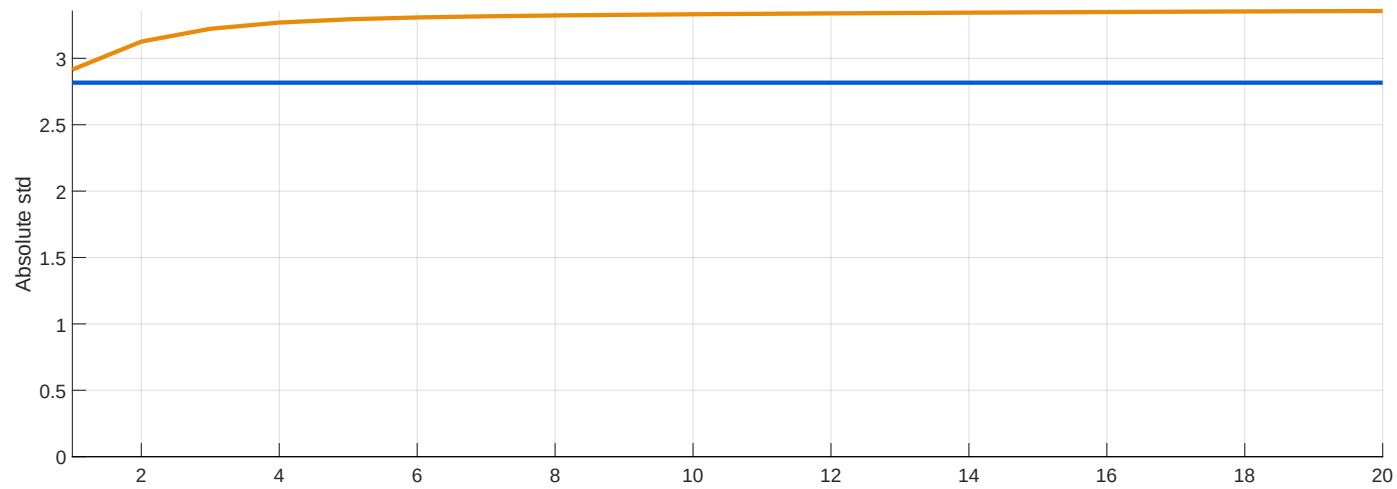
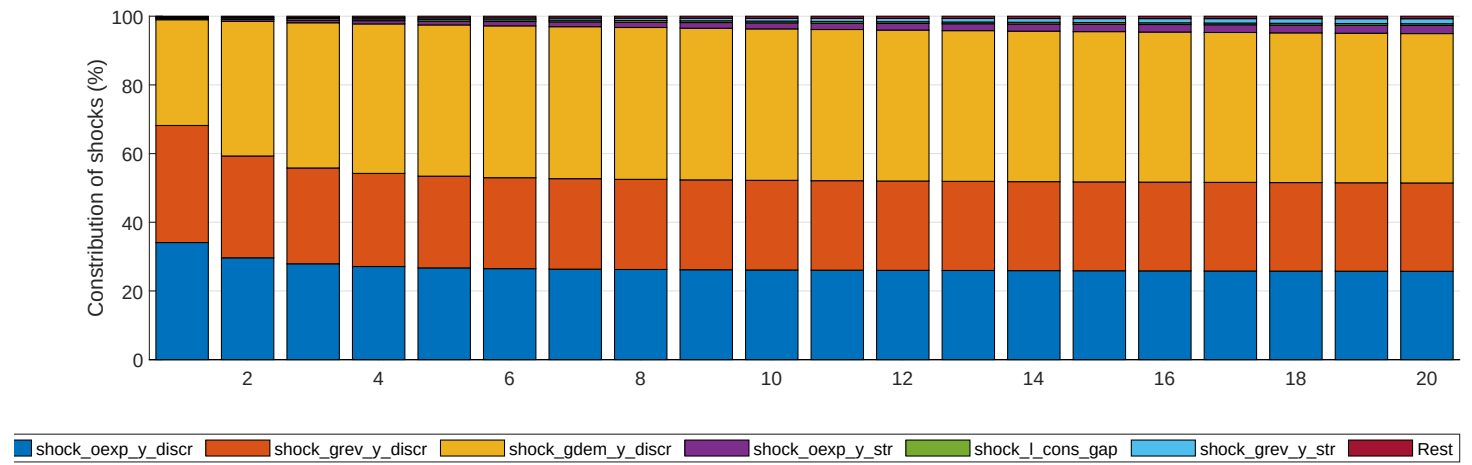


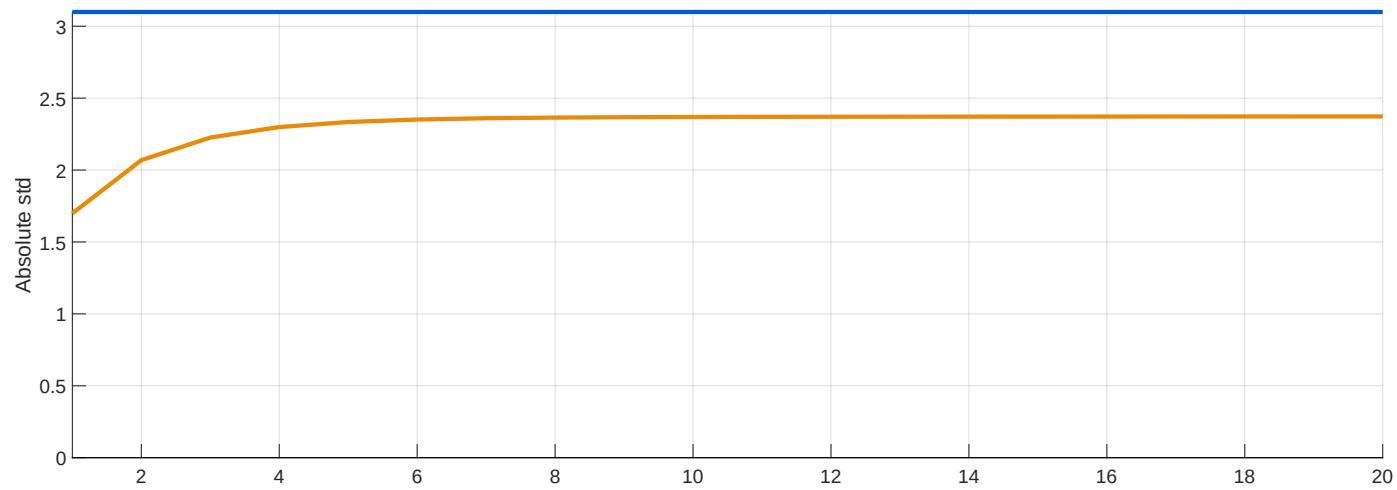
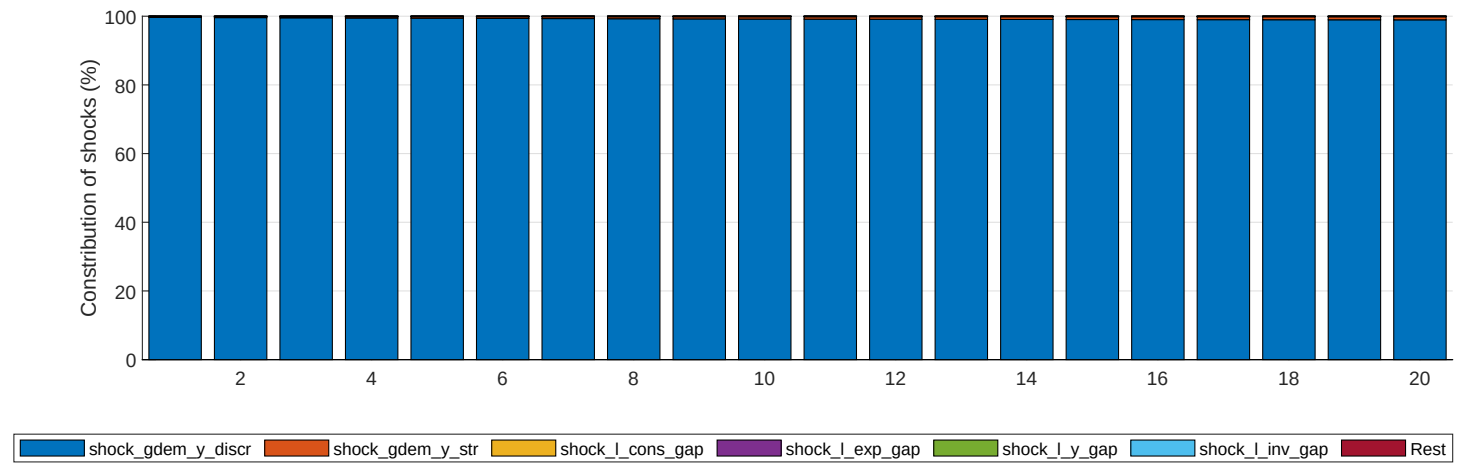
Responses to Foreign rel. ener. price gap shock, % [shock_1_rp_enerstar_gap]



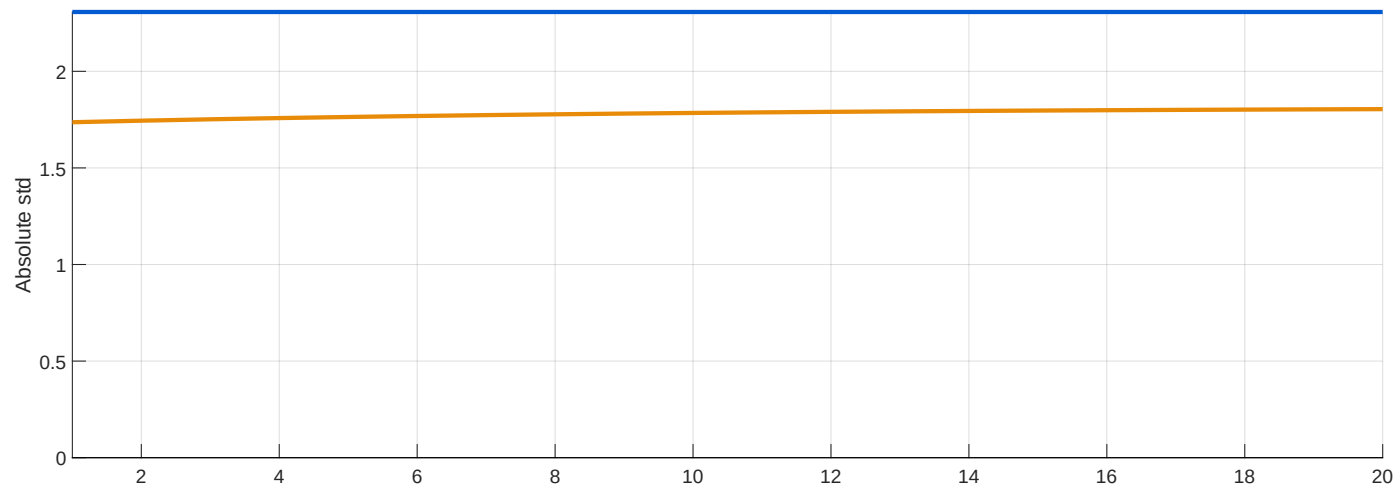
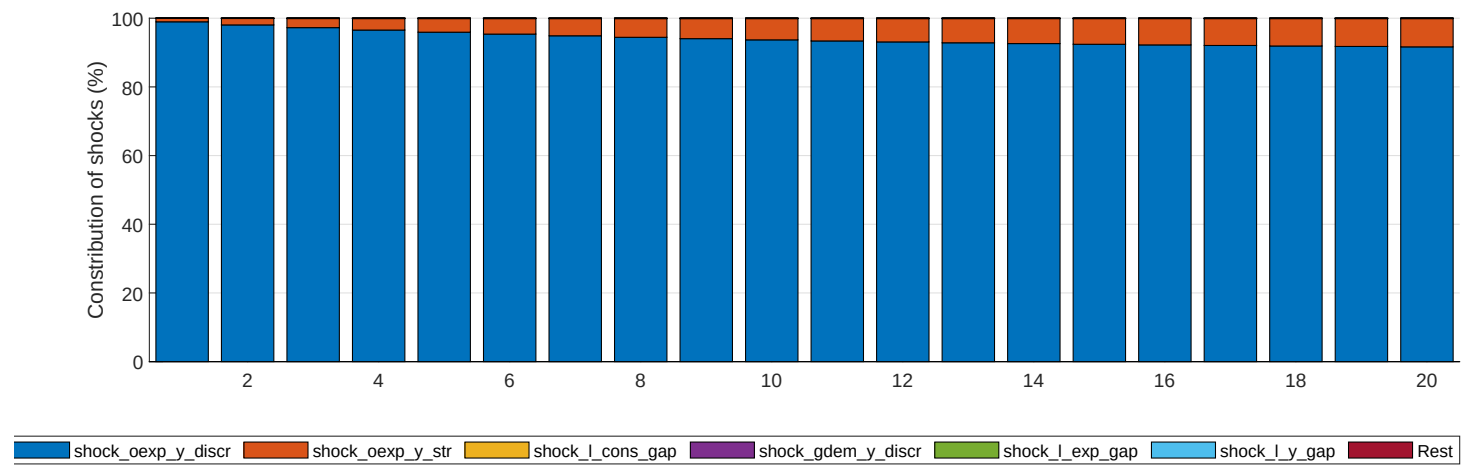
3 Variance decomposition

Budget deficit, % of GDP [def_y]

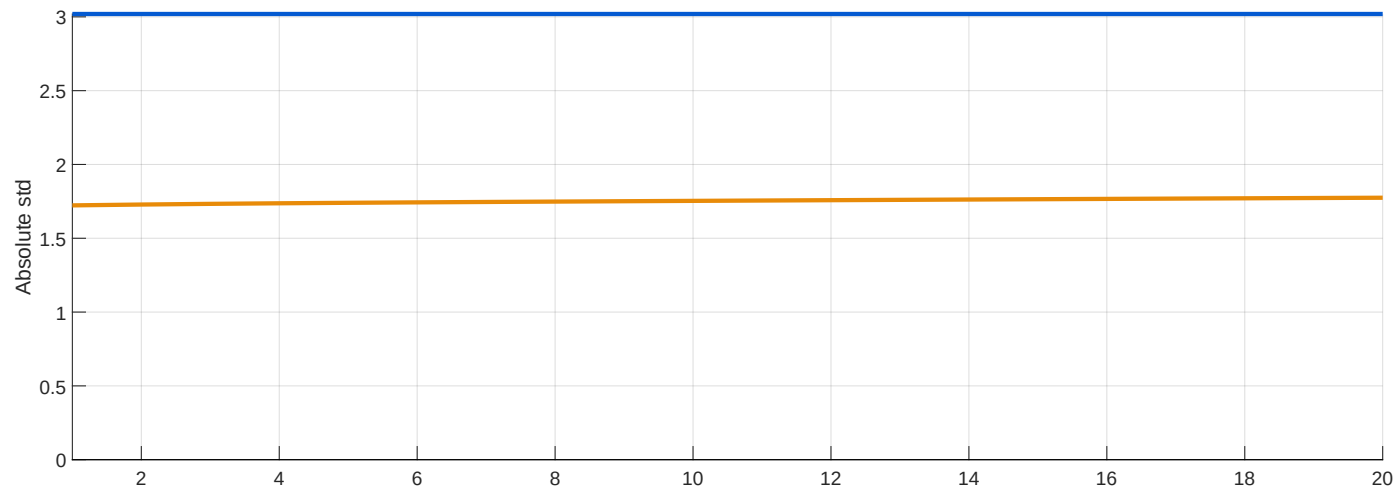
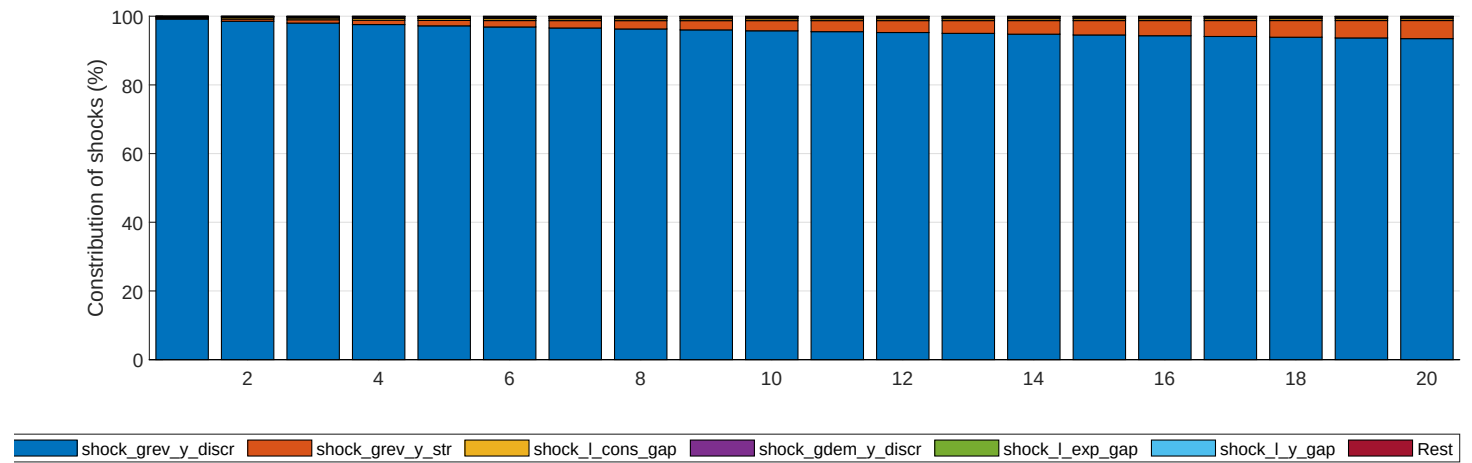


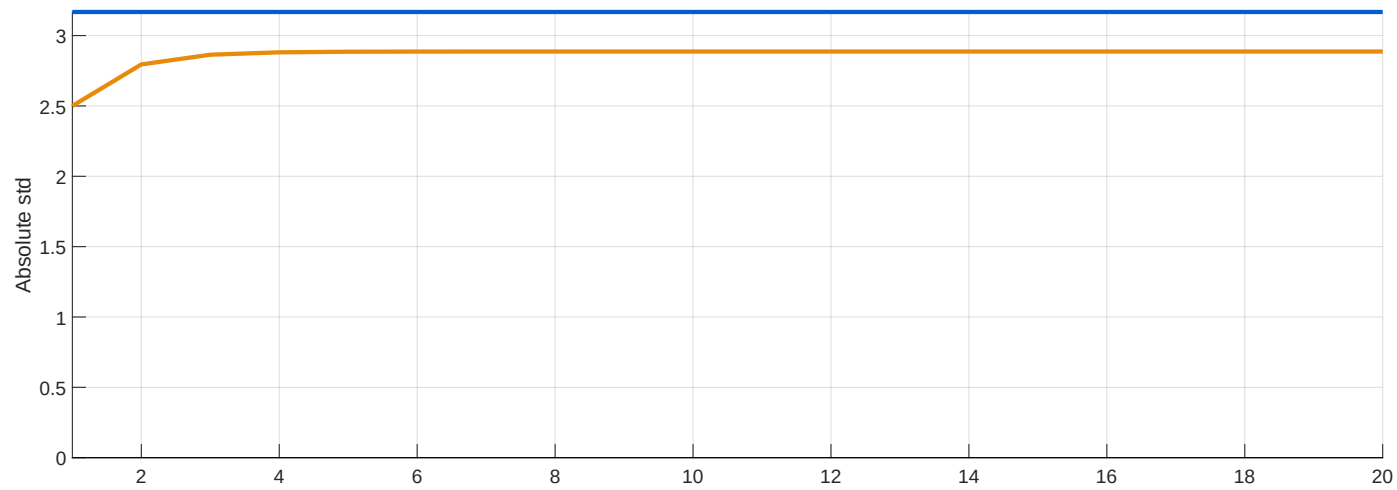
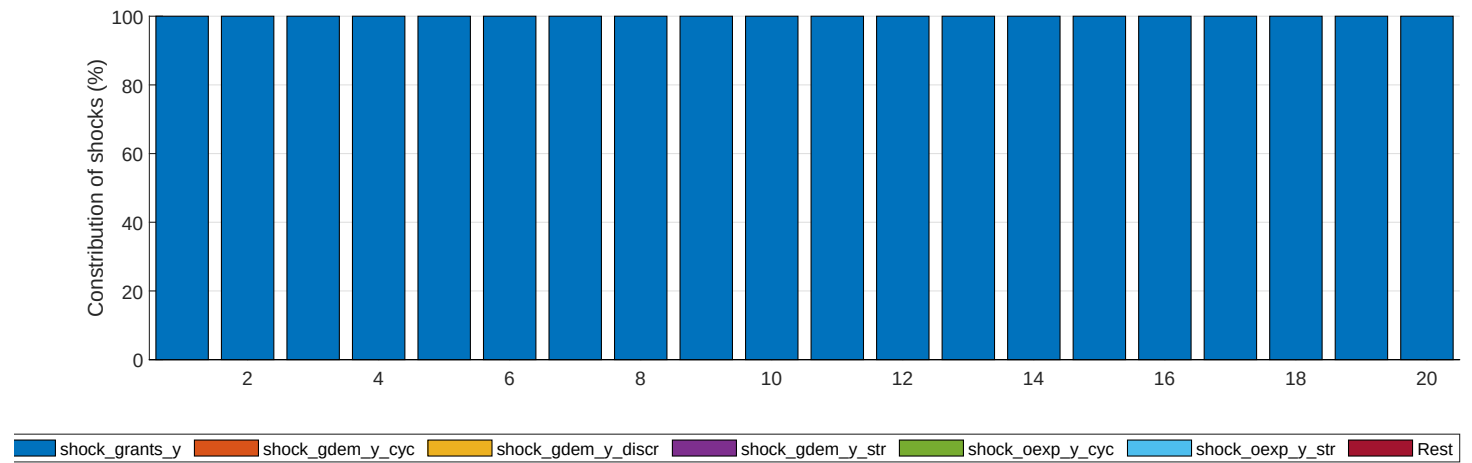
Govt. demand (G&S), % GDP [gdem_y]

Other govt. exp., % to GDP [oexp_y]

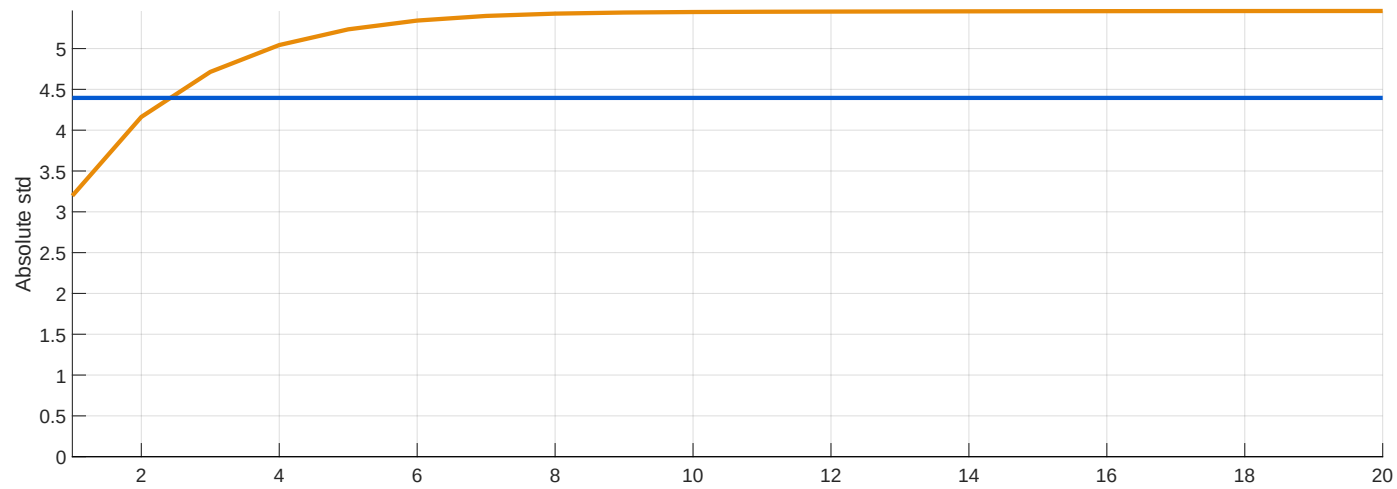
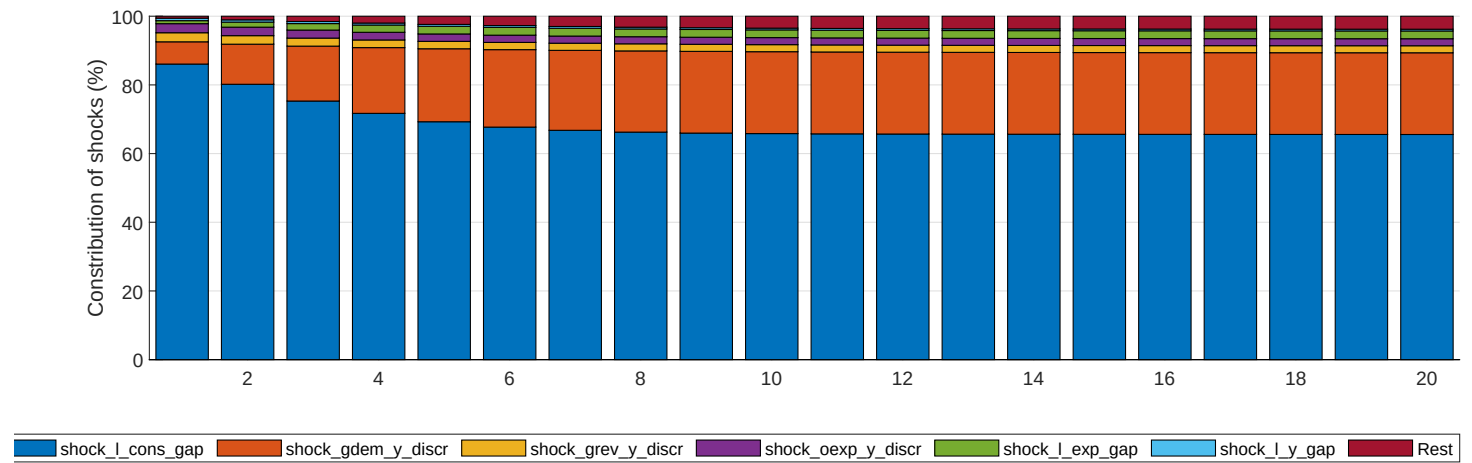


Govt. rev., % of GDP [grev_y]

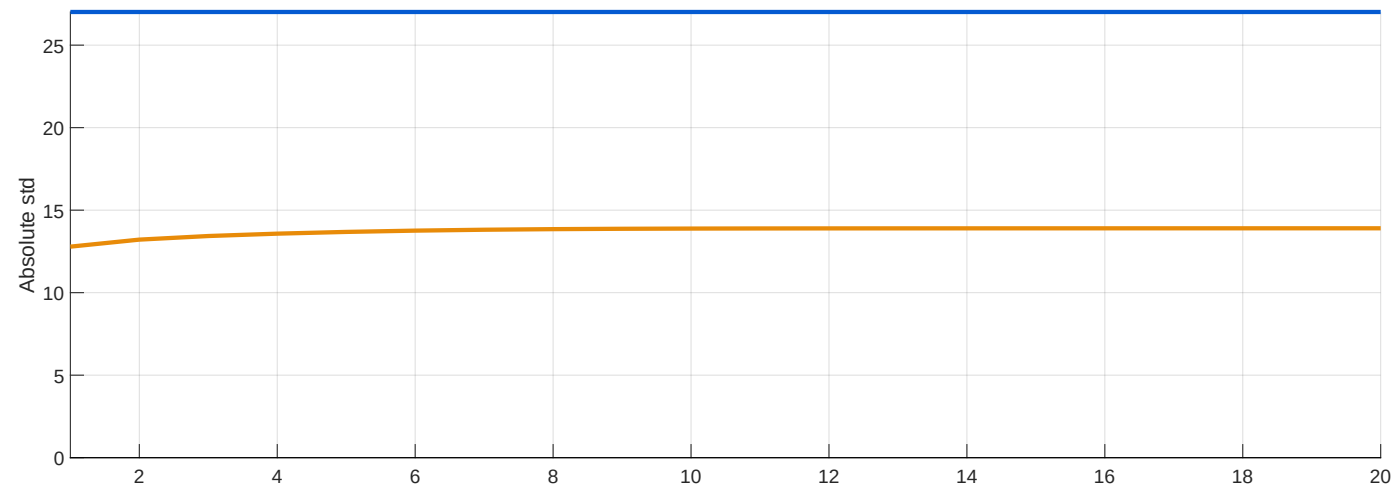
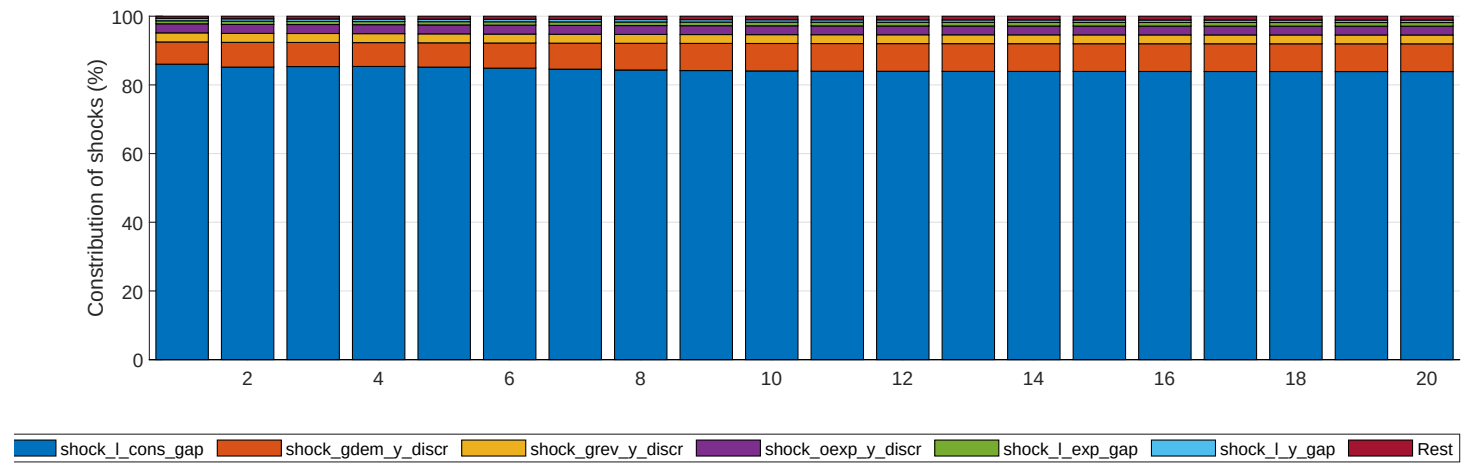


Fiscal grants, % of GDP [grants_y]

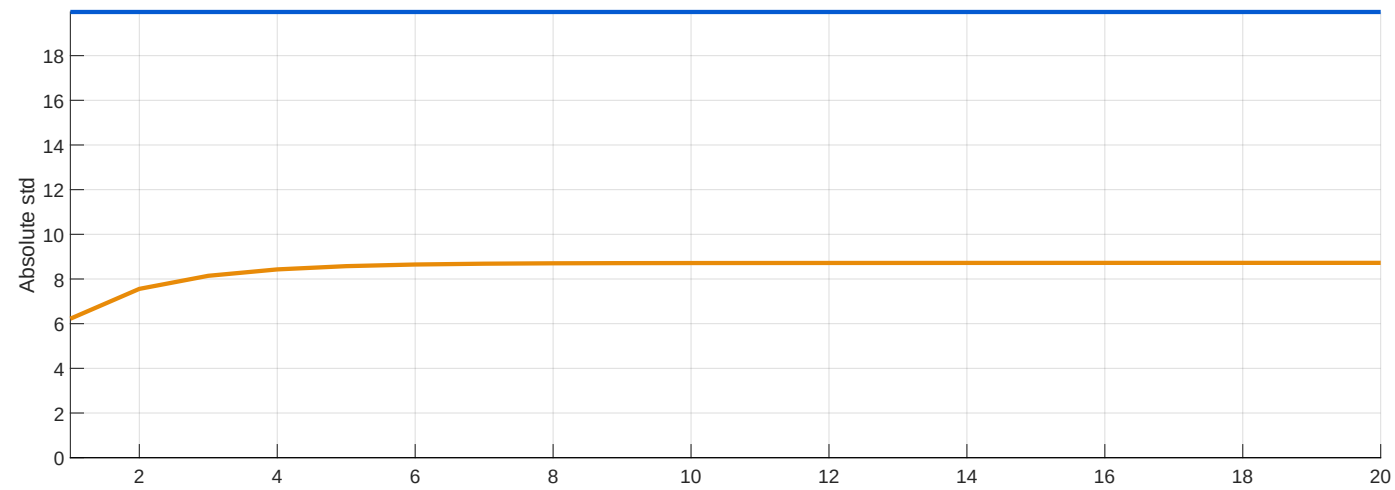
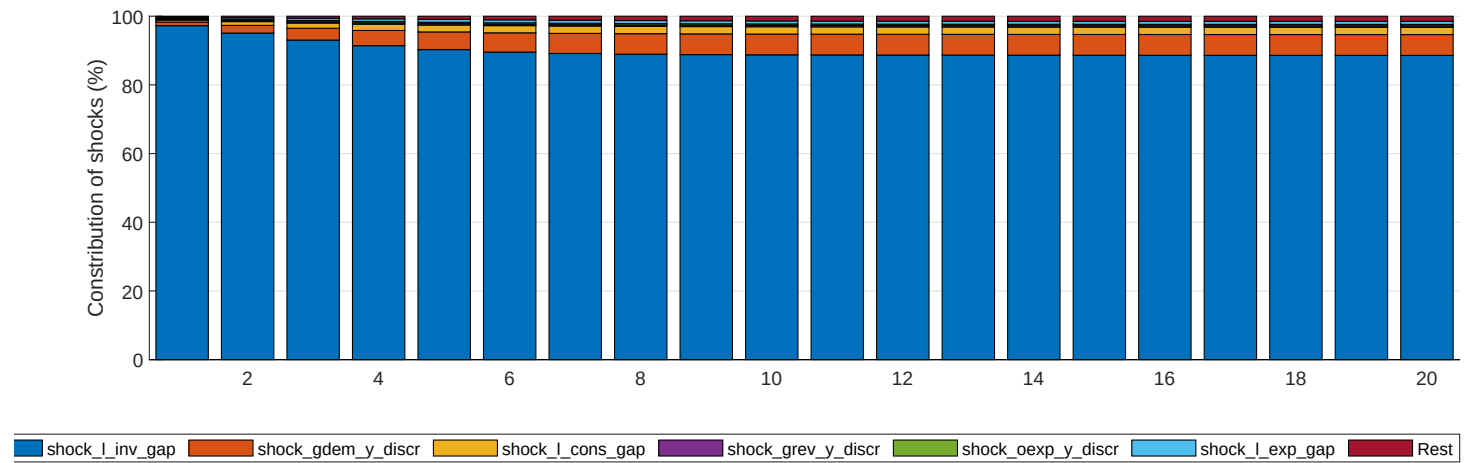
Consumption gap, % [l_cons_gap]



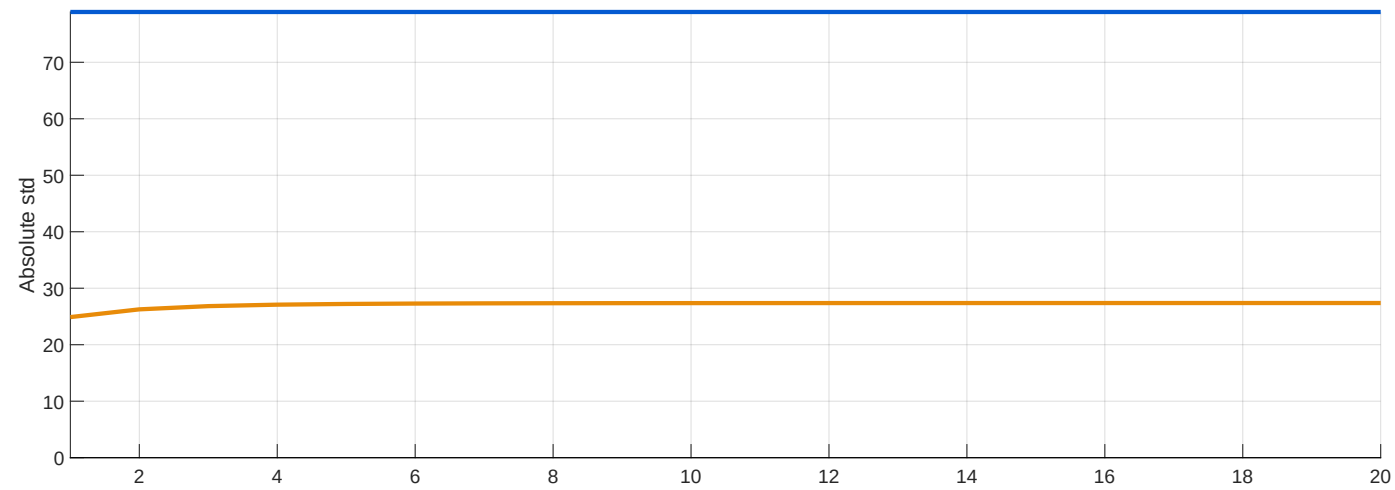
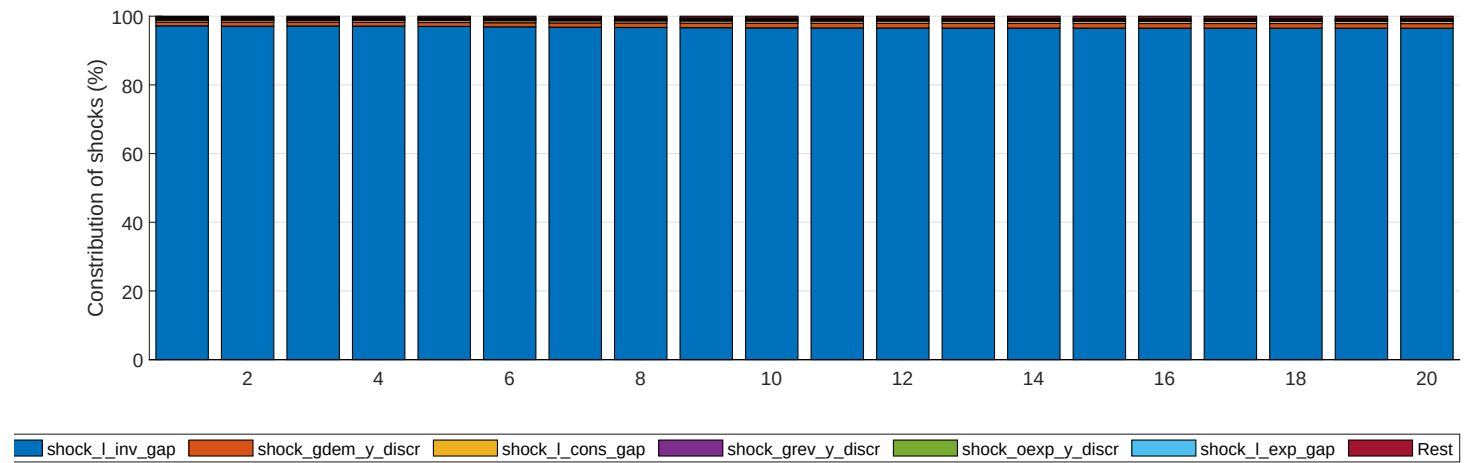
Consumption, ann. QQ % [dl_cons]



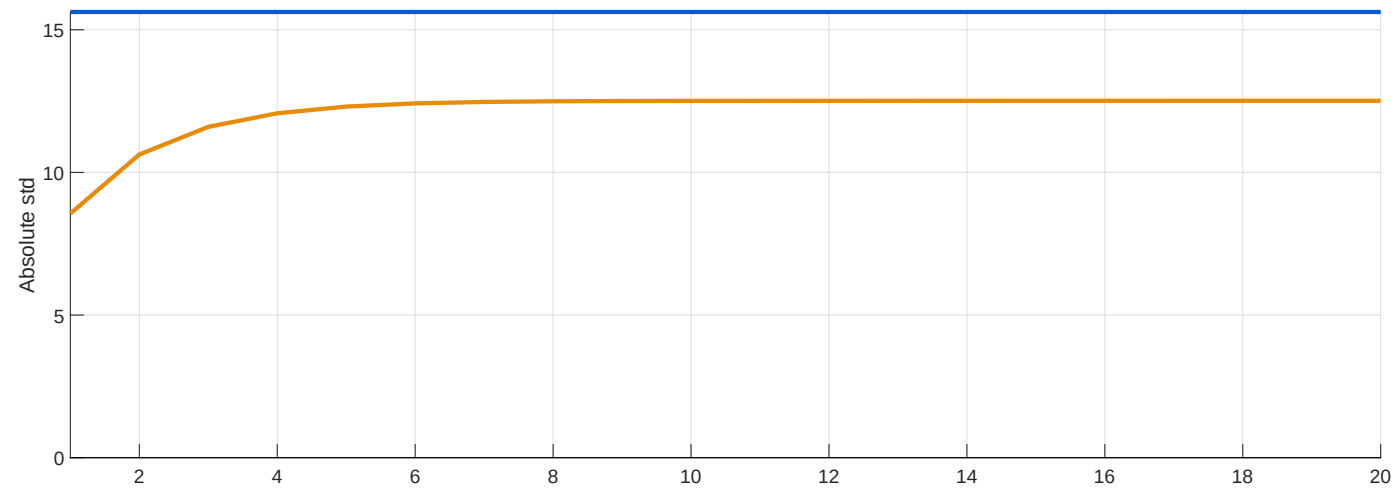
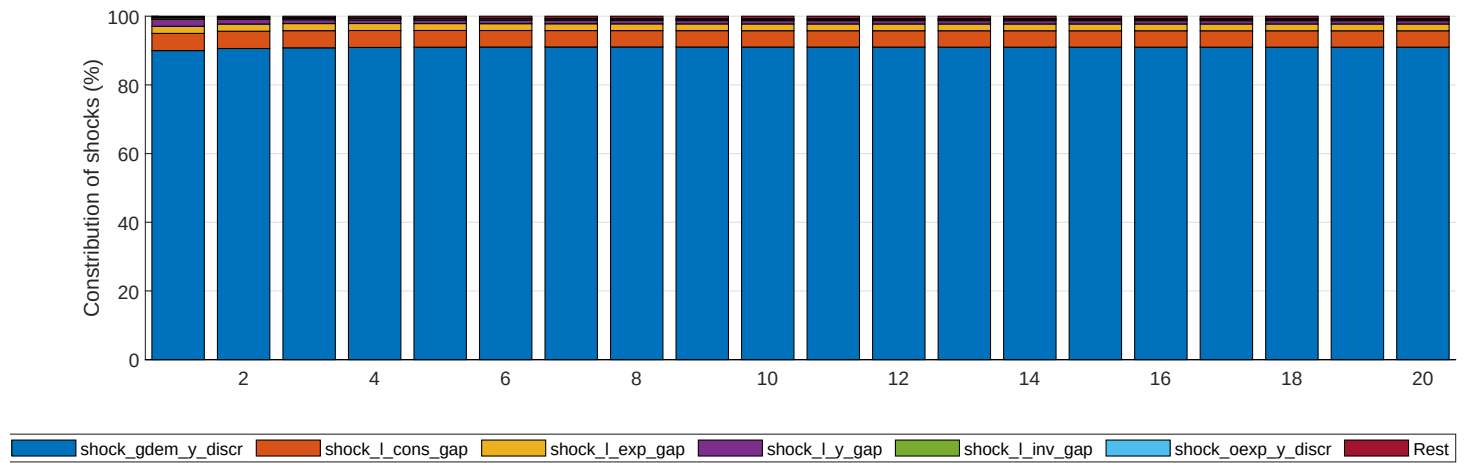
Investment gap, % [l_inv_gap]



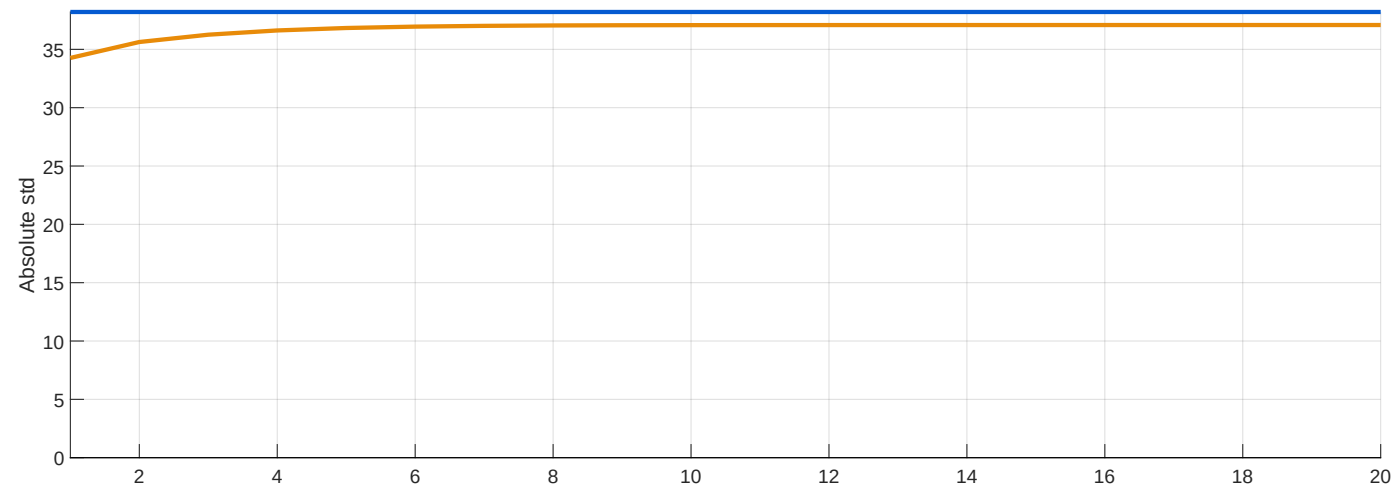
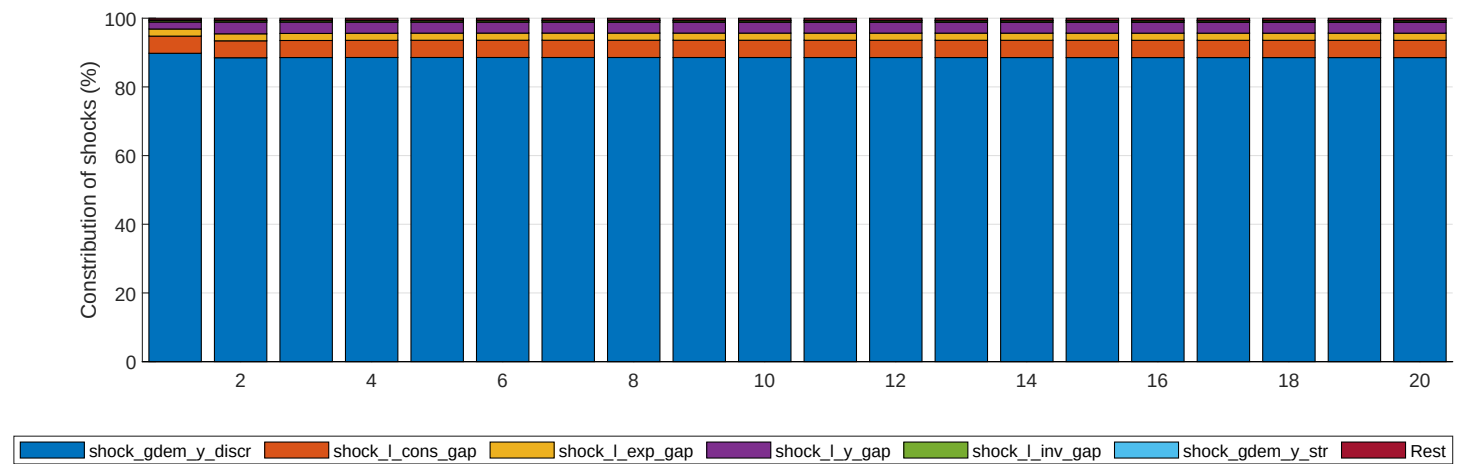
Investment, ann. QQ % [dl_inv]



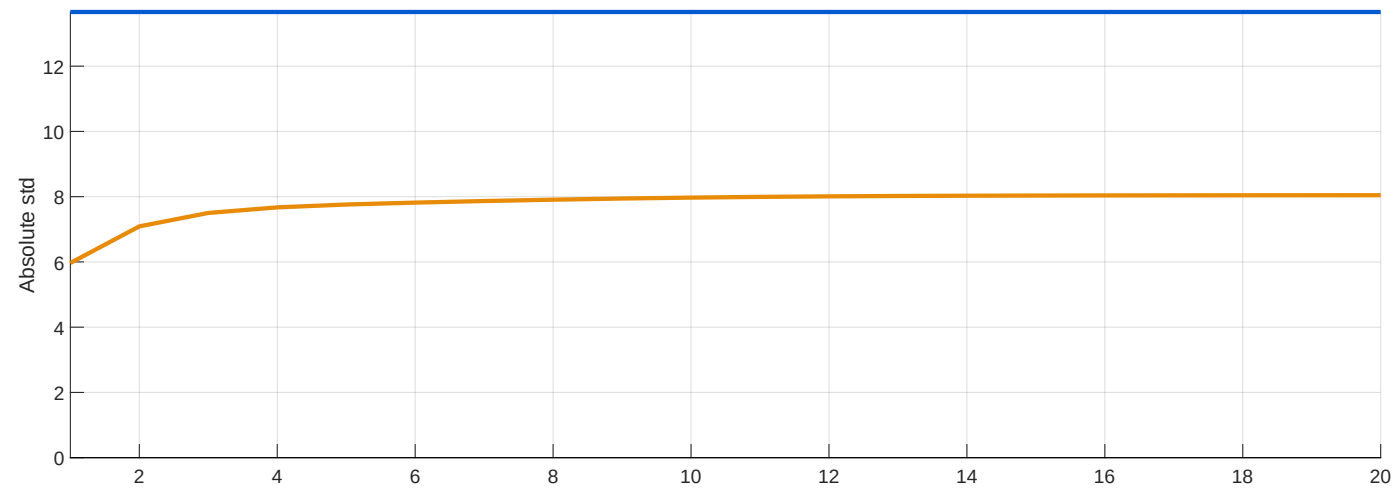
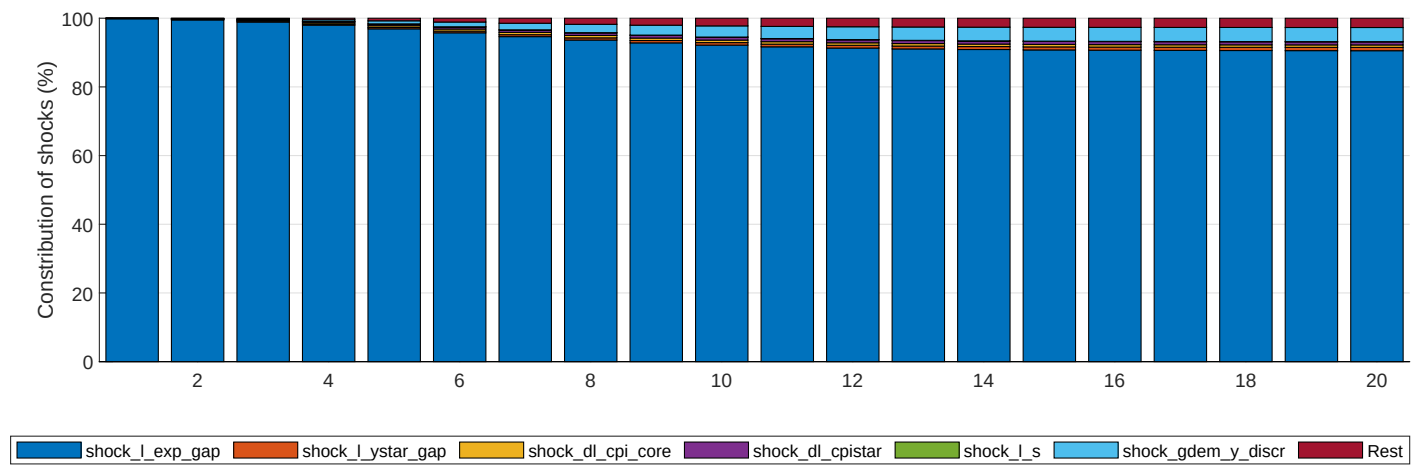
Gov. demand gap, % [l_gdem_gap]

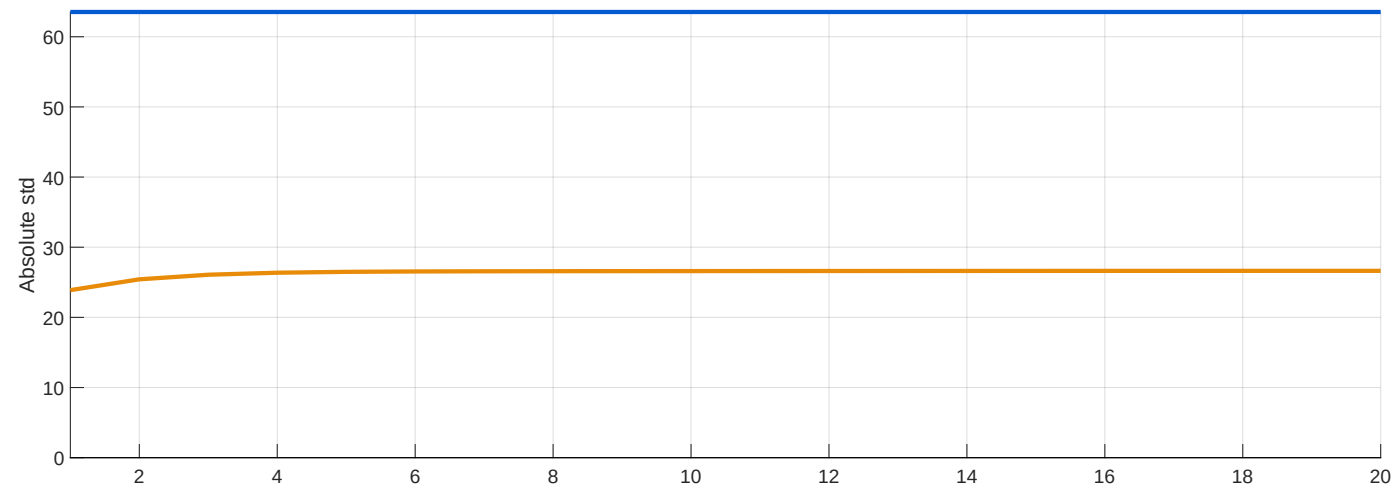
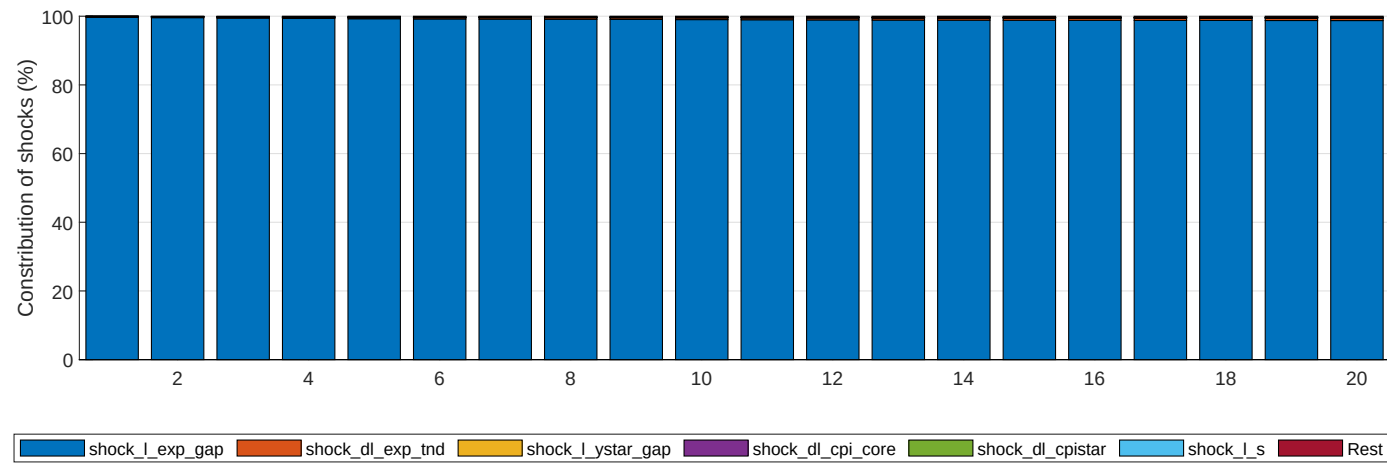


Gov. demand, ann. QQ % [dl_gdem]

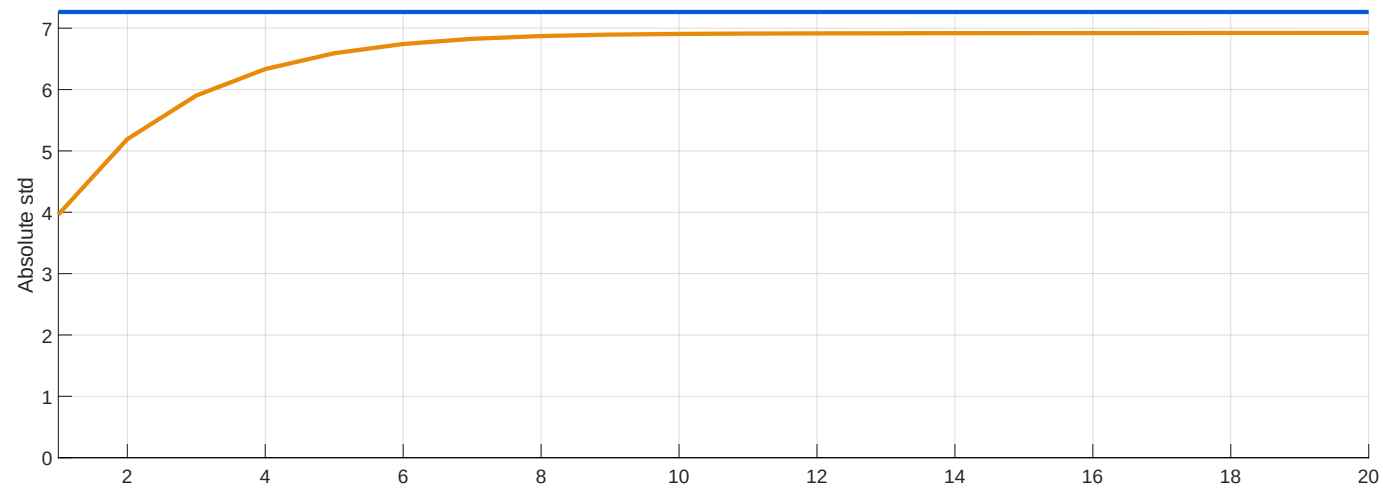
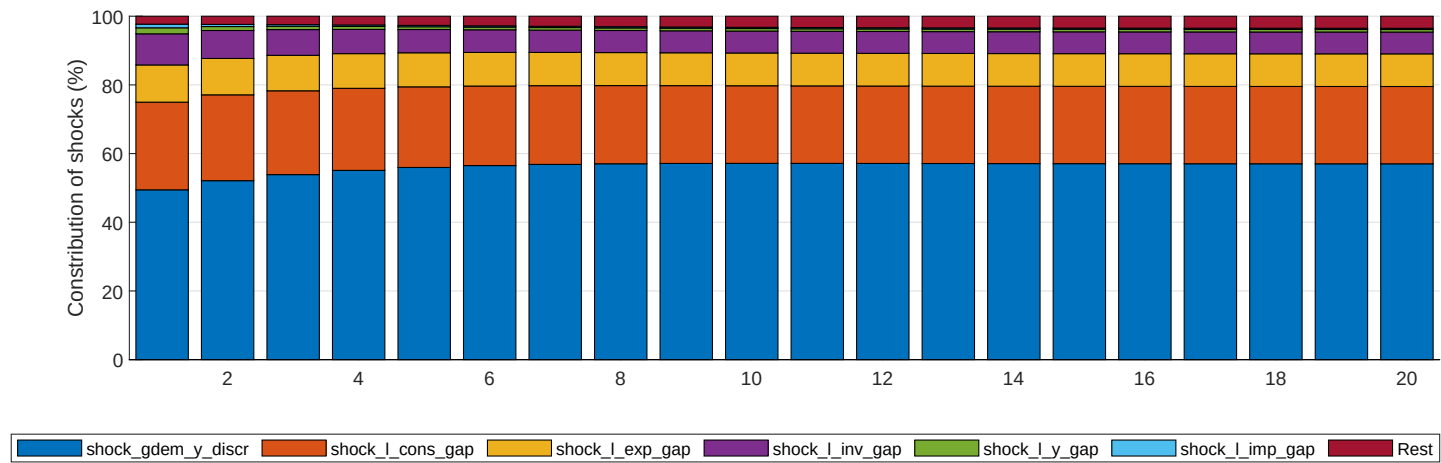


Export gap, % [l_exp_gap]

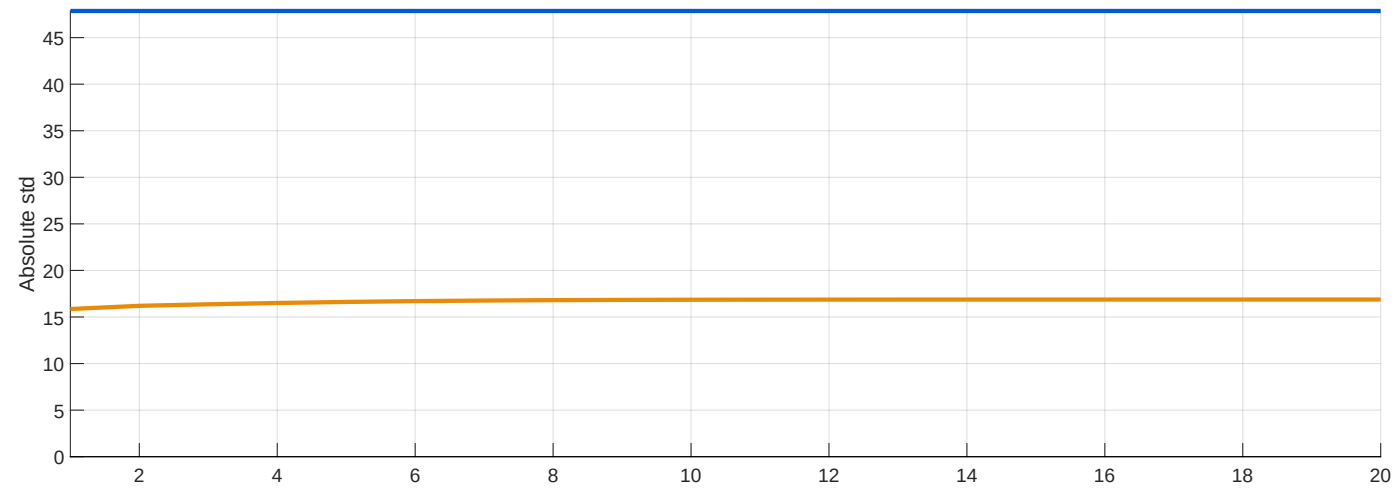
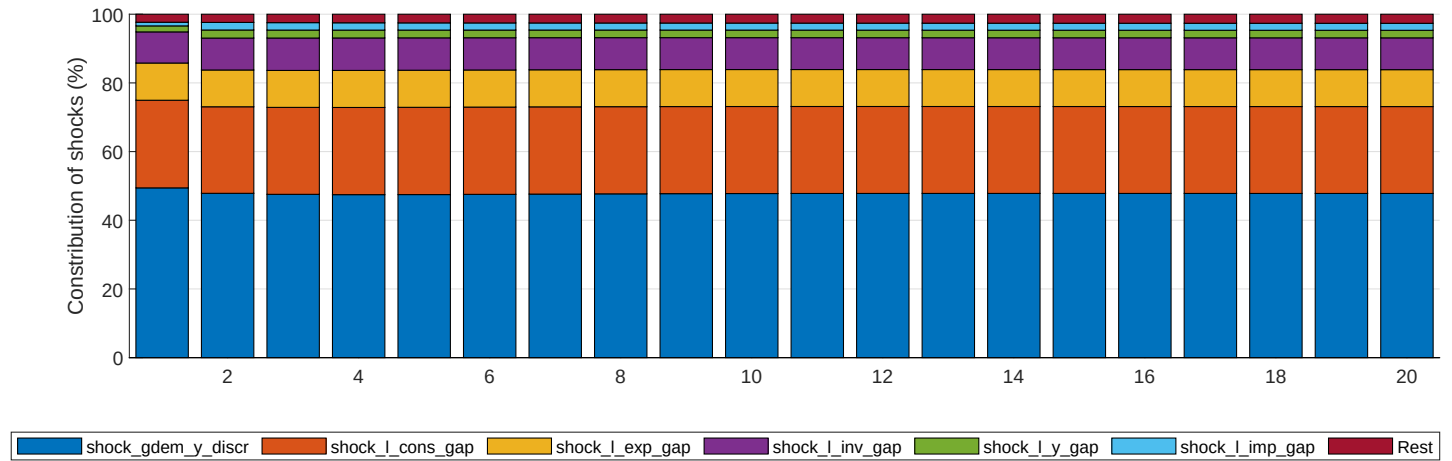


Export, ann. QQ % [dl_exp]

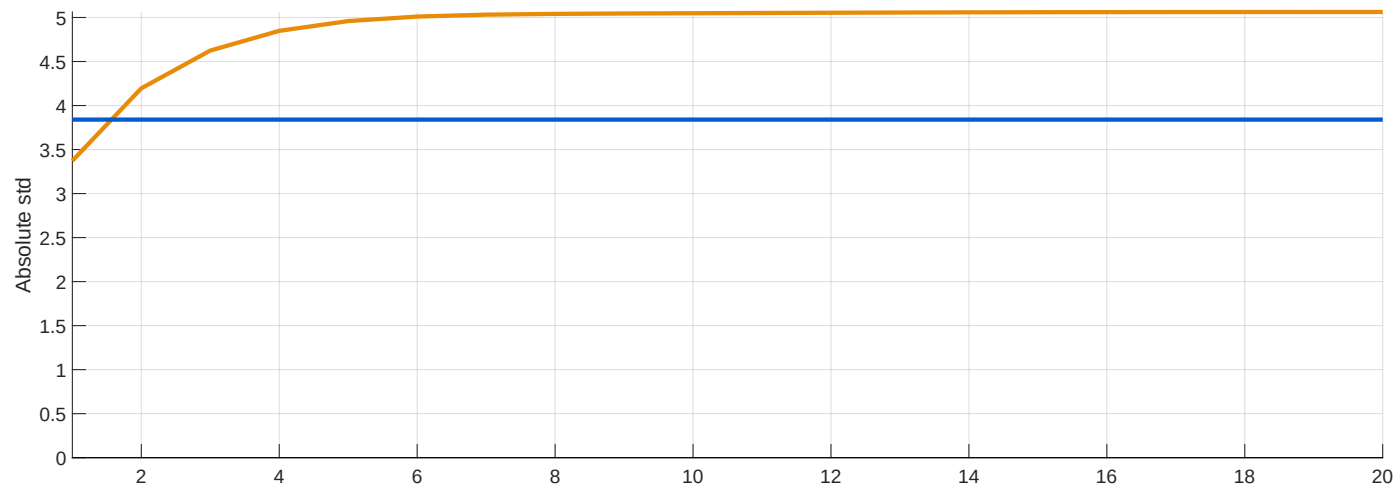
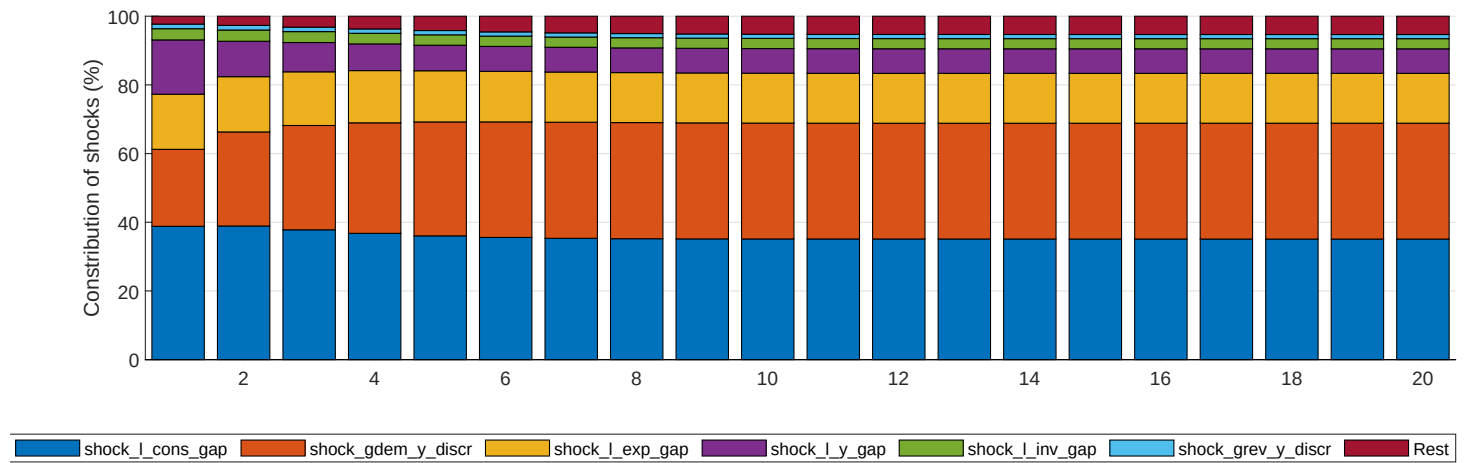
Import gap, % [l_imp_gap]



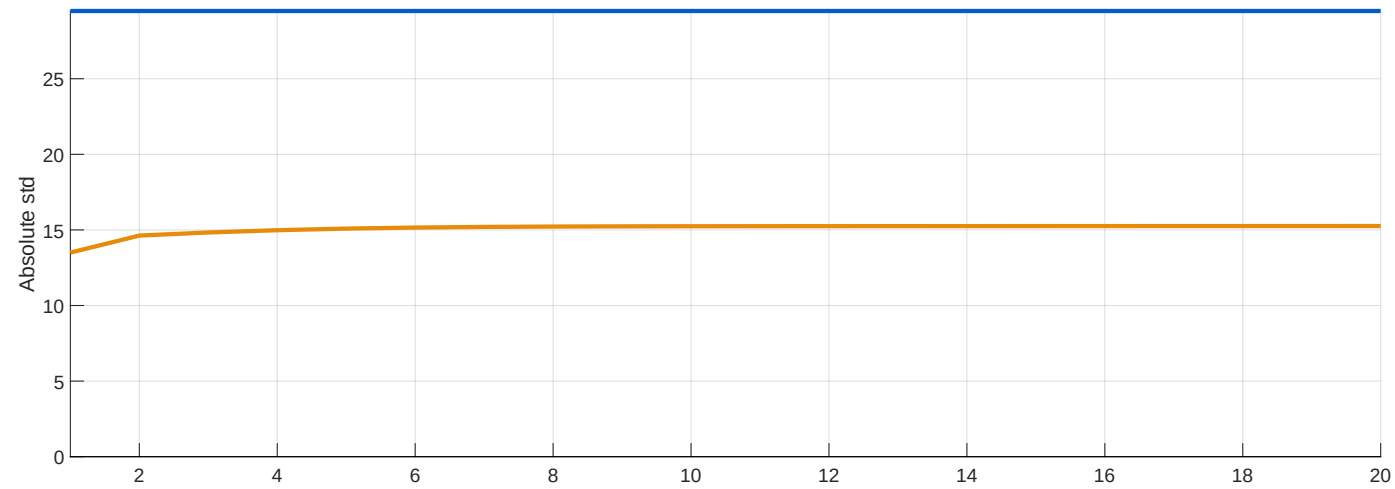
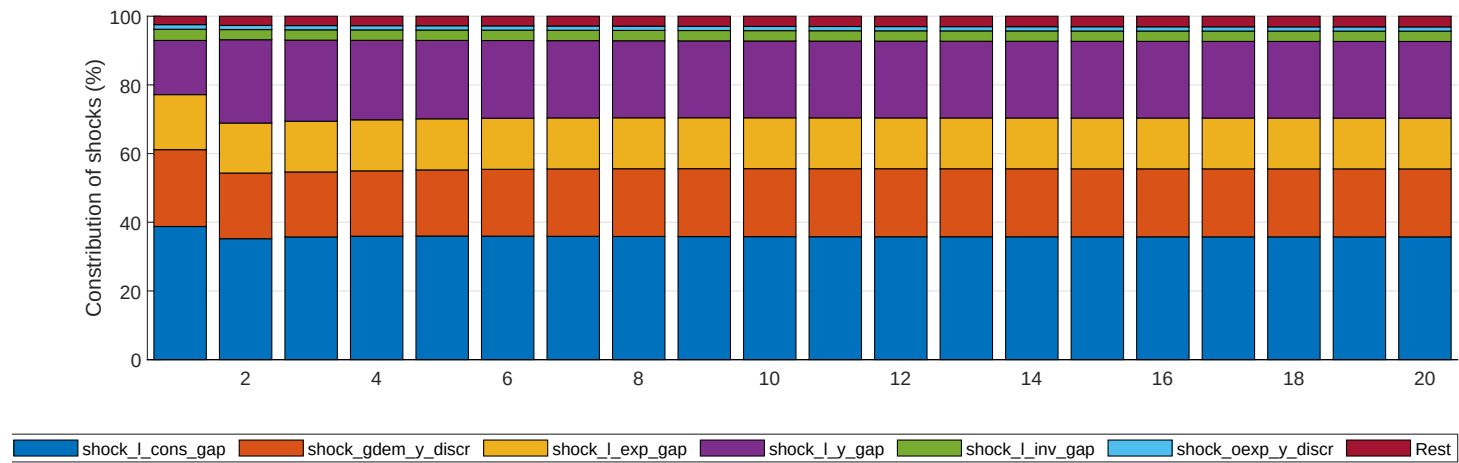
Import, ann. QQ % [dl_imp]



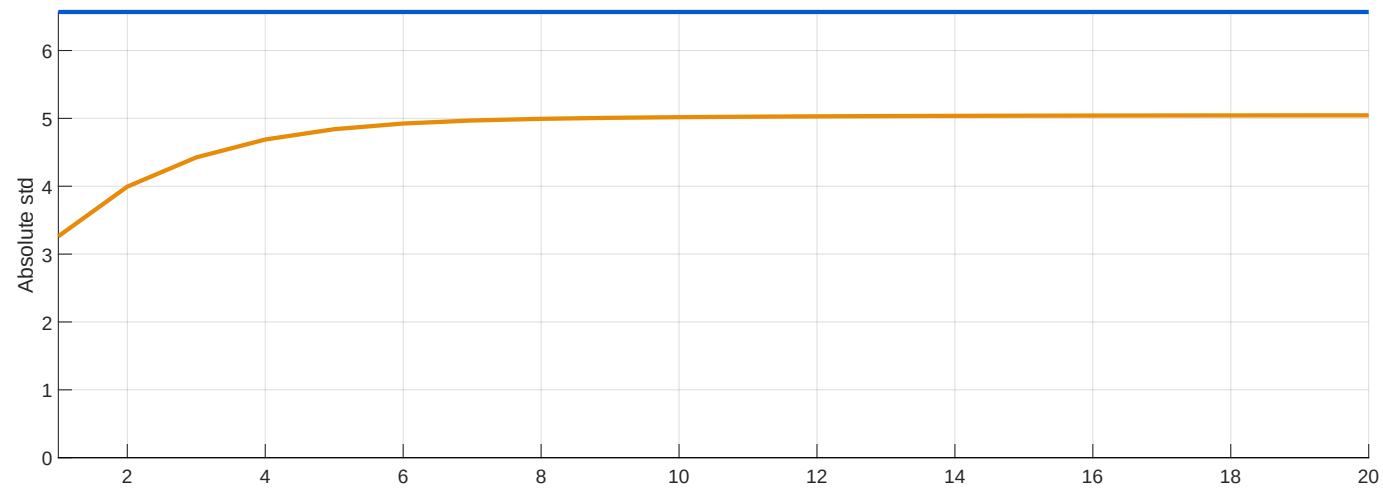
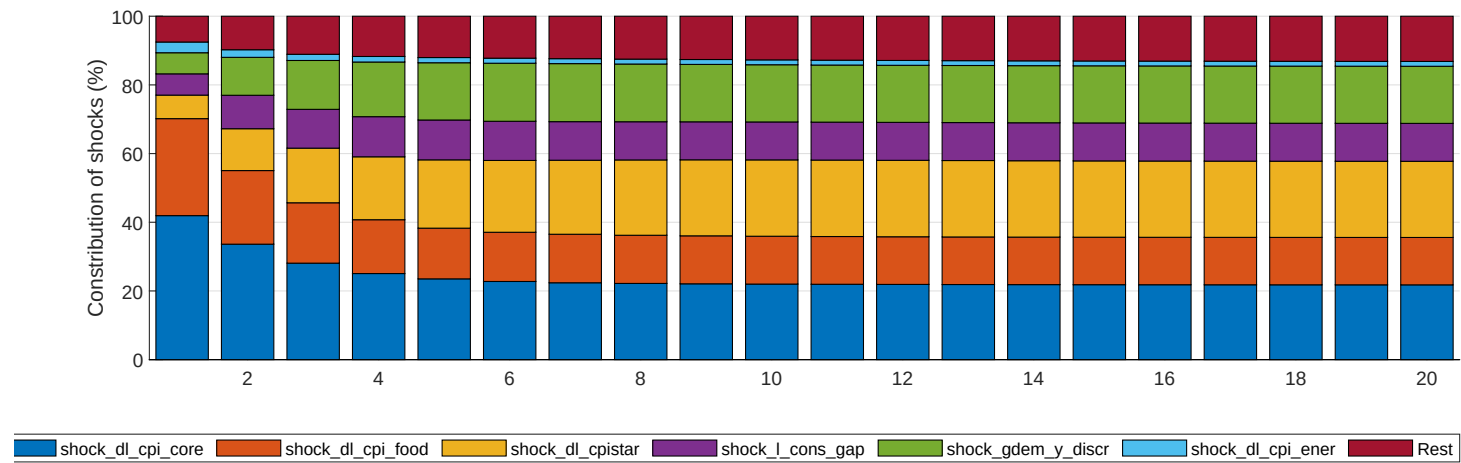
Output gap, % [l_y_gap]



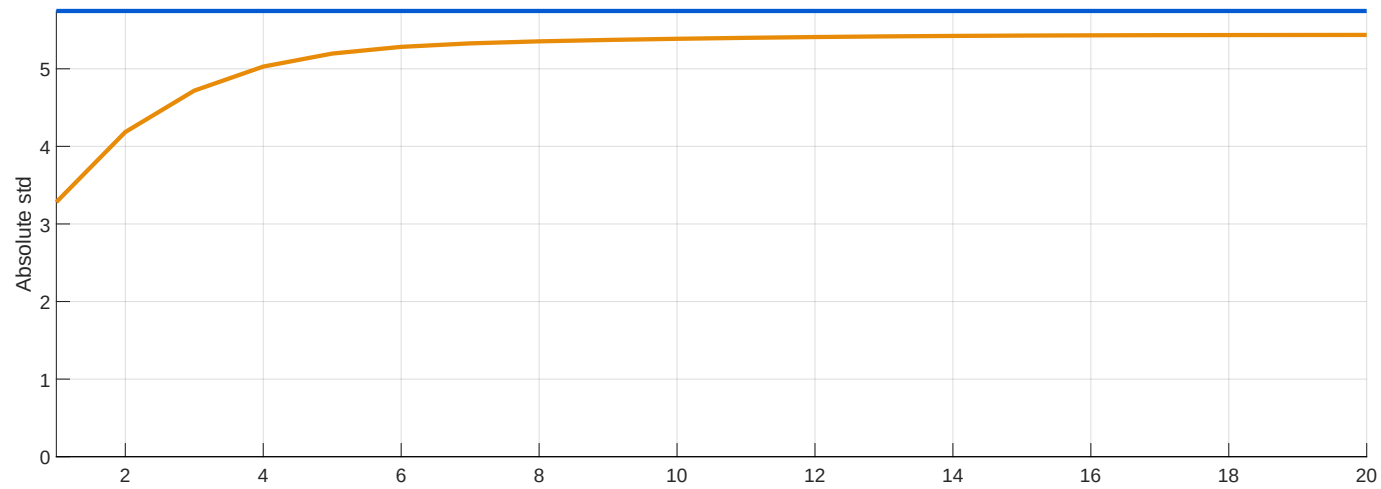
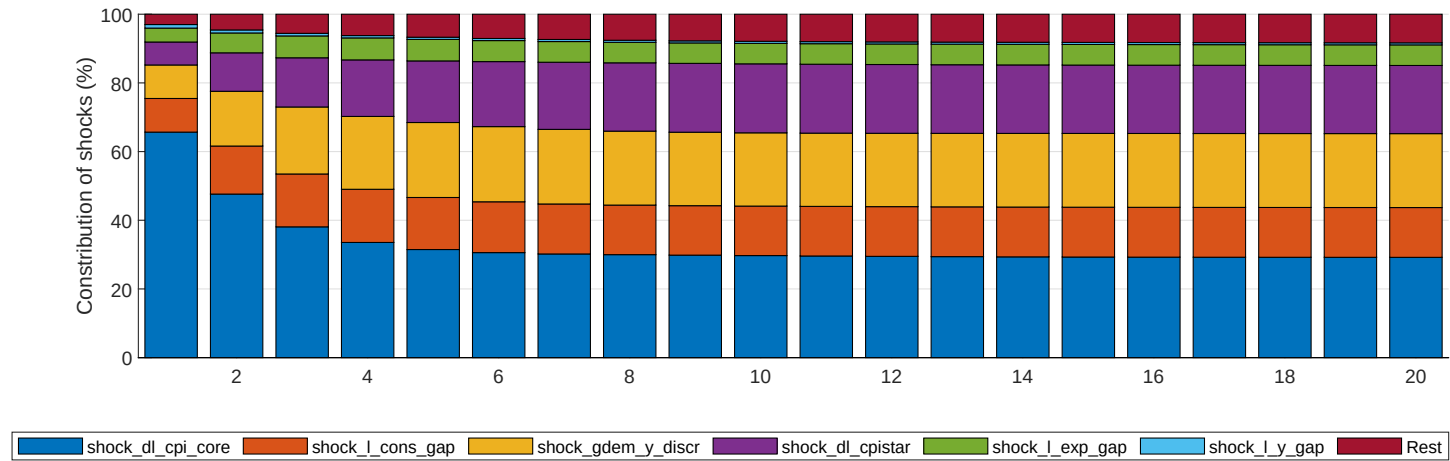
GDP, ann. QQ % [dl_y]



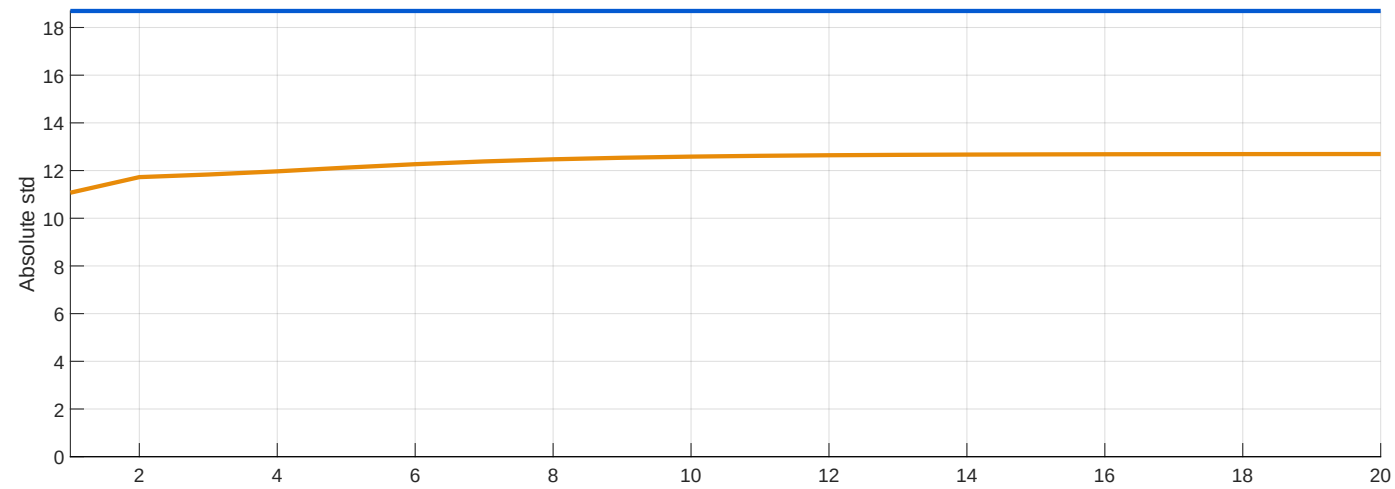
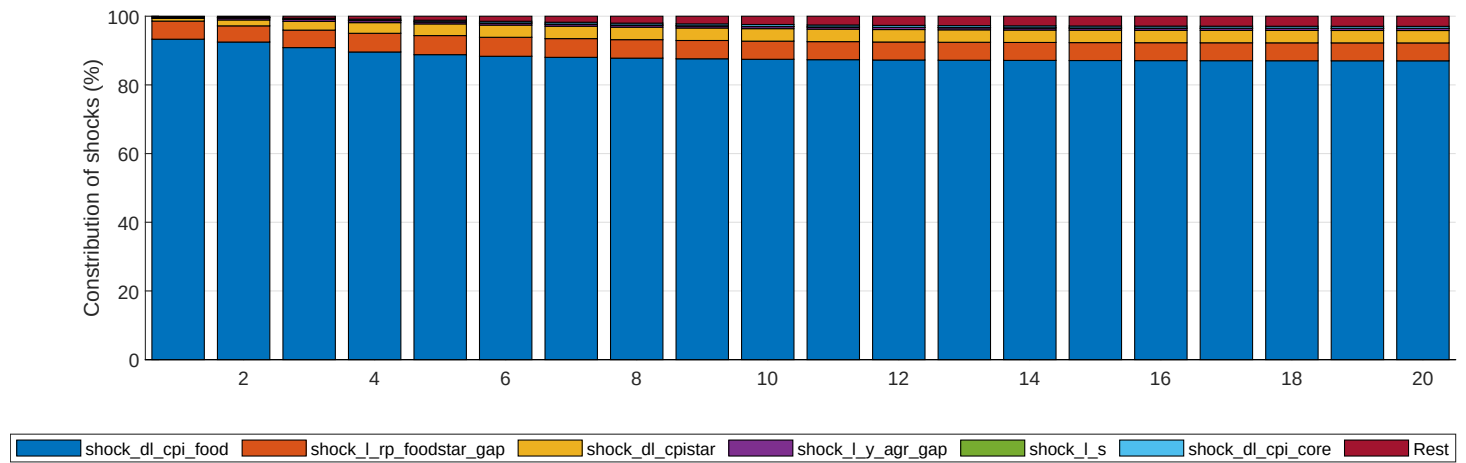
Headline CPI, ann. QQ % [dl_cpi]



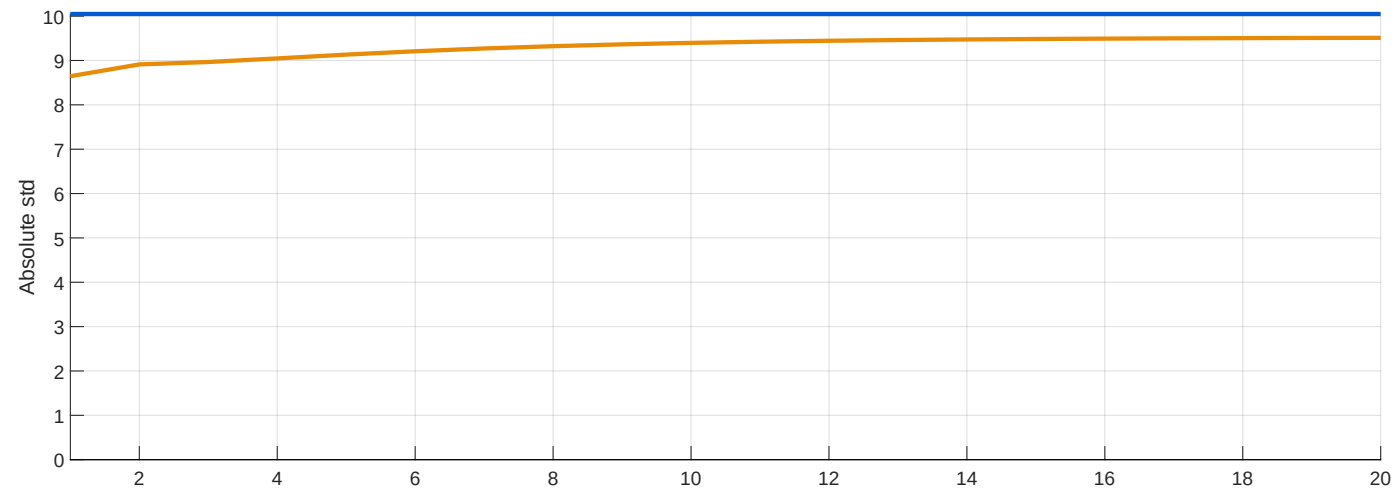
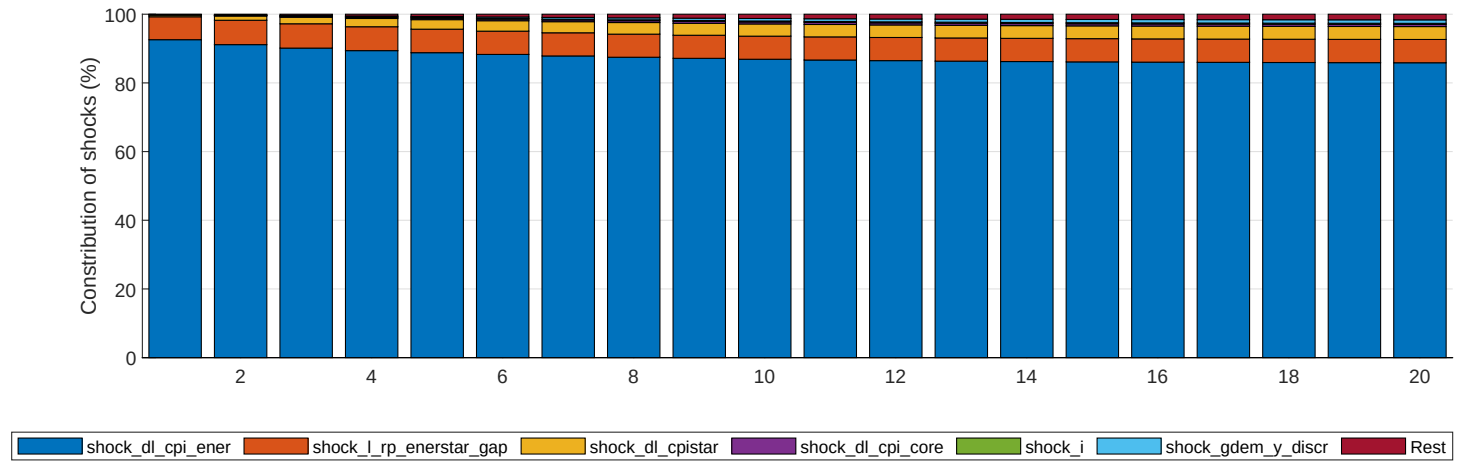
Core CPI, ann. QQ % [dl_cpi_core]



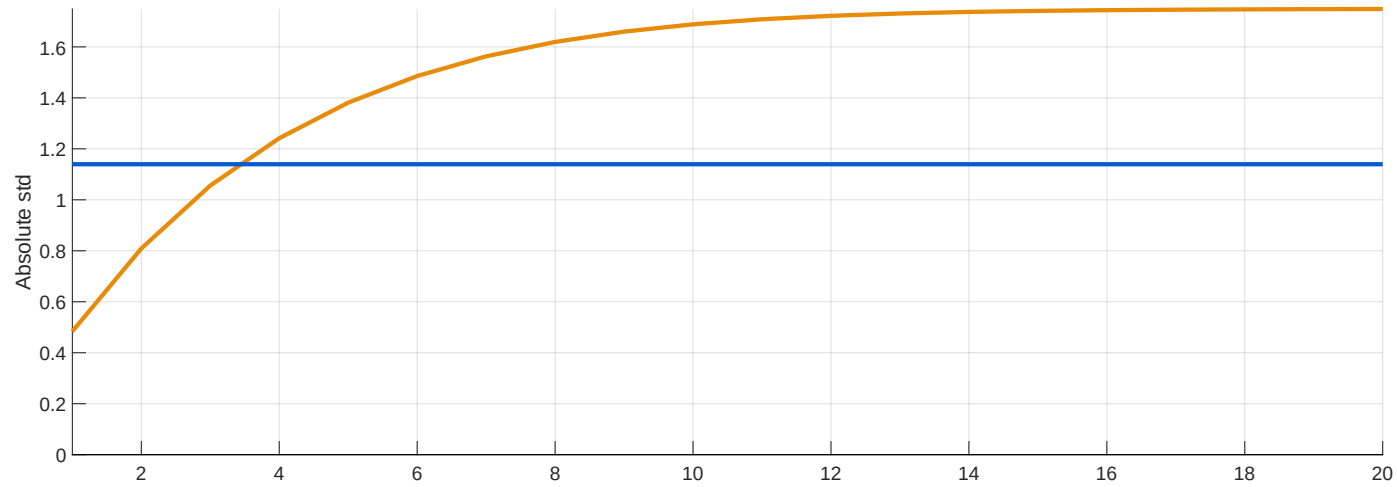
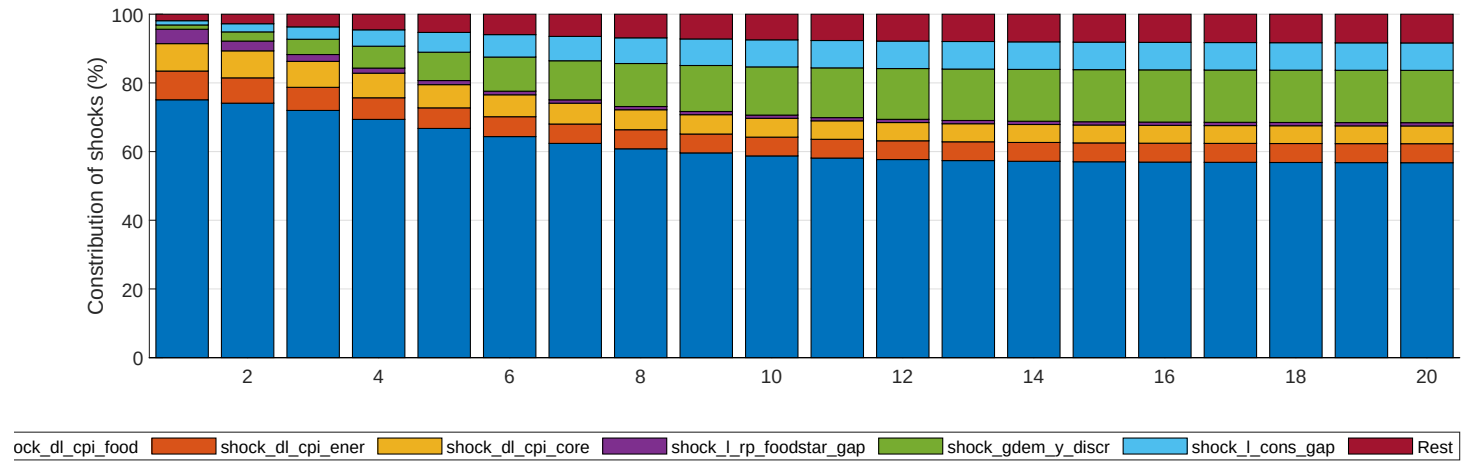
Food CPI, ann. QQ % [dl_cpi_food]



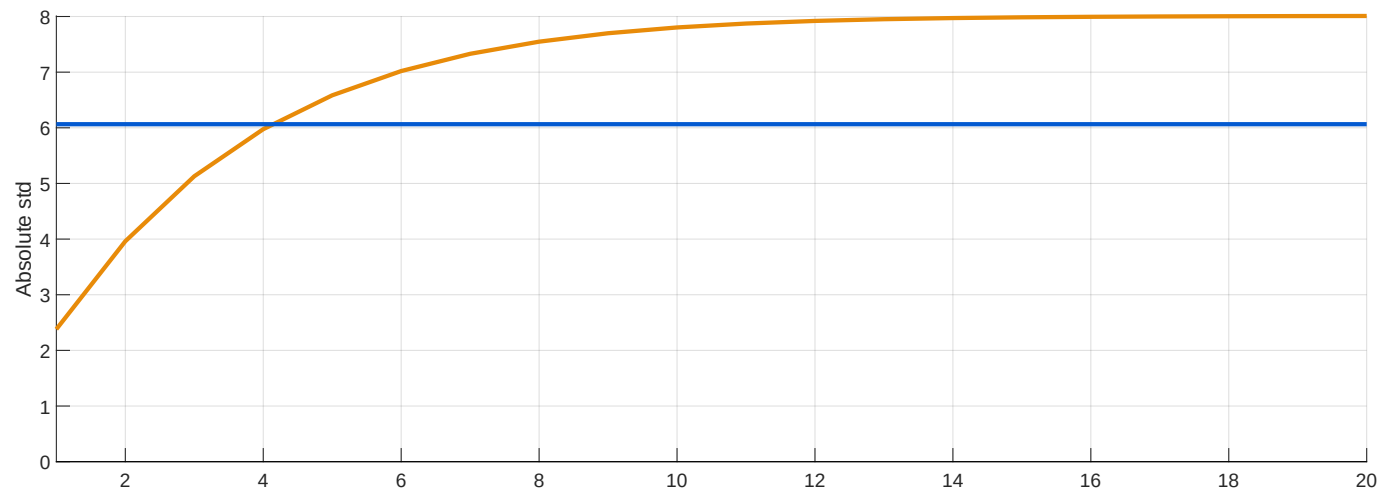
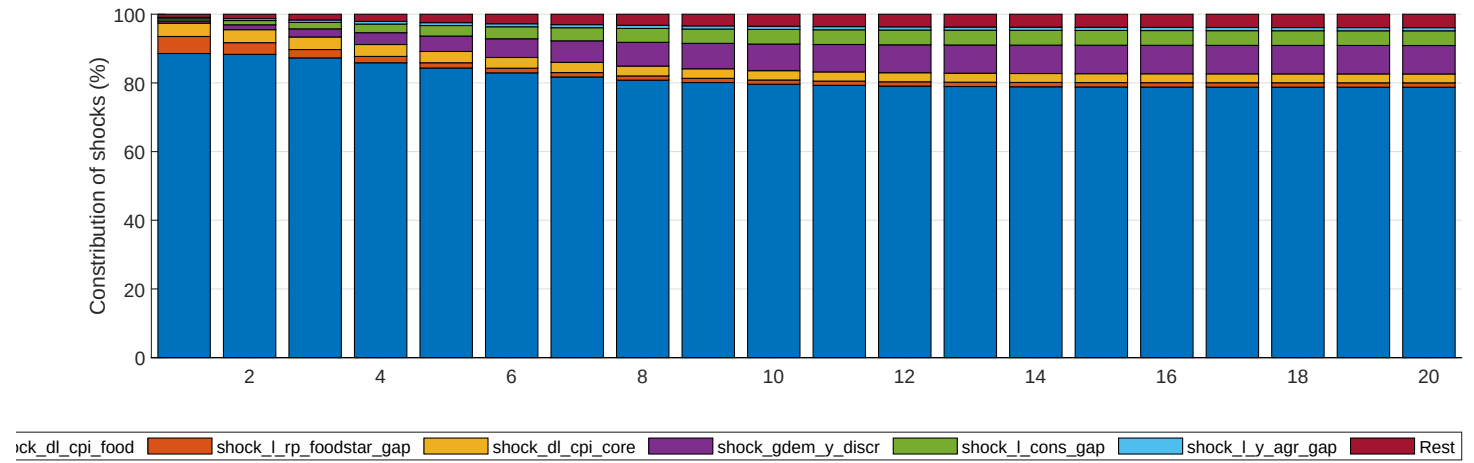
Energy CPI, ann. QQ % [dl_cpi_ener]



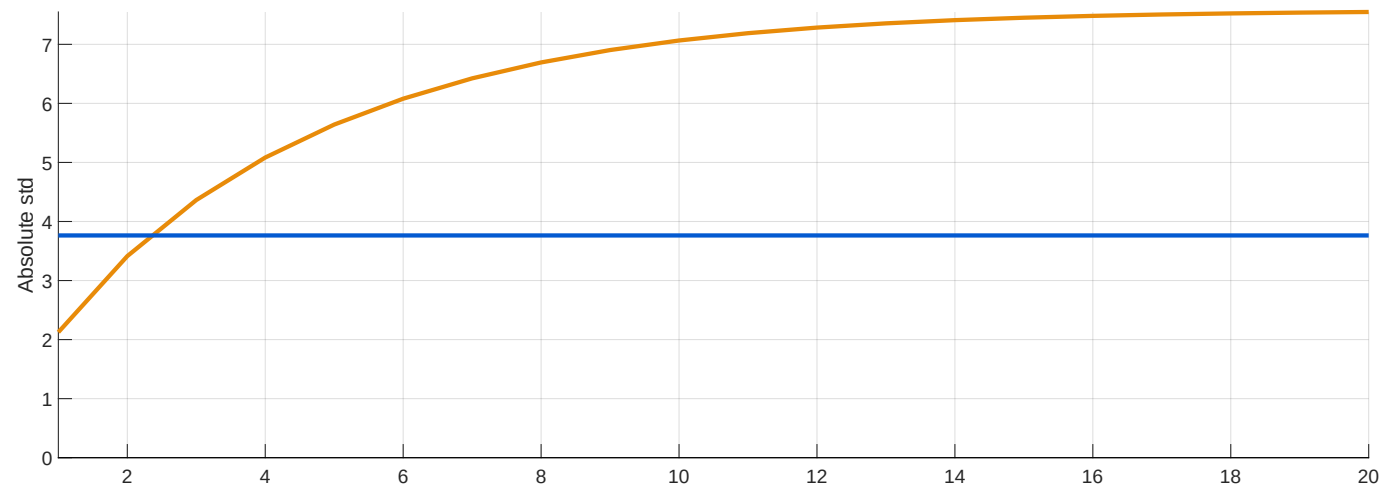
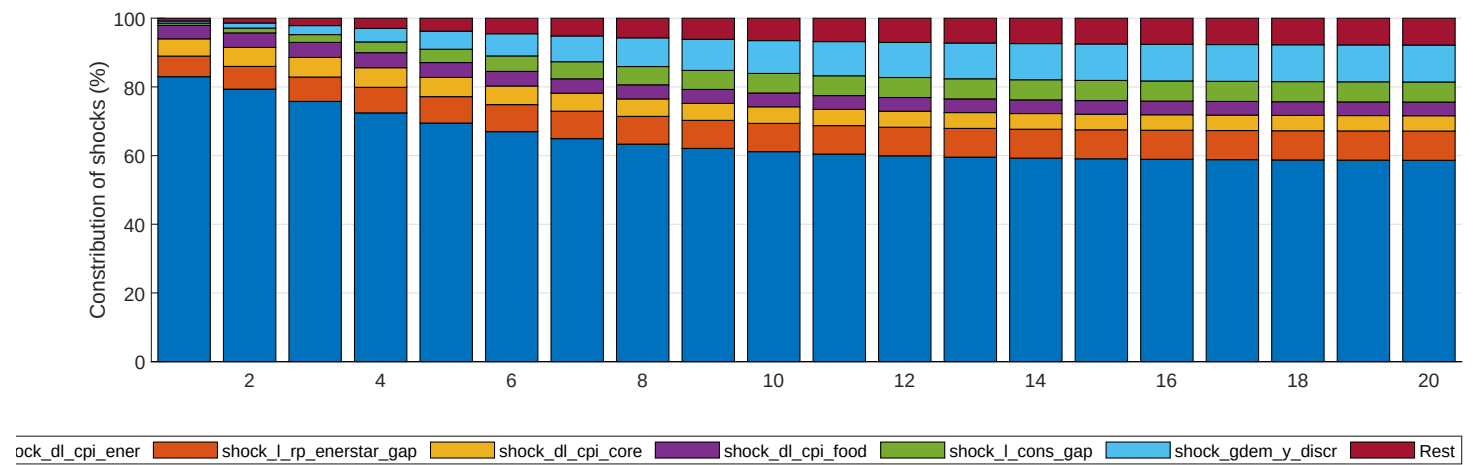
Core rel. price gap, % [l_rp_cpi_core_gap]



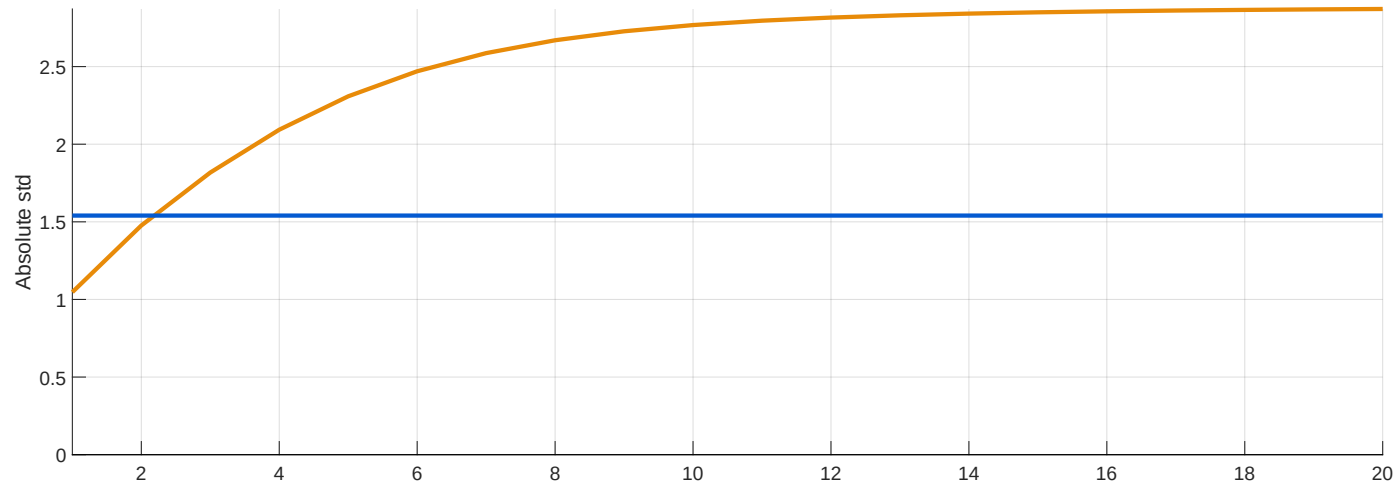
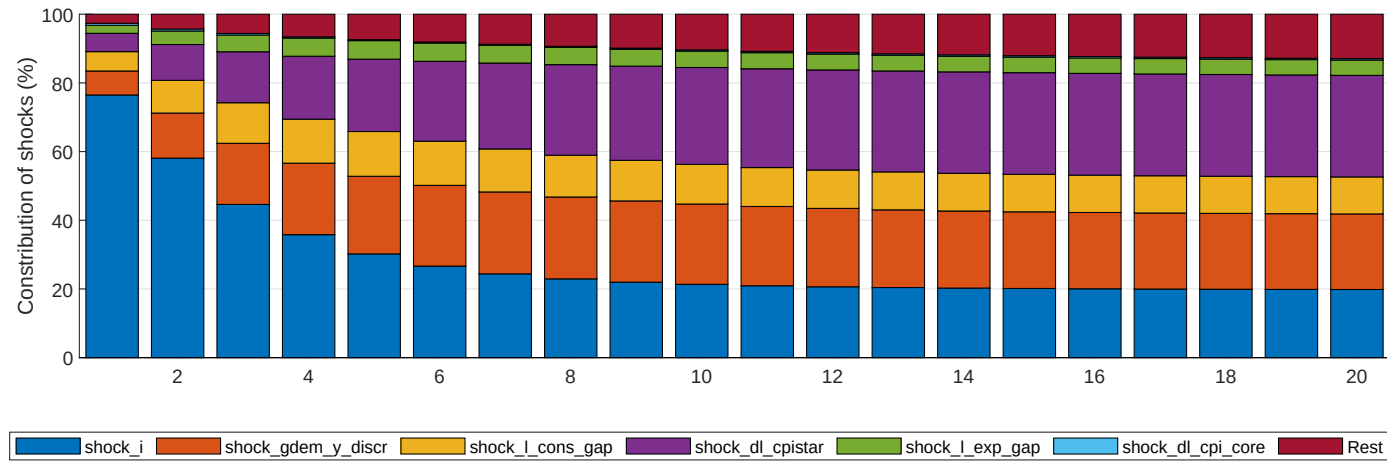
Food rel. price gap, % [l_rp_cpi_food_gap]



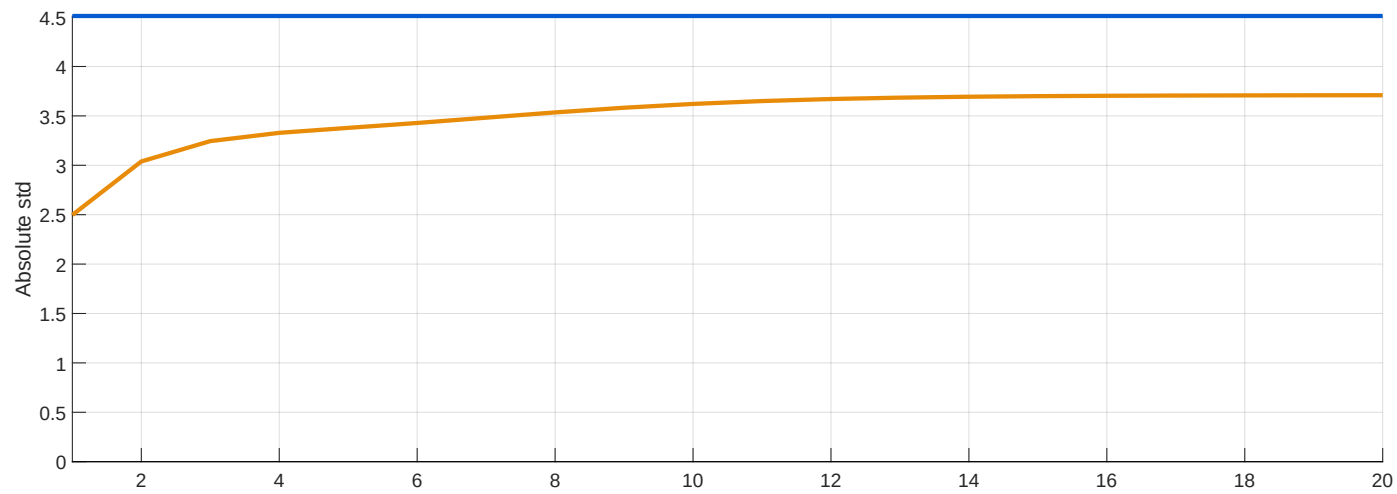
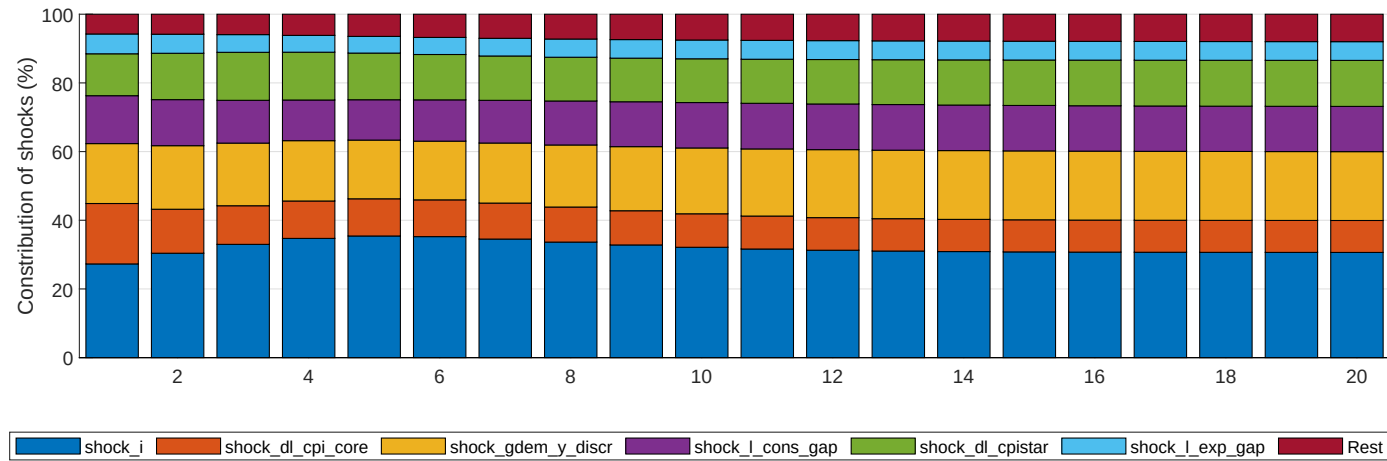
Energy rel. price gap, % [l_rp_cpi_ener_gap]

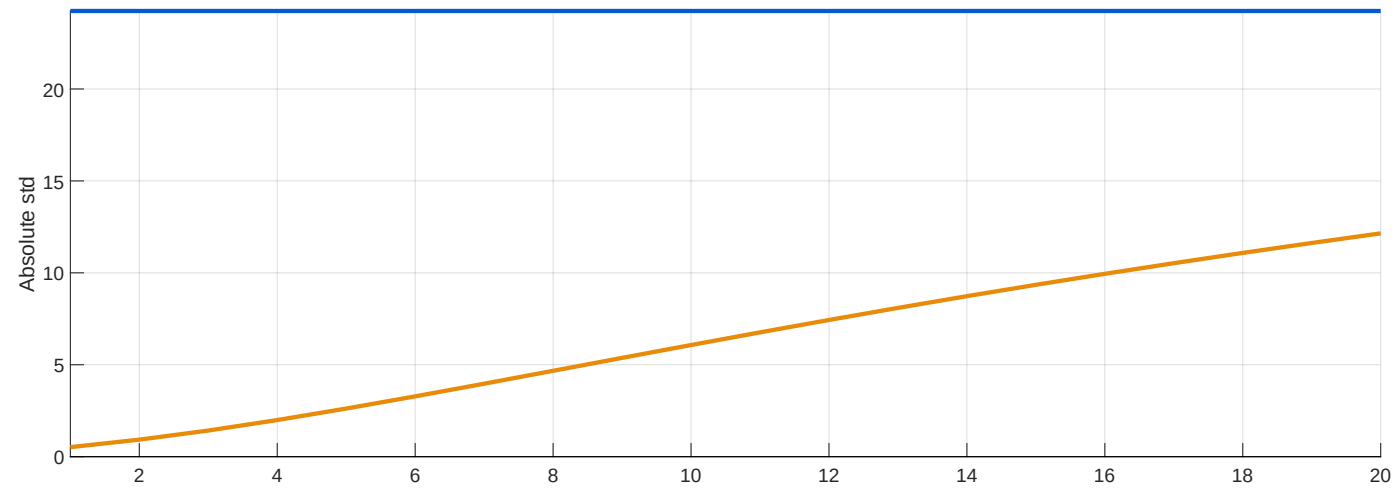
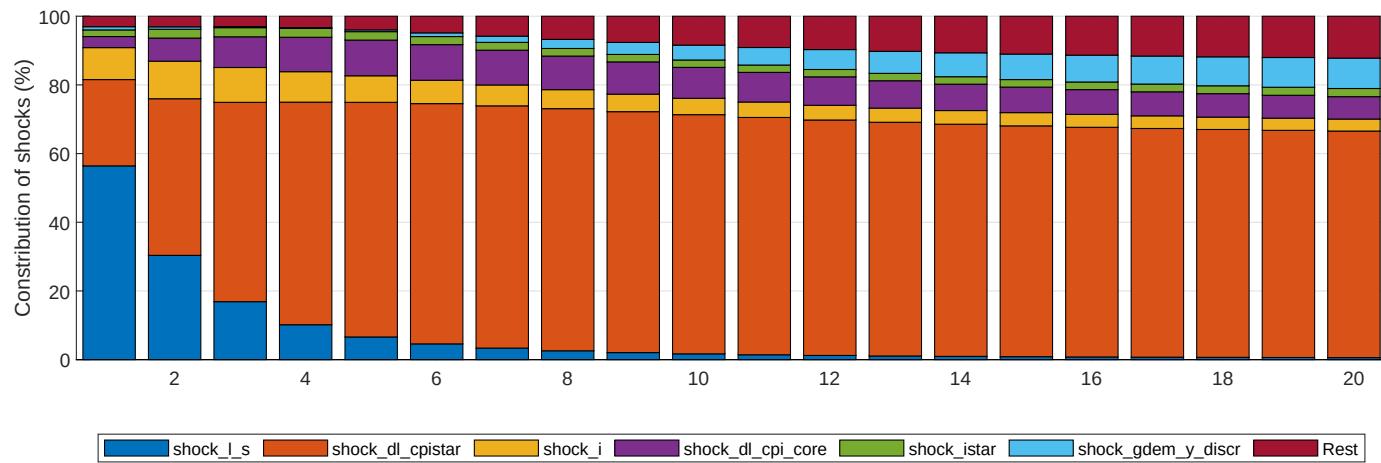


Policy rate (IB rate used), % [i]

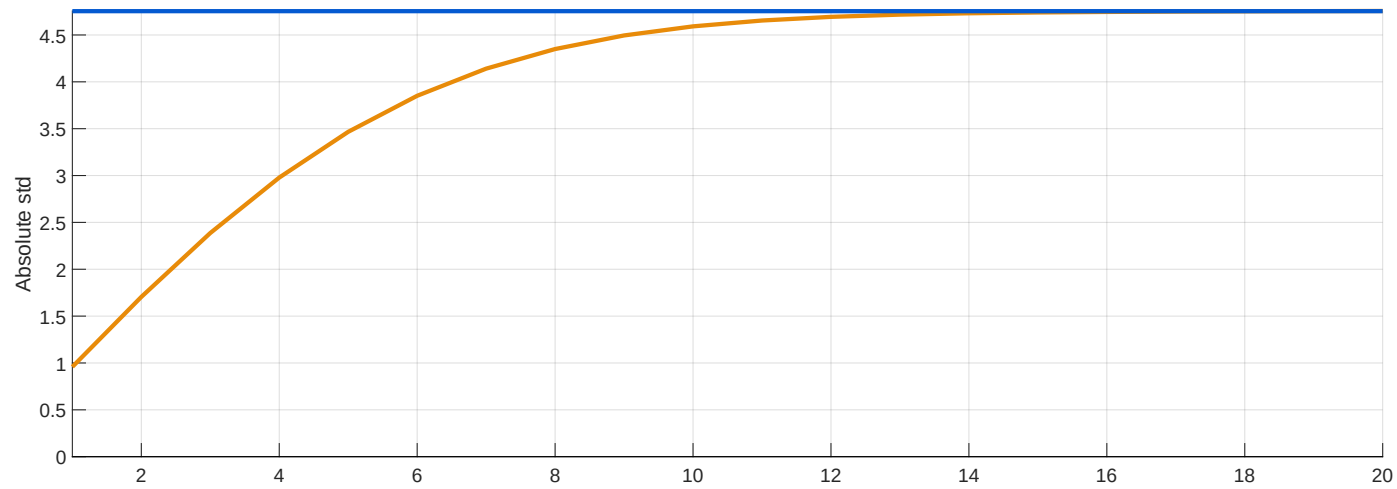
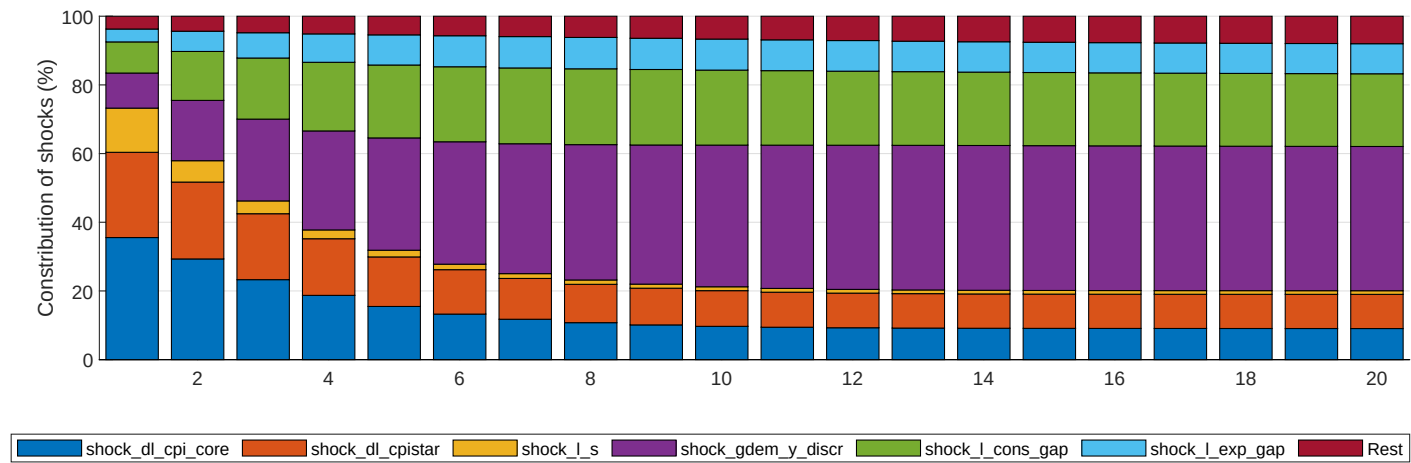


RIR (policy) gap, % [r_gap]

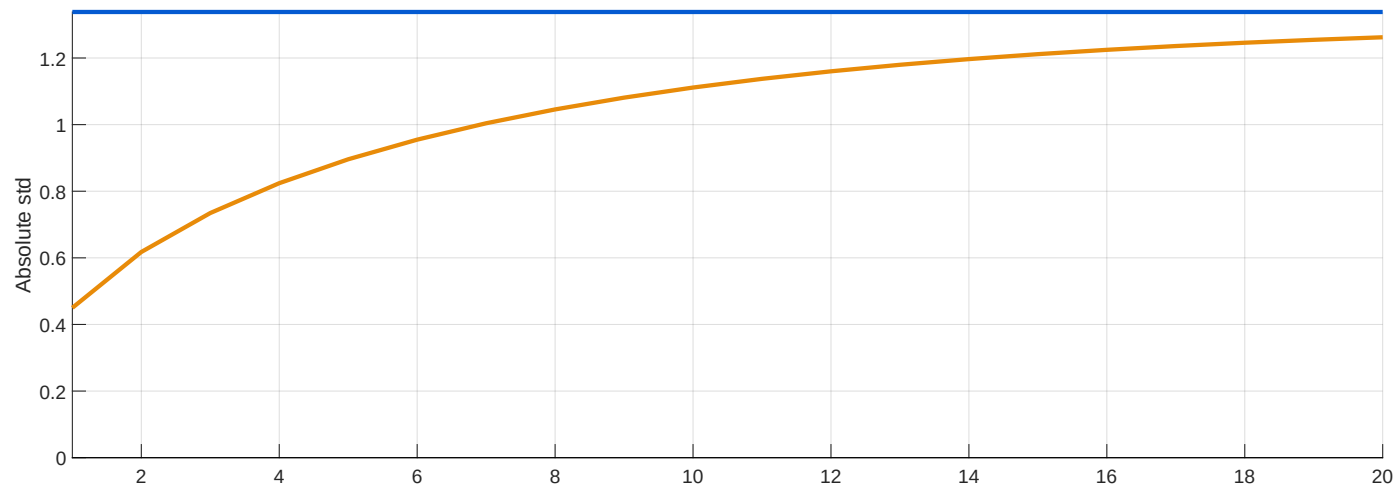
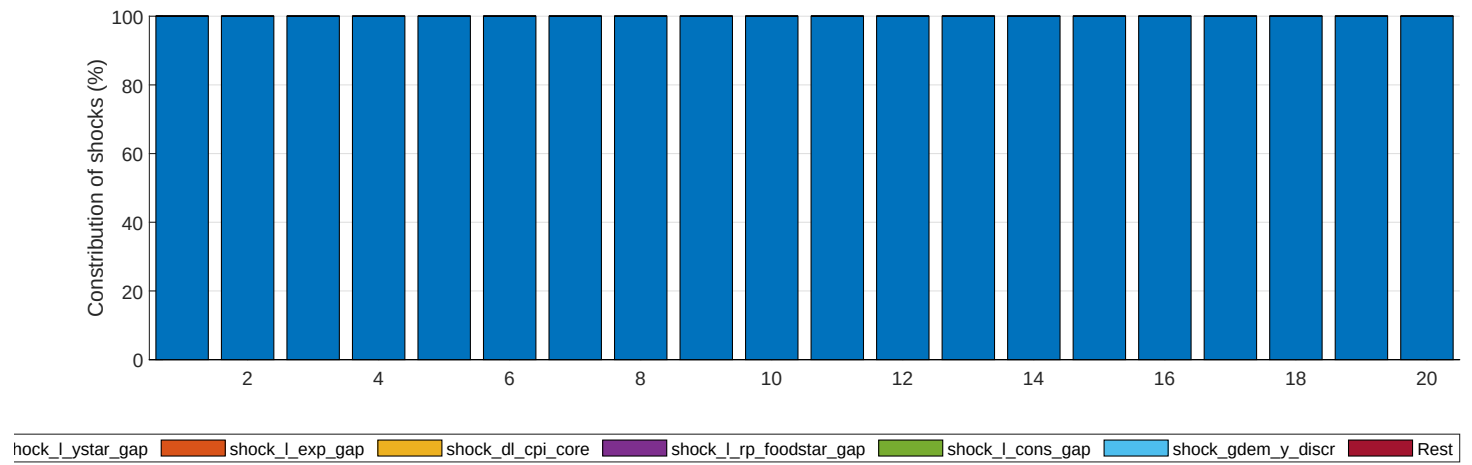


Exchang. rate, $100 \cdot \log [l_s]$ 

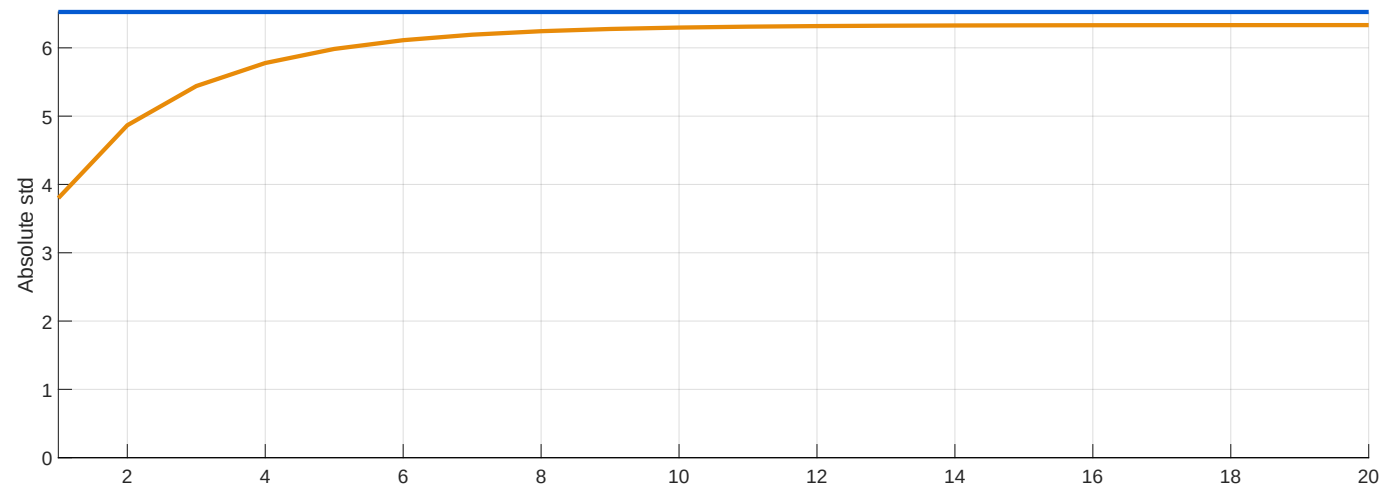
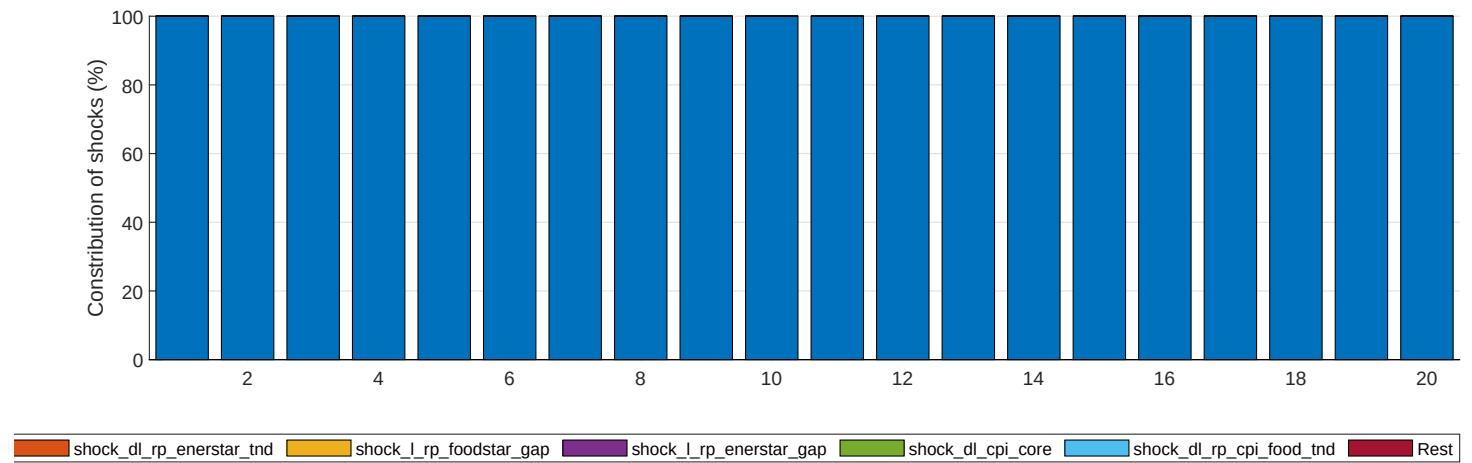
Real exchang. rate gap, % [l_z_gap]

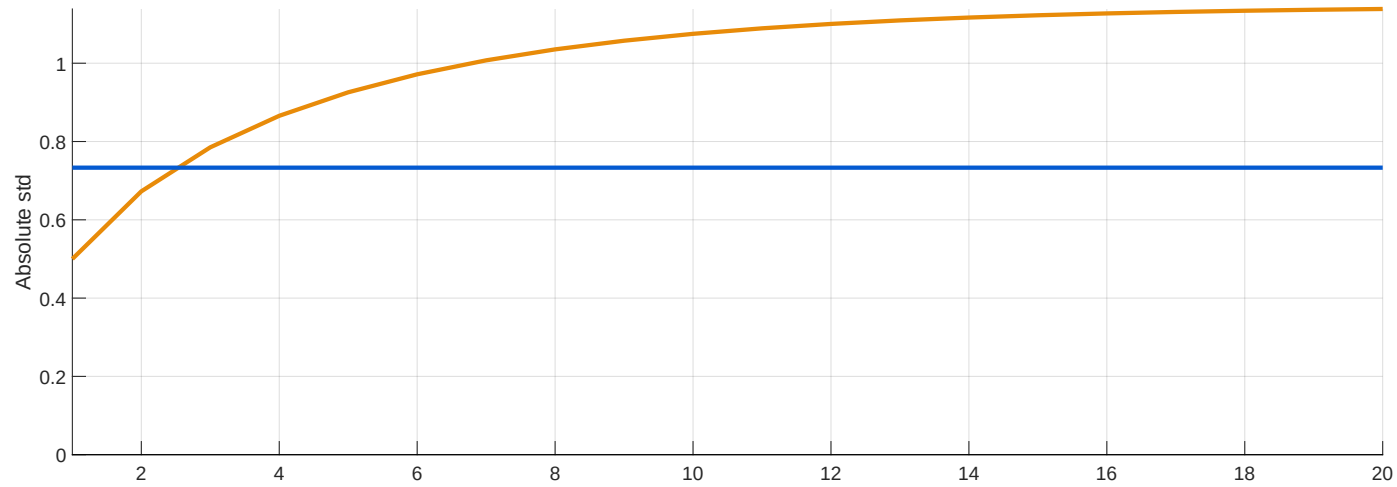
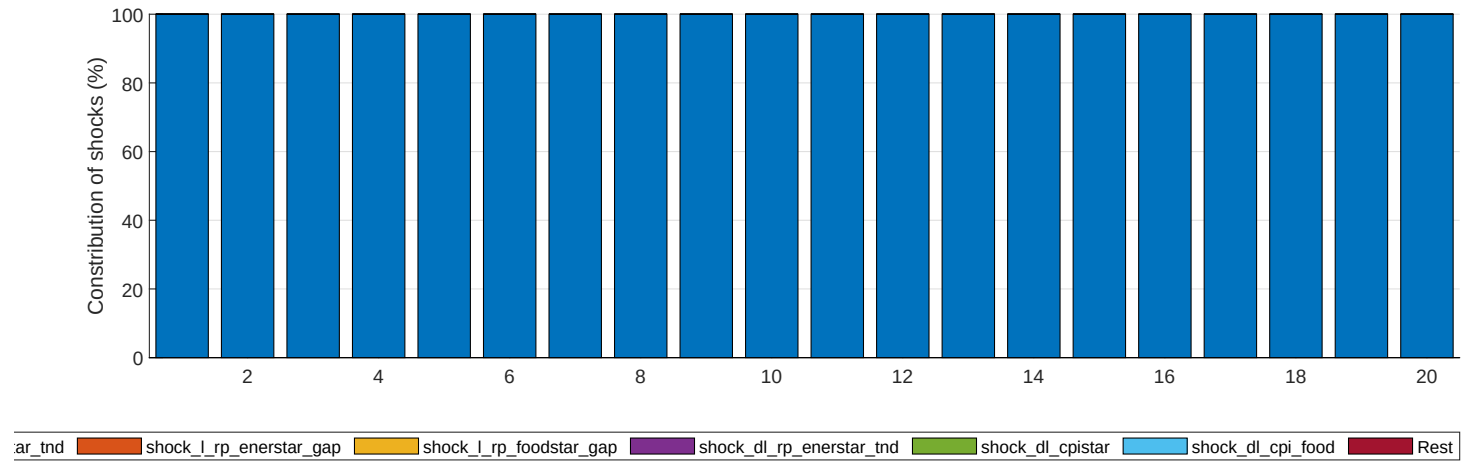


Foreign output gap, % [l_ystar_gap]

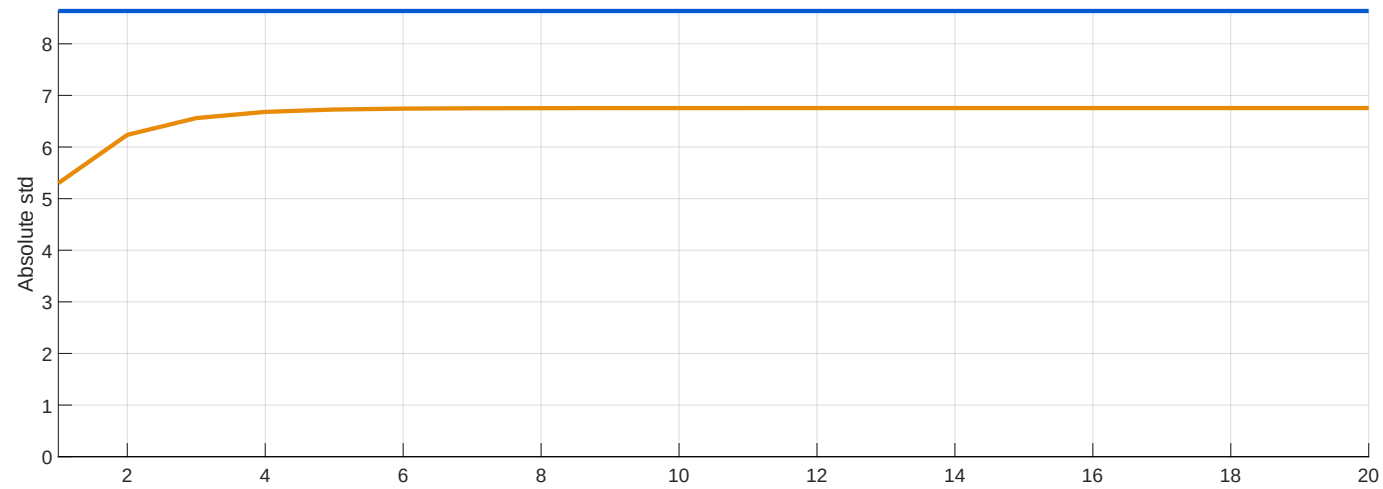
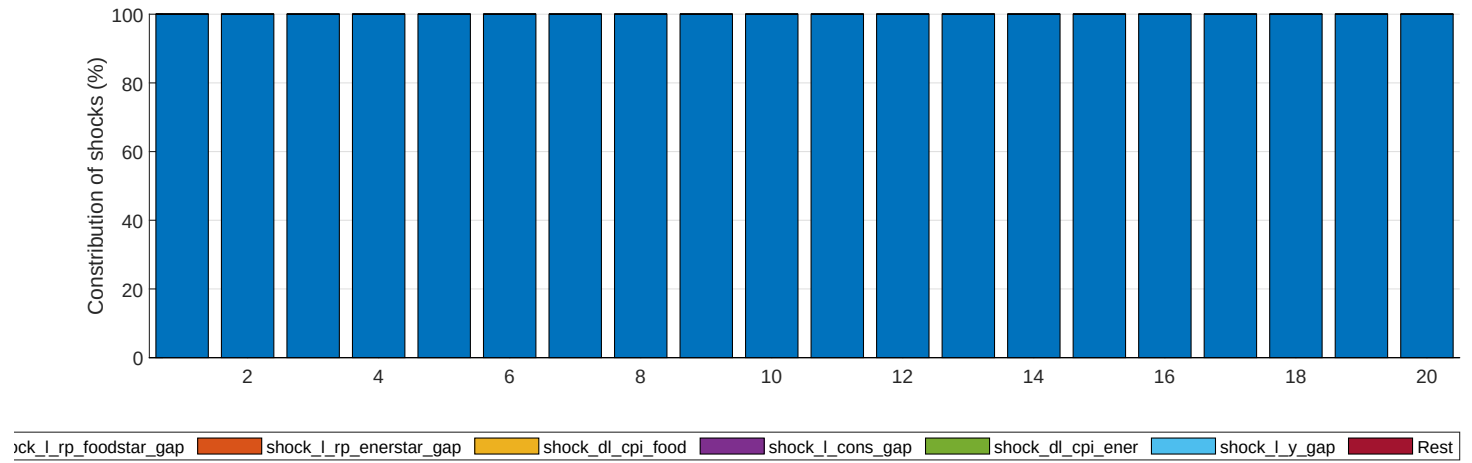


Foreign CPI, ann. QQ % [dl_cpistar]

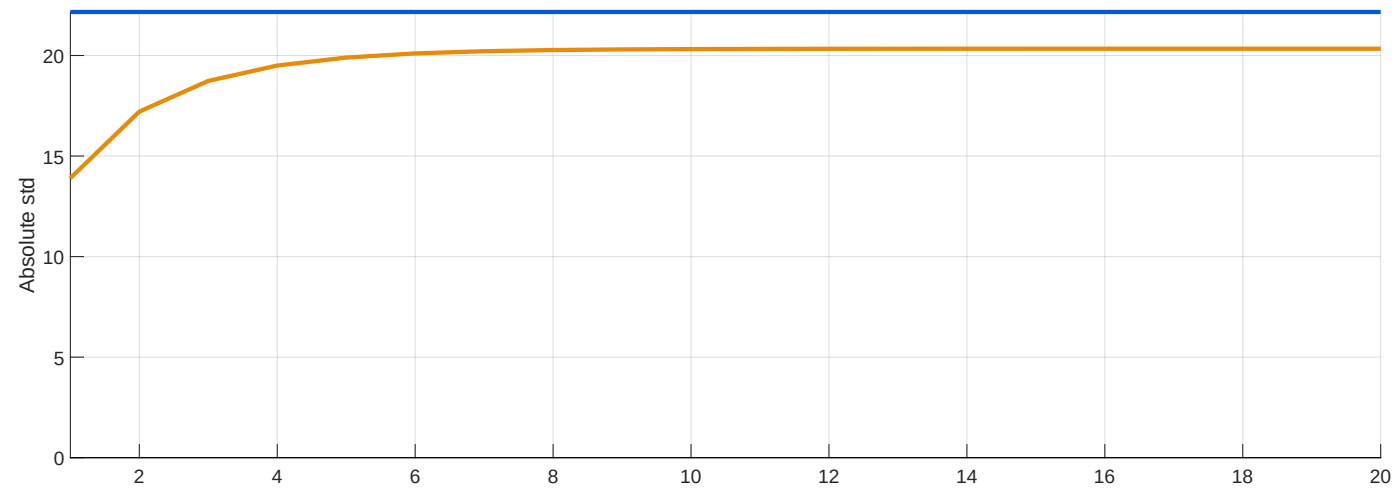
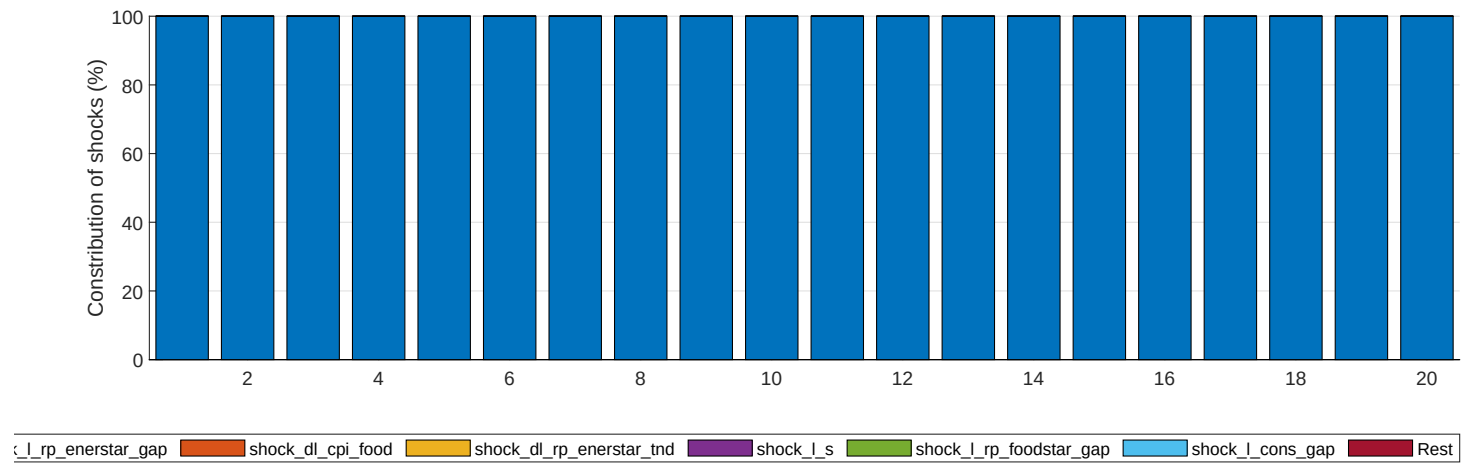


Foreign real interest rate trend, % [rstar_tnd]

Foreign rel. food price gap, % [l_rp_foodstar_gap]



Foreign rel. ener. price gap, % [l_rp_enerstar_gap]



4 Standard deviations: asymptotic model vs. data

	, data	, model
Consumption, ann. QQ % [dl_cons]	15.85	13.90
Consumption, YoY % [d4l_cons]	5.58	6.41
Investment, ann. QQ % [dl_inv]	77.77	27.39
Investment, YoY % [d4l_inv]	31.61	11.01
Gov. demand, ann. QQ % [dl_gdem]	38.88	37.09
Gov. demand, YoY % [d4l_gdem]	14.78	15.34
Export, ann. QQ % [dl_exp]	50.46	26.65
Export, YoY % [d4l_exp]	16.96	10.57
Import, ann. QQ % [dl_imp]	33.35	16.88
Import, YoY % [d4l_imp]	10.58	7.74
GDP, ann. QQ % [dl_y]	10.72	15.26
GDP, YoY % [d4l_y]	4.96	6.41
Budget deficit, % of GDP [def_y]	2.82	3.41
Govt. rev., % of GDP [grev_y]	3.02	1.87
Govt. demand (G&S), % GDP [gdem_y]	3.72	2.37
Other govt. exp., % to GDP [oexp_y]	2.27	1.82
Fiscal grants, % of GDP [grants_y]	3.17	2.89
Headline CPI, ann. QQ % [dl_cpi]	6.44	5.05
Headline CPI, YoY % [d4l_cpi]	4.86	4.16
Core CPI, ann. QQ % [dl_cpi_core]	5.71	5.44
Core CPI, YoY % [d4l_cpi_core]	4.42	4.57
Food CPI, ann. QQ % [dl_cpi_food]	18.39	12.70
Food CPI, YoY % [d4l_cpi_food]	10.99	8.31
Energy CPI, ann. QQ % [dl_cpi_ener]	9.48	9.52
Energy CPI, YoY % [d4l_cpi_ener]	5.64	5.95
Policy rate (IB rate used), % [i]	1.54	2.89
Lending premium gap, % [prem_d_gap]	1.69	1.84
Exchang. rate, ann. QQ % [dl_s]	4.34	4.37
Exchang. rate, YoY % [d4l_s]	3.59	3.91
Consumption gap, % [l_cons_gap]	4.40	5.46

Investment gap, % [l_inv_gap]	19.95	8.72
Gov. demand gap, % [l_gdem_gap]	15.62	12.51
Export gap, % [l_exp_gap]	13.66	8.05
Import gap, % [l_imp_gap]	7.26	6.92
Output gap, % [l_y_gap]	3.84	5.06
Real exchang. rate gap, % [l_z_gap]	4.75	4.76
RIR (policy) gap, % [r_gap]	4.51	3.71
Core rel. price gap, % [l_rp_cpi_core_gap]	1.14	1.75
Food rel. price gap, % [l_rp_cpi_food_gap]	6.06	8.02
Energy rel. price gap, % [l_rp_cpi_ener_gap]	3.76	7.59
Foreign output gap, % [l_ystar_gap]	1.34	1.32
Foreign CPI, ann. QQ % [dl_cpistar]	6.50	6.33
Foreign interest rate, % [istar]	1.83	1.44
Foreign real interest rate trend, % [rstar_tnd]	0.73	1.15
Foreign food price, ann. QQ % [dl_foodstar]	28.62	24.42
Foreign rel. food price gap, % [l_rp_foodstar_gap]	8.64	6.76
Foreign rel. food price tnd., ann. QQ % [dl_rp_foodstar_tnd]	3.06	1.15
Foreign ener. price, ann. QQ % [dl_enerstar]	68.50	60.39
Foreign rel. ener. price gap, % [l_rp_enerstar_gap]	22.16	20.34
Foreign rel. ener. price tnd., ann. QQ % [dl_rp_enerstar_tnd]	6.93	5.74

5 Standard deviations: filtered shocks vs. model calibration

	, filtered	, calibrated
Cons. gap shock, % [shock_l_cons_gap]	3.49	2.50
Inv. gap shock, % [shock_l_inv_gap]	16.91	6.00
Export gap shock, % [shock_l_exp_gap]	11.11	6.00
Output gap shock, % [shock_l_y_gap]	1.54	1.00
Core infl. shock, ann. QQ % [shock_dl_cpi_core]	3.42	2.00
Food infl. shock, ann. QQ % [shock_dl_cpi_food]	14.06	9.00
Energy infl. shock, ann. QQ % [shock_dl_cpi_ener]	7.10	7.10
Policy rate shock, % [shock_i]	1.08	1.00
Lending premium shock, % [shock_prem_d_gap]	0.73	0.80
Exchng. rate shock, 100*log [shock_l_s]	0.62	0.35
Gov rev. discr. shock, % of GDP [shock_grev_y_discr]	1.89	1.73
Gdem. discr. shock, % of GDP [shock_gdem_y_discr]	2.52	1.73
Gov exp. discr. shock, % of GDP [shock_oexp_y_discr]	2.12	1.73
Fiscal grants, % of GDP [shock_grants_y]	2.83	2.50

Equations (parameter values taken from setparam.m)

```

1: % -----
2: % ----- Transition variables -----
3: % -----
4:
5: !transition_variables
6:
7: % -----
8: % ----- Fiscal policy -----
9:
10: "Fiscal impulse, % of GDP"  fisc_imp
11:
12: "Budget deficit, % of GDP"   def_y
13: "Struct. deficit, % of GDP"  def_y_str
14: "Cyc. deficit, % of GDP"     def_y_cyc
15: "Discr. deficit, % of GDP"   def_y_discr
16:

```



```

17: "Govt. demand (G&S), % GDP"      gdem_y
18: "Struct. govt. demand (G&S), % GDP" gdem_y_str
19: "Cyc. govt. demand (G&S), % GDP"   gdem_y_cyc
20: "Discr. govt. demand (G&S), % GDP" gdem_y_discr
21:
22: "Other govt. exp., % to GDP"        oexp_y
23: "Cyc. other govt. exp., % to GDP"   oexp_y_cyc
24: "Struct. other govt. exp., % to GDP" oexp_y_str
25: "Discr. other govt. exp., % to GDP"  oexp_y_discr
26:
27: "Govt. rev., % of GDP"              grev_y
28: "Struct. govt. rev., % of GDP"      grev_y_str
29: "Cyc. govt. rev., % of GDP"         grev_y_cyc
30: "Discr. govt. rev., % of GDP"       grev_y_discr
31:
32: "Fiscal grants, % of GDP"           grants_y
33:
34: % -----
35: % ----- Components of real GDP -----
36:
37: % ----- Private consumption -----
38:
39: "Consumption gap, %"                l_cons_gap
40: "Exptd. cons. gap, %"               e_l_cons_gap
41: "Consumption, 100*log"              l_cons
42: "Consumption, ann. QQ %"            dl_cons
43: "Consumption, YoY %"                 d4l_cons
44: "Consumption tnd., 100*log"          l_cons_tnd
45: "Consumption tnd., ann. QQ %"        dl_cons_tnd
46: "Consumption tnd., YoY %"            d4l_cons_tnd
47:
48: % ----- Private investment -----
49:
50: "Investment gap, %"                  l_inv_gap
51: "Exptd. investment gap, %"           e_l_inv_gap
52: "Investment, 100*log"                l_inv
53: "Investment, ann. QQ %"              dl_inv
54: "Investment, YoY %"                  d4l_inv
55: "Investment tnd., 100*log"            l_inv_tnd
56: "Investment tnd., ann. QQ %"          dl_inv_tnd

```

```

57: "Investment tnd., YoY %"      d4l_inv_tnd
58:
59: % ----- Government demand for G&S -----
60:
61: "Gov. demand gap, %"         l_gdem_gap
62: "Gov. demand, 100*log"       l_gdem
63: "Gov. demand, ann. QQ %"     dl_gdem
64: "Gov. demand, YoY %"         d4l_gdem
65: "Gov. demand, tnd., 100*log" l_gdem_tnd
66: "Gov. demand, tnd., ann. QQ %" dl_gdem_tnd
67: "Gov. demand, tnd., YoY %"   d4l_gdem_tnd
68:
69: % ----- Exports of G&S -----
70:
71: "Export gap, %"              l_exp_gap
72: "Exptd. export gap, %"      e_l_exp_gap
73: "Export RMCi, %"            rmci_exp
74: "Export, 100*log"           l_exp
75: "Export, ann. QQ %"         dl_exp
76: "Export, YoY %"             d4l_exp
77: "Export tnd., 100*log"       l_exp_tnd
78: "Export tnd., ann. QQ %"     dl_exp_tnd
79: "Export tnd., YoY %"         d4l_exp_tnd
80:
81: % ----- Imports of G&S -----
82:
83: "Import gap, %"              l_imp_gap
84: "Import, 100*log"            l_imp
85: "Import, ann. QQ %"          dl_imp
86: "Import, YoY %"              d4l_imp
87: "Import tnd., 100*log"       l_imp_tnd
88: "Import tnd., ann. QQ %"     dl_imp_tnd
89: "Import tnd., YoY %"         d4l_imp_tnd
90:
91: % ----- GDP -----
92:
93: "Output gap, %"              l_y_gap
94: "Exptd. output gap, %"      e_l_y_gap
95: "GDP, 100*log"               l_y
96: "GDP, ann. QQ %"             dl_y

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```

97: "GDP, YoY %"          d4l_y
98: "GDP tnd., 100*log"   l_y_tnd
99: "GDP tnd., ann. QQ %" dl_y_tnd
100: "GDP tnd., YoY %"    d4l_y_tnd
101:
102: % ----- Agricultural output -----
103:
104: "Agric. output gap, %"      l_y_agr_gap
105: "Exptd. agric. output gap, %" e_l_y_agr_gap
106: "Agric. GDP, 100*log"      l_y_agr
107: "Agric. GDP, ann. QQ %"    dl_y_agr
108: "Agric. GDP, YoY %"       d4l_y_agr
109: "Agric. GDP tnd., 100*log" l_y_agr_tnd
110: "Agric. GDP tnd., ann. QQ %" dl_y_agr_tnd
111: "Agric. GDP tnd., YoY %"   d4l_y_agr_tnd
112:
113: % -----
114: % ----- Prices -----
115:
116: % ----- Headline CPI -----
117:
118: "Headline CPI, 100*log"      l_cpi
119: "Headline CPI, ann. QQ %"    dl_cpi
120: "Exptd. Headline CPI, ann. QQ %" e_dl_cpi
121: "Headline CPI, YoY %"       d4l_cpi
122:
123: % ----- Core CPI -----
124:
125: "Core CPI, 100*log"          l_cpi_core
126: "Core CPI, ann. QQ %"       dl_cpi_core
127: "Exptd. Core CPI, ann. QQ %" e_dl_cpi_core
128: "Core CPI, YoY %"           d4l_cpi_core
129:
130: % ----- Food CPI -----
131:
132: "Food CPI, 100*log"          l_cpi_food
133: "Food CPI, ann. QQ %"       dl_cpi_food
134: "Exptd. Food CPI, ann. QQ %" e_dl_cpi_food
135: "Food CPI, YoY %"           d4l_cpi_food
136:

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```

137: % ----- Energy CPI -----
138:
139: "Energy CPI, 100*log"          l_cpi_ener
140: "Energy CPI, ann. QQ %"       dl_cpi_ener
141: "Exptd. energy CPI, ann. QQ %" e_dl_cpi_ener
142: "Energy CPI, YoY %"          d4l_cpi_ener
143:
144: % -----
145: % ----- Relative prices -----
146:
147: % ----- Real marginal cost gap -----
148:
149: "RMC, %"    rmc
150:
151: % ----- Core/headline relative prices -----
152:
153: "Core rel. price, 100*log"      l_rp_cpi_core
154: "Core rel. price, ann. QQ %"   dl_rp_cpi_core
155: "Core rel. price, YoY %"       d4l_rp_cpi_core
156: "Core rel. price gap, %"       l_rp_cpi_core_gap
157: "Core rel. price tnd., 100*log" l_rp_cpi_core_tnd
158: "Core rel. price tnd., ann. QQ %" dl_rp_cpi_core_tnd
159: "Core rel. price tnd., YoY %"   d4l_rp_cpi_core_tnd
160:
161: "Dir. ext. eff. in core, ann. QQ %" dl_cpi_core_direct
162:
163: % ----- Food/headline relative prices -----
164:
165: "Food rel. price, 100*log"      l_rp_cpi_food
166: "Food rel. price, ann QQ. %"   dl_rp_cpi_food
167: "Food rel. price, YoY %"       d4l_rp_cpi_food
168: "Food rel. price gap, %"       l_rp_cpi_food_gap
169: "Food rel. price tnd., 100*log" l_rp_cpi_food_tnd
170: "Food rel. price tnd., ann. QQ %" dl_rp_cpi_food_tnd
171: "Food rel. price tnd., YoY %"   d4l_rp_cpi_food_tnd
172:
173: "Dir. ext. eff. in food, ann. QQ %" dl_cpi_food_direct
174:
175: % ----- Energy/headline relative prices -----
176:

```

```

177: "Energy rel. price, 100*log"          l_rp_cpi_ener
178: "Energy rel. price, ann. QQ %"        dl_rp_cpi_ener
179: "Energy rel. price, YoY %"            d4l_rp_cpi_ener
180: "Energy rel. price gap, %"            l_rp_cpi_ener_gap
181: "Energy rel. price tnd., 100*log"      l_rp_cpi_ener_tnd
182: "Energy rel. price tnd., ann. QQ %"    dl_rp_cpi_ener_tnd
183: "Energy rel. price tnd., YoY %"        d4l_rp_cpi_ener_tnd
184:
185: "Dir. ext. eff. in energy, ann. QQ %" dl_cpi_ener_direct
186:
187: % -----
188: % ----- Monetary policy -----
189:
190: % ----- Nominal interest rate -----
191:
192: "Policy rate (IB rate used), %"        i
193: "Policy rate tnd (IB rate used), %"    i_tnd
194:
195: % ----- Real interest rate -----
196:
197: "Real interest (policy) rate, %"        r
198: "RIR (policy) gap, %"                  r_gap
199: "Exptd. RIR (policy) gap, %"           e4_r_gap
200: "Exptd. RIR (lending) gap, 4-Q ahead av., %" r4_gap
201: "RIR (policy) trend, %"                r_tnd
202:
203: % ----- Lending premium -----
204:
205: "Lending premium, %"                  prem_d
206: "Lending premium gap, %"              prem_d_gap
207:
208: % ----- Inflation target -----
209:
210: "Inflation target, YoY %"              d4l_cpi_tar
211: "Headline CPI dev, 4-Q ahead YoY %"    d4l_cpi_dev
212:
213: "Impl. core infl. target, YoY %"       d4l_cpi_core_tar
214: "Impl. food infl. target, YoY %"       d4l_cpi_food_tar
215: "Impl. ener. infl. target, YoY %"      d4l_cpi_ener_tar
216:

```

```

217: % ----- Real monetary condition index -----
218:
219: "RMCI cons., %"      rmci_cons
220: "RMCI invest., %"    rmci_inv
221:
222: % -----
223: % ----- Exchange rate -----
224:
225: % ----- Nominal exchange rate -----
226:
227: "Exchang. rate, 100*log"      l_s
228: "Exchang. rate, ann. QQ %"    dl_s
229: "Exchang. rate, YoY %"        d4l_s
230: "Exptd. exchang. rate, 100*log" e_l_s
231:
232: "Exchang. rate target, ann. QQ %" dl_s_tar
233:
234: % ----- Sovereign risk premium -----
235:
236: "Risk premium, %"    prem
237:
238: % ----- Real exchange rate -----
239:
240: "Real exchang. rate, 100*log"      l_z
241: "Real exchang. rate, ann. QQ %"    dl_z
242: "Real exchang. rate, YoY %"        d4l_z
243: "Exptd. RER, ann. QQ %"            e_dl_z
244: "Real exchang. rate trend, 100*log" l_z_tnd
245: "Real exchang. rate trend, ann. QQ %" dl_z_tnd
246: "Real exchang. rate trend, YoY %"    d4l_z_tnd
247: "Exp. RER trend., ann. QQ %"        e_dl_z_tnd
248: "Real exchang. rate gap, %"          l_z_gap
249:
250: % ----- Money demand -----
251:
252: "Money, 100*log"      l_md
253: "Real money, 100*log" l_rmd
254: "Real money, ann. QQ %" dl_rmd
255: "Real money, YoY %"    d4l_rmd
256: "Velocity, ann. QQ %"  dl_v

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```

257:
258: % -----
259: % ----- External variables -----
260:
261: % ----- External demand -----
262:
263: "Foreign output gap, %"      l_ystar_gap
264:
265: % ----- Foreign prices -----
266:
267: "Foreign CPI, 100*log"      l_cpistar
268: "Foreign CPI, ann. QQ %"    dl_cpistar
269: "Foreign CPI, YoY %"        d4l_cpistar
270: "Exptd. foreign CPI, 100*log" e_dl_cpistar
271:
272: % ----- Foreign interest rate -----
273:
274: "Foreign interest rate, %"      istar
275: "Foreign real interest rate trend, %" rstar_tnd
276:
277: % ----- World food prices -----
278:
279: "Foreign food price, 100*log"    l_foodstar
280: "Foreign food price, ann. QQ %"  dl_foodstar
281: "Foreign food price, YoY %"      d4l_foodstar
282:
283: "Foreign rel. food price, 100*log" l_rp_foodstar
284: "Foreign rel. food price, ann. QQ %" dl_rp_foodstar
285: "Foreign rel. food price, YoY %"   d4l_rp_foodstar
286:
287: "Foreign rel. food price gap, %" l_rp_foodstar_gap
288:
289: "Foreign rel. food price tnd., 100*log" l_rp_foodstar_tnd
290: "Foreign rel. food price tnd., ann. QQ %" dl_rp_foodstar_tnd
291: "Foreign rel. food price tnd., YoY %"   d4l_rp_foodstar_tnd
292:
293: % ----- World energy prices -----
294:
295: "Foreign ener. price, 100*log"    l_enerstar
296: "Foreign ener. price, ann. QQ %"  dl_enerstar

```

```

297: "Foreign ener. price, YoY %"      d4l_enerstar
298:
299: "Foreign rel. ener. price, 100*log"  l_rp_enerstar
300: "Foreign rel. ener. price, ann. QQ %" dl_rp_enerstar
301: "Foreign rel. ener. price, YoY %"    d4l_rp_enerstar
302:
303: "Foreign rel. ener. price gap, %"    l_rp_enerstar_gap
304:
305: "Foreign rel. ener. price tnd., 100*log" l_rp_enerstar_tnd
306: "Foreign rel. ener. price tnd., ann. QQ %" dl_rp_enerstar_tnd
307: "Foreign rel. ener. price tnd., YoY %" d4l_rp_enerstar_tnd
308:
309: % -----
310: % ----- Transition shocks -----
311: % -----
312:
313: !transition_shocks
314:
315: "Fiscal grants, % of GDP"  shock_grants_y< $\sigma=2.5$ >
316:
317: "Gdem. cyc. shock, % of GDP"  shock_gdem_y_str< $\sigma=0.075$ >
318: "Gdem. str. shock, % of GDP"  shock_gdem_y_cyc< $\sigma=0$ >
319: "Gdem. discr. shock, % of GDP" shock_gdem_y_discr< $\sigma=1.7321$ >
320:
321: "Gov exp. cyc. shock, % of GDP"  shock_oexp_y_str< $\sigma=0.1732$ >
322: "Gov exp. str. shock, % of GDP"  shock_oexp_y_cyc< $\sigma=0$ >
323: "Gov exp. discr. shock, % of GDP" shock_oexp_y_discr< $\sigma=1.7321$ >
324:
325: "Gov rev. str. shock, % of GDP"  shock_grev_y_str< $\sigma=0.1$ >
326: "Gov rev. cyc. shock, % of GDP"  shock_grev_y_cyc< $\sigma=0$ >
327: "Gov rev. discr. shock, % of GDP" shock_grev_y_discr< $\sigma=1.7321$ >
328:
329: "Cons. gap shock, %"      shock_l_cons_gap< $\sigma=2.5$ >
330: "Inv. gap shock, %"      shock_l_inv_gap< $\sigma=6$ >
331: "Export gap shock, %"    shock_l_exp_gap< $\sigma=6$ >
332: "Import gap shock, %"    shock_l_imp_gap< $\sigma=0.5$ >
333: "Output gap shock, %"    shock_l_y_gap< $\sigma=1$ >
334: "Agric. output gap shock, %" shock_l_y_agr_gap< $\sigma=0.3$ >
335:
336: "Core infl. shock, ann. QQ %"  shock_dl_cpi_core< $\sigma=2$ >

```



```

337: "Food infl. shock, ann. QQ %"      shock_dl_cpi_food< $\sigma=9$ >
338: "Energy infl. shock, ann. QQ %"    shock_dl_cpi_ener< $\sigma=7.1$ >
339:
340: "Policy rate shock, %"      shock_i< $\sigma=1$ >
341: "Lending premium shock, %"  shock_prem_d_gap< $\sigma=0.8$ >
342:
343: "Exchnng. rate shock, 100*log"      shock_l_s< $\sigma=0.35$ >
344: "Exchnng. rate target shock, ann. QQ %"  shock_dl_s_tar< $\sigma=0.3$ >
345:
346: "Real money dem. shock, ann. QQ %"  shock_dl_rmd< $\sigma=1$ >
347: "Chng. of vel. shock, ann. QQ %"    shock_dl_v< $\sigma=0.1$ >
348:
349: "Cons. tnd. shock, ann. QQ %"      shock_dl_cons_tnd< $\sigma=0.25$ >
350: "Inv. tnd. shock, ann. QQ %"      shock_dl_inv_tnd< $\sigma=0.5$ >
351: "Export shock, ann. QQ %"          shock_dl_exp_tnd< $\sigma=0.75$ >
352: "Import tnd. shock, ann. QQ %"     shock_dl_imp_tnd< $\sigma=0.25$ >
353:
354: "CPI target shocck, YoY %"          shock_d4l_cpi_tar< $\sigma=0.1$ >
355: "Food rel. price. tnd. shock, ann. QQ %"  shock_dl_rp_cpi_food_tnd< $\sigma=0.5$ >
356: "Energy rel. price. tnd. shock, ann. QQ %"  shock_dl_rp_cpi_ener_tnd< $\sigma=0.15$ >
357: "RER tnd. shock, ann. QQ %"          shock_dl_z_tnd< $\sigma=0.1$ >
358: "Premium shock, %"                  shock_prem< $\sigma=0.1$ >
359: "Agric. output tnd. shock, %"        shock_dl_y_agr_tnd< $\sigma=0.06$ >
360:
361: "Foreign output gap shock, %"        shock_l_ystar_gap< $\sigma=0.45$ >
362: "Foreign CPI shock, ann. QQ %"      shock_dl_cpistar< $\sigma=3.8$ >
363: "Foreign intr. rate shock, %"        shock_istar< $\sigma=0.45$ >
364: "Foreign rel. food price gap shock, %"  shock_l_rp_foodstar_gap< $\sigma=5.3$ >
365: "Foreign rel. ener. price gap shock, %"  shock_l_rp_enerstar_gap< $\sigma=13.9$ >
366: "Foreign rel intr. rate rtnd. shock, %"  shock_rstar_tnd< $\sigma=0.5$ >
367: "Foreign rel. food price tnd. shock, ann. QQ %"  shock_dl_rp_foodstar_tnd< $\sigma=0.5$ >
368: "Foreign rel. ener. price tnd. shock, ann. QQ %"  shock_dl_rp_enerstar_tnd< $\sigma=2.5$ >
369:
370: % -----
371: % ----- Parameters -----
372: % -----
373:
374: !parameters
375:
376: % -----

```

```

377: % ----- Steady states -----
378:
379: "steady state govt rev in % of GDP"      ss_grev_y_str<21>
380: "steady state govt other exp in % of GDP" ss_oexp_y_str<6>
381: "steady state govt other exp in % of GDP" ss_gdem_y_str<26>
382:
383: "SS of borrowing ratio, % of GDP"  ss_bor_str<6>
384: "SS of FCY debt share"              ss_debt_fcy_rat<0.8462>
385: "SS of fiscal grants ratio, % GDP"  ss_grants_y<5>
386:
387: "SS of FCY gov. debt. intr. prem., %" ss_prem_debt_fcy<NaN>
388: "SS of LCY gov. debt. intr. prem., %" ss_prem_debt_lcy<NaN>
389:
390: "SS of GDP growth, ann. QQ %"      ss_dl_y_tnd<7.23207>
391: "SS of agric.GDP growth, ann. QQ %" ss_dl_y_agr_tnd<7.23207>
392:
393: "CPI target, YoY %"                ss_d4l_cpi_tar<4.87902>
394: "SS of food rel. price, ann. QQ %" ss_dl_rp_cpi_food_tnd<1.98026>
395: "SS of ener. rel. price, ann. QQ %" ss_dl_rp_cpi_ener_tnd<0>
396:
397: "SS of RER depr., ann. QQ %"      ss_dl_z_tnd<0>
398: "SS of risk premium, %"          ss_prem<2>
399: "SS of lending premium, %"       ss_prem_d<8.5>
400:
401: "SS of chng. of vel., ann. QQ %"  ss_dl_v<-2.02027>
402:
403: % -----
404: % ----- Cyclical coefficients -----
405:
406: % ----- Fiscal parameters -----
407:
408: "Fiscal, grants. persist."        d7<0.5>
409:
410: " grev_y_cyc response to cons * rev/GDP" v1<0.012>
411: " grev_y_cyc response to imports * rev/GDP" v2<0.006>
412: " grev_y_cyc response outputgap * rev/GDP" v3<0.02>
413: " grev_y_str, lag"                v4<0.99>
414:
415: "oexp_y_cyc,lag %"                u1<0>
416: "oexp_y_cyc response output gap * oexp/GDP" u2<0.012>

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417: "oexp_y_str, lag %"                                u3<0.95>
418:
419: "gdem_y_cyc (Govt demand G&S), lag"                t1<0>
420: "gdem_y_cyc response outputgap * gdem/GDP"          t2<0.023>
421: "gdem_y_str (Govt demand G&S), lag"                t3<0.95>
422: "gdem_y_discr (Govt demand G&S), lag"              t4<0.7>
423:
424: "gdem_y_discr in fiscal impulse" s1<1>
425:
426: % ----- GDP components -----
427:
428: "Consumption gap, lag"                             a1_cons<0.65>
429: "Consumption gap, expect."                         a2_cons<0>
430: "Consumption gap, RMCI"                            a3_cons<0.1>
431: "Consumption gap, output gap"                      a4_cons<0.2>
432: "Consumption gap, fisc. imp."                      a5_cons<0.25>
433: "Consumption gap, domestic in RMCI" a6_cons<1>
434:
435: "Investment gap, lag"                             a1_inv<0.65>
436: "Investment gap, expect."                         a2_inv<0>
437: "Investment gap, RMCI(RIR, RER)"                   a3_inv<0.1>
438: "Investment gap, output gap"                      a4_inv<0.2>
439: "Investment gap, fisc. imp."                      a5_inv<0.15>
440: "Investment gap, domestic in RMCI" a6_inv<1>
441:
442: "Export gap, lag"                                 a1_exp<0.65>
443: "Export gap, expect."                             a2_exp<0>
444: "Export gap, RMCI(RIR, REER)"                     a3_exp<0.2>
445: "Export gap, foreign dem."                         a5_exp<0.3>
446: "Export gap, domestic in RMCI" a6_exp<0>
447:
448: "Import gap, REER coefficient"                     a1_imp<0.3>
449:
450: "Agric.output gap AR(1)"                          r1_y_agr<0.4>
451:
452: % ----- Phillips curves -----
453:
454: "Core Phillips curve, lag"                        b1<0.35>
455: "Core Phillips curve, RMC"                        b2<0.2>
456: "Core Phillips curve, direct imp."                b3<0.05>

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457: "Core Phillips curve, domestic in RMC"  b4(0.8)
458:
459: "Food Phillips c., lag"                    bf1(0.35)
460: "Food Phillips c.,RMC indirect importprice" bf2(0.1)
461: "Food Phillips c.,direct importprice"      bf3(0.1)
462: "+Food Phillips c., agric.output gap neg." bf4(1.5)
463:
464: "Energy Phillips curve, lag"               be1(0.25)
465: "Energy Phillips curve, RMC"               be2(0.04)
466: "Energy Phillips curve, direct imp."       be3(0.02)
467:
468: % ----- Monetary policy rule -----
469:
470: "Policy rule, lag"                         c1(0.8)
471: "Policy rule, inflation"                   c2(1.2)
472: "Policy rule, output gap"                  c3(0.2)
473: "Policy rule, FX target"                   c4(0)
474: "Inflation target, persist."               c5(0.9)
475: "Lending premium, lag"                    c6(0.9)
476:
477: % ----- Exchange rate -----
478:
479: "Exchng rate, UIP"                         e1(0.5)
480: "Exchng rate expect., forward"             e2(0.5)
481: "Premium, persist."                       e3(0.9)
482: "Exchng rate target, persist."             e4(0)
483: "Exchng rate target, infl. dev."           e5(0.3)
484: "Exchng rate target, REER gap"             e6(0.85)
485:
486: % ----- Money demand -----
487:
488: "Real money dem., lag"                     m1(0.7)
489: "Real money dem., interest rate"           m2(0.5)
490: "Change velocity,lag"                     m3(0.9)
491:
492: % -----
493: % ----- Weights -----
494:
495: "Import gap, cons.share imports"           w_imp_cons(0.445714)
496: "Import gap, inv.share imports"            w_imp_inv(0.156)

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497: "Import gap, govdemand share imports"    w_imp_gdem<0.276>
498: "Import gap, exp.share imports"           w_imp_exp<0.122286>
499:
500: "GDP gap, cons. share GDP"                 w_y_cons<0.78>
501: "GDP gap, inv. share GDP"                 w_y_inv<0.13>
502: "GDP gap, govdemand share GDP"            w_y_gdem<0.23>
503: "GDP gap, exp. share GDP"                 w_y_exp<0.21>
504: "GDP gap, imp. share GDP"                 w_y_imp<0.35>
505:
506: "Weight of core in CPI"                    w_core<0.7747>
507: "Weight of food in CPI"                   w_food<0.1577>
508: "Weight of energy in CPI"                  w_ener<0.0676>
509:
510: % -----
511: % ----- Trend persistences -----
512:
513: "Cons. tnd. persist."                     r_cons<0.95>
514: "Inv. tnd. persist."                     r_inv<0.95>
515: "Gov. demand, tnd. persist."              r_gdem<0.95>
516: "Export tnd. persist."                   r_exp<0.95>
517: "Import tnd. persist."                   r_imp<0.95>
518: "Agric.output tnd. persist."              r2_y_agr<0.8>
519: "RER tnd. persist."                      r_z<0.95>
520: "Food rel. price tnd. persist."           r_rp_food<0.9>
521: "Energy rel. price tnd. persist."         r_rp_ener<0.9>
522:
523: % -----
524: % ----- Foreign block parameters -----
525:
526: "SS of foreign CPI, ann. QQ %"            ss_dl_cpistar<1.98026>
527: "SS of foreign real intr. rate, %"        ss_rstar_tnd<0>
528: "SS of foreign rel. food price, ann. QQ %" ss_dl_rp_foodstar_tnd<0>
529: "SS of foreign rel. ener. price, ann. QQ %" ss_dl_rp_enerstar_tnd<0>
530:
531: "foreign demand persist."                 r_ystar<0.94>
532: "foreign CPI persist."                   r_cpistar<0.8>
533: "foreign interest rate persist."          r_istar<0.95>
534: "for. interest rate tnd. persist."        r_rstar_tnd<0.9>
535: "for. rp food persist."                  r_rp_foodstar_gap<0.62>
536: "for. rp energy persist."                 r_rp_enerstar_gap<0.73>

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537: "for. rp food tnd. persist."      r_rp_foodstar_tnd<0.9>
538: "for. rp energy tnd. persist."    r_rp_enerstar_tnd<0.9>
539:
540: % -----
541: % ----- Transition equations -----
542: % -----
543:
544: !transition_equations
545:
546: % -----
547: % ----- Fiscal policy -----
548:
549: grants_y    = d7<0.5> * grants_y {-1} + (1-d7<0.5>) * ss_grants_y<5> + shock_grants_y< $\sigma=2.5$ >;
550:
551: grev_y = grev_y_str + grev_y_cyc + grev_y_discr;
552: grev_y_cyc = v1<0.012> * l_cons_gap + v2<0.006> * l_imp_gap + v3<0.02> * l_y_gap + shock_grev_y_cyc< $\sigma=0$ >;
553: grev_y_str = ...
554:           + v4<0.99> * grev_y_str{-1} ...
555:           + (1-v4<0.99>) * ss_grev_y_str<21> ...
556:           + shock_grev_y_str< $\sigma=0.1$ >;
557: grev_y_discr = shock_grev_y_discr< $\sigma=1.7321$ >;
558:
559: oexp_y = oexp_y_str + oexp_y_cyc + oexp_y_discr;
560: oexp_y_cyc = u1<0> * oexp_y_cyc{-1} - u2<0.012> * l_y_gap + shock_oexp_y_cyc< $\sigma=0$ >;
561: oexp_y_str = ...
562:           + u3<0.95> * oexp_y_str{-1} ...
563:           + (1-u3<0.95>) * ss_oexp_y_str<6> ...
564:           + shock_oexp_y_str< $\sigma=0.1732$ >;
565: oexp_y_discr = shock_oexp_y_discr< $\sigma=1.7321$ >;
566:
567: gdem_y = gdem_y_str + gdem_y_cyc + gdem_y_discr;
568: gdem_y_cyc = t1<0> * gdem_y_cyc{-1} - t2<0.023> * l_y_gap + shock_gdem_y_cyc< $\sigma=0$ >;
569: gdem_y_str = ...
570:           + t3<0.95> * gdem_y_str{-1} ...
571:           + (1-t3<0.95>) * ss_gdem_y_str<26> ...
572:           + shock_gdem_y_str< $\sigma=0.075$ >;
573: gdem_y_discr = t4<0.7> * gdem_y_discr{-1} + shock_gdem_y_discr< $\sigma=1.7321$ >;
574:
575: l_gdem/100 = log(gdem_y / 100) + l_y/100;
576: l_gdem_tnd/100 = log(gdem_y_str / 100) + l_y_tnd/100;

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577:
578: def_y      = gdem_y      + oexp_y      - grev_y;
579: def_y_str  = gdem_y_str  + oexp_y_str  - grev_y_str;
580: def_y_cyc  = gdem_y_cyc  + oexp_y_cyc  - grev_y_cyc;
581: def_y_discr = gdem_y_discr + oexp_y_discr - grev_y_discr;
582:
583: fisc_imp = ...
584:   + s1(1) * gdem_y_discr + oexp_y_discr - grev_y_discr ...
585:   + def_y_str - def_y_str{-1};
586:
587: % -----
588: % ----- Real demand -----
589:
590: % ----- Consumption gap -----
591: % (note fisc_imp is indirect effect of l_gdem_gap)
592:
593: l_cons_gap = ...
594:   + a1_cons(0.65) * l_cons_gap{-1} ...
595:   + a2_cons(0) * e_l_cons_gap ...
596:   - a3_cons(0.1) * rmci_cons ...
597:   + a4_cons(0.2) * l_y_gap ...
598:   + a5_cons(0.25) * fisc_imp ...
599:   + shock_l_cons_gap( $\sigma=2.5$ );
600:
601: rmci_cons = a6_cons(1) * r4_gap + (1 - a6_cons(1)) * -l_z_gap;
602:
603: e_l_cons_gap = l_cons_gap{+1};
604:
605: % ----- Investment gap -----
606: % (note fisc_imp is indirect effect of l_gdem_gap)
607:
608: l_inv_gap = ...
609:   + a1_inv(0.65) * l_inv_gap{-1} ...
610:   + a2_inv(0) * e_l_inv_gap ...
611:   - a3_inv(0.1) * rmci_inv ...
612:   + a4_inv(0.2) * l_y_gap ...
613:   + a5_inv(0.15) * fisc_imp ...
614:   + shock_l_inv_gap( $\sigma=6$ );
615:
616: rmci_inv = a6_inv(1) * r4_gap + (1 - a6_inv(1)) * -l_z_gap;

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617:
618: e_l_inv_gap = l_inv_gap{+1};
619:
620: % ----- Export gap -----
621:
622: l_exp_gap = ...
623:   + a1_exp(0.65) * l_exp_gap{-1} ...
624:   + a2_exp(0) * e_l_exp_gap ...
625:   - a3_exp(0.2) * rmci_exp ...
626:   + a5_exp(0.3) * l_ystar_gap ...
627:   + shock_l_exp_gap( $\sigma=6$ );
628:
629: rmci_exp = a6_exp(0) * r4_gap + (1 - a6_exp(0)) * - l_z_gap;
630:
631: e_l_exp_gap = l_exp_gap{+1};
632:
633: % ----- Import gap -----
634: % (RER added same for all importdemands,coeff REER>0,so -a1_imp)
635:
636: l_imp_gap = ...
637:   + w_imp_cons(0.445714) * l_cons_gap ...
638:   + w_imp_inv(0.156) * l_inv_gap ...
639:   + w_imp_gdem(0.276) * l_gdem_gap ...
640:   + w_imp_exp(0.122286) * l_exp_gap ...
641:   - a1_imp(0.3) * l_z_gap ...
642:   + shock_l_imp_gap( $\sigma=0.5$ );
643:
644: % ----- Output gap -----
645: % (note direct effect of l_gdem_gap on l_y_gap)
646:
647: l_y_gap = ...
648:   + w_y_cons(0.78) * l_cons_gap ...
649:   + w_y_inv(0.13) * l_inv_gap ...
650:   + w_y_gdem(0.23) * l_gdem_gap ...
651:   + w_y_exp(0.21) * l_exp_gap ...
652:   - w_y_imp(0.35) * l_imp_gap ...
653:   + shock_l_y_gap( $\sigma=1$ );
654:
655: e_l_y_gap = l_y_gap{+1};
656:

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657: % -----
658: % ----- Agricultural output gap -----
659:
660: l_y_agr_gap = r1_y_agr<0.4> * l_y_agr_gap{-1} + shock_l_y_agr_gap< $\sigma=0.3$ >;
661:
662: e_l_y_agr_gap = l_y_agr_gap{+1};
663:
664: % -----
665: % ----- Prices -----
666:
667: % ----- Core Phillips curve -----
668:
669: dl_cpi_core = ...
670:   + b1<0.35> * dl_cpi_core{-1} ...
671:   + (1 - b1<0.35> - b3<0.05>) * (e_dl_cpi_core) ...
672:   + b3<0.05> * dl_cpi_core_direct ...
673:   + b2<0.2> * rmc ...
674:   + shock_dl_cpi_core< $\sigma=2$ >;
675:
676: rmc = b4<0.8> * l_y_gap + (1 - b4<0.8>) * l_z_gap;
677:
678: dl_cpi_core_direct = dl_cpistar + dl_s - dl_z_tnd;
679:
680: e_dl_cpi_core = dl_cpi_core{+1};
681:
682: l_rp_cpi_core = l_cpi_core - l_cpi;
683:
684: % ----- Food Phillips curve -----
685: % agric. output gap in food inflation, with coeff as BNR-IMF -0.5
686:
687: dl_cpi_food = ...
688:   + bf1<0.35> * dl_cpi_food{-1} ...
689:   + (1 - bf1<0.35> - bf3<0.1>) * (e_dl_cpi_food) ...
690:   + bf3<0.1> * dl_cpi_food_direct ...
691:   + bf2<0.1> * (l_rp_foodstar_gap + l_z_gap + l_rp_cpi_core_gap - l_rp_cpi_food_gap) ...
692:   - bf4<1.5> * l_y_agr_gap ...
693:   + shock_dl_cpi_food< $\sigma=9$ >;
694:
695: dl_cpi_food_direct = dl_foodstar - dl_rp_foodstar_tnd + dl_s - dl_z_tnd ...
696:   - dl_rp_cpi_core_tnd + dl_rp_cpi_food_tnd;

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697:
698: e_dl_cpi_food = dl_cpi_food{+1};
699:
700: l_rp_cpi_food = l_cpi_food - l_cpi;
701:
702: % ----- Energy Phillips curve -----
703:
704: dl_cpi_ener = ...
705:   + be1<0.25> * dl_cpi_ener{-1} ...
706:   + (1 - be1<0.25> - be3<0.02>) * (e_dl_cpi_ener)...
707:   + be3<0.02> * dl_cpi_ener_direct ...
708:   + be2<0.04> * (l_rp_enerstar_gap + l_z_gap + l_rp_cpi_core_gap - l_rp_cpi_ener_gap)
709:   + shock_dl_cpi_ener< $\sigma=7.1$ >;
710:
711: dl_cpi_ener_direct = dl_enerstar - dl_rp_enerstar_tnd + dl_s - dl_z_tnd ...
712:   - dl_rp_cpi_core_tnd + dl_rp_cpi_ener_tnd;
713:
714: e_dl_cpi_ener = dl_cpi_ener{+1};
715:
716: l_rp_cpi_ener = l_cpi_ener - l_cpi;
717:
718: % ----- Headline CPI level -----
719:
720: l_cpi = w_core<0.7747> * l_cpi_core + w_food<0.1577> * l_cpi_food + w_ener<0.0676> * l_cpi_ener;
721:
722: e_dl_cpi = dl_cpi{+1};
723:
724: 0 = w_core<0.7747> * l_rp_cpi_core_gap + w_food<0.1577> * l_rp_cpi_food_gap + w_ener<0.0676> * l_rp_cpi_ener_gap;
725:
726: % -----
727: % ----- Monetary policy -----
728: % (standard, except ER deviation, but coeff=0)
729:
730: % ----- Inflation forecast based rule -----
731:
732: i = c1<0.8> * i{-1} + (1 - c1<0.8>) * ( ...
733:   + i_tnd ...
734:   + c2<1.2> * d4l_cpi_dev ...
735:   + c3<0.2> * l_y_gap ...
736:   + c4<0> * (dl_s - dl_s_tar) ...

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737: ) ...
738: + shock_i( $\sigma=1$ );
739:
740: d4l_cpi_dev = d4l_cpi{+4} - d4l_cpi_tar;
741:
742: i_tnd = r_tnd + d4l_cpi_tar + dl_rp_cpi_core_tnd;
743:
744: % ----- Real interest rate -----
745:
746: r = i - e_dl_cpi_core;
747:
748: r_tnd = rstar_tnd + prem + e_dl_z_tnd;
749:
750: r4_gap = (r_gap + r_gap{+1} + r_gap{+2} + r_gap{+3}) / 4 + prem_d_gap;
751:
752: e4_r_gap = (r_gap + r_gap{+1} + r_gap{+2} + r_gap{+3}) / 4;
753:
754: % ----- Lending premium -----
755:
756: prem_d = prem_d_gap + ss_prem_d(8.5);
757:
758: prem_d_gap = c6(0.9) * prem_d_gap{-1} + shock_prem_d_gap( $\sigma=0.8$ );
759:
760: % ----- Inflation target -----
761:
762: d4l_cpi_tar = c5(0.9) * d4l_cpi_tar{-1} + (1 - c5(0.9)) * ss_d4l_cpi_tar(4.87902) + shock_d4l_cpi_tar( $\sigma=0.1$ );
763:
764: d4l_cpi_core_tar = d4l_cpi_tar + d4l_rp_cpi_core_tnd;
765:
766: d4l_cpi_food_tar = d4l_cpi_tar + d4l_rp_cpi_food_tnd;
767:
768: d4l_cpi_ener_tar = d4l_cpi_tar + d4l_rp_cpi_ener_tnd;
769:
770: % -----
771: % ----- Exchange rate -----
772: % (UIP & move to ER-target, det. by RER-tnd & target infl.diff.)
773:
774: l_s = ...
775: + e1(0.5) * (e_l_s - (i - istar - prem)/4) ...
776: + (1 - e1(0.5)) * (l_s{-1} + dl_s_tar/4) ...

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777:   + shock_l_s( $\sigma=0.35$ );
778:
779: e_l_s = ...
780:   + e2(0.5) * l_s{+1} ...
781:   + (1 - e2(0.5)) * (l_s{-1} + 2*(dl_z_tnd + d4l_cpi_tar + dl_rp_cpi_core_tnd - ss_dl_cpistar(1.98026))/4);
782:
783: prem = e3(0.9) * prem{-1} + (1 - e3(0.9)) * ss_prem(2) + shock_prem( $\sigma=0.1$ );
784:
785: dl_s_tar = e4(0) * dl_s_tar{-1} + (1 - e4(0)) * ( ...
786:   dl_z_tnd + d4l_cpi_tar + dl_rp_cpi_core_tnd - ss_dl_cpistar(1.98026) - e5(0.3) * d4l_cpi_dev - e6(0.85) * l_z_gap ...
787: ) + shock_dl_s_tar( $\sigma=0.3$ );
788:
789: l_z = l_s + l_cpistar - l_cpi_core;
790:
791: e_dl_z = dl_z{+1};
792:
793: e_dl_z_tnd = dl_z_tnd{+1};
794:
795: % -----
796: % ----- Real money demand -----
797: % (function of nominal i dev., cp. BNR: change in r-tnd)
798:
799: l_rmd = l_md - l_cpi;
800:
801: dl_rmd = ...
802:   + m1(0.7) * dl_rmd{-1} ...
803:   + (1 - m1(0.7)) * (dl_y - dl_v + m2(0.5) * (i - i_tnd)) ...
804:   + shock_dl_rmd( $\sigma=1$ );
805:
806: dl_v = m3(0.9) * dl_v{-1} + (1 - m3(0.9)) * ss_dl_v(-2.02027) + shock_dl_v( $\sigma=0.1$ );
807:
808: % -----
809: % ----- Trends -----
810: % (weights for output-tnd are same as for output-gap)
811:
812: dl_cons_tnd = r_cons(0.95) * dl_cons_tnd{-1} + (1 - r_cons(0.95)) * ss_dl_y_tnd(7.23207) + shock_dl_cons_tnd( $\sigma=0.25$ );
813: dl_inv_tnd = r_inv(0.95) * dl_inv_tnd{-1} + (1 - r_inv(0.95)) * ss_dl_y_tnd(7.23207) + shock_dl_inv_tnd( $\sigma=0.5$ );
814: dl_exp_tnd = r_exp(0.95) * dl_exp_tnd{-1} + (1 - r_exp(0.95)) * ss_dl_y_tnd(7.23207) + shock_dl_exp_tnd( $\sigma=0.75$ );
815: dl_imp_tnd = r_imp(0.95) * dl_imp_tnd{-1} + (1 - r_imp(0.95)) * ss_dl_y_tnd(7.23207) + shock_dl_imp_tnd( $\sigma=0.25$ );
816:

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817: dl_y_tnd = ...
818:   + w_y_cons(0.78) * dl_cons_tnd ...
819:   + w_y_inv(0.13) * dl_inv_tnd ...
820:   + w_y_gdem(0.23) * dl_gdem_tnd ...
821:   + w_y_exp(0.21) * dl_exp_tnd ...
822:   - w_y_imp(0.35) * dl_imp_tnd;
823:
824: dl_y_agr_tnd = r2_y_agr(0.8) * dl_y_agr_tnd{-1} + (1 - r2_y_agr(0.8)) * ss_dl_y_agr_tnd(7.23207) + shock_dl_y_agr_tnd( $\sigma=0.06$ ); % no role yet
825:
826: dl_z_tnd      = r_z(0.95) * dl_z_tnd{-1} + (1 - r_z(0.95)) * ss_dl_z_tnd(0) + shock_dl_z_tnd( $\sigma=0.1$ );
827:
828: dl_rp_cpi_food_tnd = r_rp_food(0.9) * dl_rp_cpi_food_tnd{-1} + (1 - r_rp_food(0.9)) * ss_dl_rp_cpi_food_tnd(1.98026) + shock_dl_rp_cpi_food_tnd( $\sigma=0.05$ );
829: dl_rp_cpi_ener_tnd = r_rp_ener(0.9) * dl_rp_cpi_ener_tnd{-1} + (1 - r_rp_ener(0.9)) * ss_dl_rp_cpi_ener_tnd(0) + shock_dl_rp_cpi_ener_tnd( $\sigma=0.05$ );
830:
831: % -----
832: % ----- External sector -----
833:
834: l_ystar_gap = r_ystar(0.94) * l_ystar_gap{-1} + shock_l_ystar_gap( $\sigma=0.45$ );
835:
836: dl_cpistar = r_cpistar(0.8) * dl_cpistar{-1} + (1 - r_cpistar(0.8)) * ss_dl_cpistar(1.98026) + shock_dl_cpistar( $\sigma=3.8$ );
837: e_dl_cpistar = dl_cpistar{+1};
838:
839: istar = r_istar(0.95) * istar{-1} + (1 - r_istar(0.95)) * (ss_rstar_tnd(0) + ss_dl_cpistar(1.98026)) + shock_istar( $\sigma=0.45$ );
840: rstar_tnd = r_rstar_tnd(0.9) * rstar_tnd{-1} + (1 - r_rstar_tnd(0.9)) * ss_rstar_tnd(0) + shock_rstar_tnd( $\sigma=0.5$ );
841:
842: l_rp_foodstar = l_foodstar - l_cpistar;
843: l_rp_enerstar = l_enerstar - l_cpistar;
844:
845: l_rp_foodstar_gap = r_rp_foodstar_gap(0.62) * l_rp_foodstar_gap{-1} + shock_l_rp_foodstar_gap( $\sigma=5.3$ );
846: l_rp_enerstar_gap = r_rp_enerstar_gap(0.73) * l_rp_enerstar_gap{-1} + shock_l_rp_enerstar_gap( $\sigma=13.9$ );
847:
848: dl_rp_foodstar_tnd = r_rp_foodstar_tnd(0.9) * dl_rp_foodstar_tnd{-1} + (1 - r_rp_foodstar_tnd(0.9)) * ss_dl_rp_foodstar_tnd(0) + shock_dl_rp_foodstar_tnd( $\sigma=0.05$ );
849: dl_rp_enerstar_tnd = r_rp_enerstar_tnd(0.9) * dl_rp_enerstar_tnd{-1} + (1 - r_rp_enerstar_tnd(0.9)) * ss_dl_rp_enerstar_tnd(0) + shock_dl_rp_enerstar_tnd( $\sigma=0.05$ );
850:
851: % -----
852: % ----- Trend/gap identities -----
853:
854: !for
855:   l_y, l_y_agr, l_cons, l_inv, l_exp, l_imp, l_gdem,
856:   r, l_z

```

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857:   l_rp_cpi_food, l_rp_cpi_ener, l_rp_cpi_core
858:   l_rp_foodstar, l_rp_enerstar
859: !do
860:   !transition_equations
861:   ? = ?_tnd + ?_gap;
862: !end
863:
864: % -----
865: % ----- Growth rates -----
866:
867: !for
868:   y, y_tnd, cons, cons_tnd, inv, inv_tnd, gdem, gdem_tnd, exp, exp_tnd, imp, imp_tnd
869:   y_agr, y_agr_tnd,
870:   cpi, cpi_core, cpi_food, cpi_ener
871:   rp_cpi_food, rp_cpi_ener, rp_cpi_core, rp_cpi_food_tnd, rp_cpi_ener_tnd, rp_cpi_core_tnd
872:   s, z, z_tnd
873:   rmd
874:   cpistar, foodstar, enerstar
875:   rp_foodstar, rp_enerstar, rp_foodstar_tnd, rp_enerstar_tnd
876: !do
877:   !transition_equations
878:   dl_? = 4 * (l_? - l_?{-1}); % ak we should *4 for flow variables
879:   d4l_? = (l_? - l_?{-4});
880: !end
881:
882: % -----
883: % ----- Measurement variables/equations -----
884: % -----
885:
886: % -----
887: % ----- Hard observations -----
888:
889: !for
890:   def_y, grants_y, grev_y
891:   l_y, l_y_agr, l_cons, l_gdem, l_inv, l_exp, l_imp
892:   l_cpi, l_cpi_food, l_cpi_ener
893:   i, prem_d, l_s, l_md
894:   l_ystar_gap, l_cpistar, istar, rstar_tnd
895:   l_foodstar, l_enerstar
896:   l_rp_foodstar_gap, l_rp_enerstar_gap

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```

897: !do
898:   !measurement_variables
899:   obs_?
900:   !measurement_equations
901:   obs_? = ?;
902: !end
903:
904: % -----
905: % ----- Expert/judgemental tunes -----AK added Oct 13, def_y, grev_y
906:
907: !for
908:   gdem_y, oexp_y, def_y, grev_y
909:   gdem_y_str, oexp_y_str, grev_y_str
910:   gdem_y_cyc, oexp_y_cyc, grev_y_cyc
911:   grev_y_discr, oexp_y_discr, gdem_y_discr
912:   l_y_tnd, l_cons_tnd, l_inv_tnd, l_gdem_tnd, l_exp_tnd, l_imp_tnd
913:   d4l_cpi_tar, l_rp_cpi_food_tnd, l_rp_cpi_ener_tnd
914:   dl_cpi_core, dl_cpi_food, dl_cpi_ener, dl_cpi
915:   r_tnd, l_z_tnd, prem, dl_s_tar
916:   def_y_str, def_y_discr, grants_y
917:   l_y_gap, l_cons_gap, l_inv_gap, l_gdem_gap, l_exp_gap, l_imp_gap
918:   l_y_agr_gap
919:   l_z_gap
920:   l_y, d4l_y
921:   dl_s, l_s
922:   l_cpi_core
923: !do
924:   !measurement_variables
925:   tune_?
926:   !measurement_equations
927:   tune_? = ?;
928: !end
929:
930: % -----
931: % ----- Auxiliary variables/equations for shock tuning in the filter -----
932:
933: !for
934:   shock_dl_cons_tnd( $\sigma=0.25$ )
935:   shock_dl_cpi_ener( $\sigma=7.1$ )
936:   shock_dl_cpi_core( $\sigma=2$ )

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937:   shock_l_inv_gap( $\sigma=6$ )
938: !do
939:   !transition_variables
940:     aux_?
941:   !transition_equations
942:     aux_? = ?;
943:   !measurement_variables
944:     tune_?
945:   !measurement_equations
946:     tune_? = aux_?;
947: !end
948:
949: % -----
950: % ----- Reporting equations -----
951: % -----
952:
953: !parameters
954:
955: "Share of import prices in CPI"      mu_pimp(0.35)
956: "Share of export prices in GDP deflator" mu_pexp(0.2)
957:
958: "share of fcy debt in total debt"    r_debt_fcy_rat(0.9)
959: "interest rate fcy debt persistence" r_debt_fcy_intrate_pers(0.9)
960: "interest rate lcy debt persistence"  r_debt_lcy_intrate_pers(0.9)
961:
962: gamma_r(0.5)
963: gamma_k(0.5)
964: gamma_BP_tnd(0.9)
965: % AK 7/14/23 replace k_bar by l_BP_tnd; % log(BP)=k, BP cum inflows mln$,
966: % BP does not go to fixed ss level k_bar=log(6500), but to trend
967: % with initial value trend set equal to actual in readData
968:
969: "Import share in consumption" lam_imp_cons(0.2)
970: "Import share in investment"   lam_imp_inv(0.42)
971: "Import share in govdemand"    lam_imp_gdem(0.42)
972: "Import share in exports"      lam_imp_exp(0.20381)
973:
974: !reporting_equations
975:
976: % True percentages (added Oct 28, '22)

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977: !for
978:   y, cons, inv, gdem, exp, imp
979:   cpi, cpi_core, cpi_food, cpi_ener
980:   s, z
981: !do
982:   pct_?   = exp(dl_?/100)*100 - 100; % annualized: correct for stocks, incorrect for flows
983:   pct4_?  = exp(d4l_?/100)*100 - 100;
984: !end
985:
986: !for
987:   i, r
988: !do
989:   pct_?   = exp(?/100)*100 - 100;
990: !end
991:
992: % Levels exchange rate and nominal money demand
993: s   = exp(l_s/100);
994: md  = exp(l_md/100);
995: dl_md = 4*(l_md - l_md{-1}); % ak 7/15/23 must be annualized
996:
997: % Deflators (recall: all annualized, *4)
998:
999: dl_pexp = dl_pexpstar + dl_s;
1000: dl_pimp = dl_pimpstar + dl_s;
1001:
1002: dl_pdom = (dl_cpi - mu_pimp(0.35) * dl_pimp) / (1 - mu_pimp(0.35));
1003:
1004: dl_py   = mu_pexp(0.2) * dl_pexp + (1 - mu_pexp(0.2)) * dl_pdom;
1005: % deflator for govt demand assumed equal to dl_cpi_core to simplify
1006:
1007: % Nominal growth rates (recall:all annualized, *4)
1008:
1009: dl_nexp = dl_exp  + dl_pexp;
1010: dl_nimp = dl_imp  + dl_pimp;
1011: dl_ny   = dl_y    + dl_py;
1012: dl_ngdem = dl_gdem + dl_cpi_core;
1013: dl_ncons = dl_cons + dl_cpi_core; % 4/19 AK for ratios (% to nom GDP)
1014: dl_ninv  = dl_inv  + dl_cpi_core;
1015:
1016: % Nominal levels exports, imports, GDP

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1017:
1018: nexp      = exp(log(nexp{-1}) + dl_nexp/100/4);
1019: nimp      = exp(log(nimp{-1}) + dl_nimp/100/4);
1020: ny        = exp(log(ny{-1}) + dl_ny/100/4); % get ny{-1} etc from dbAUX to have start value forecast
1021: ngdem     = exp(log(ngdem{-1}) + (dl_ngdem/100)/4); % idem
1022: ncons     = exp(log(ncons{-1}) + dl_ncons/100/4); % 4/19/23 AK for ratios
1023: ninv      = exp(log(ninv{-1}) + dl_ninv/100/4);
1024:
1025: %tb_rat   = (nexp - nimp) / ny *100; ak 3/29/2025: alt. in real terms:
1026: tb_rat    = 100*(exp(l_exp/100) - exp(l_imp/100))/exp(l_y/100);
1027:
1028: % Capital account (Ostry/Ghosh/Chamon,2012:log(BP)=k; define rstar here, not done in model)
1029: % 7/14/23 trend eq for ss cumul capital inflows in line with real GDP, instead of k_bar
1030: rstar = istar - e_dl_cpistar;
1031: % first evaluate trend
1032: dl_BP_tnd = gamma_BP_tnd(0.9) * dl_BP_tnd{-1} + (1-gamma_BP_tnd(0.9)) * ss_dl_y_tnd(7.23207);
1033: l_BP_tnd = dl_BP_tnd/4 + l_BP_tnd{-1};
1034: % then evaluate level
1035: l_BP      = l_BP{-1} + gamma_r(0.5) * ( (r - rstar - prem)/4 - e_dl_z ) ...
1036:         - gamma_k(0.5) * (l_BP{-1} - l_BP_tnd);
1037: dBP_usd   = exp(l_BP/100) - exp(l_BP{-1}/100);
1038:
1039: % evaluate debt and deficit first before rest of BOP and MON
1040: % ak debt changes with deficit-/grants, nb def_y and grants are over Q-GDP, as is debt
1041: debt_y = ...
1042:     + debt_lcy_y{-1} * (1) / exp(dl_ny/400) ...
1043:     + debt_fcy_y{-1} * exp(dl_s/400) / exp(dl_ny/400) ...
1044:     + def_y - grants_y;
1045:
1046: debt_fcy_rat = r_debt_fcy_rat(0.9) * debt_fcy_y{-1} / debt_y{-1} + (1 - r_debt_fcy_rat(0.9)) * ss_debt_fcy_rat(0.8462);
1047:
1048: debt_fcy_y = debt_fcy_rat * debt_y;
1049:
1050: debt_lcy_y = debt_y - debt_fcy_y; % residual
1051:
1052: def_lcy_y = debt_lcy_y - debt_lcy_y{-1} * (1) / exp(dl_ny/400);
1053:
1054: def_fcy_y = debt_fcy_y - debt_fcy_y{-1} * exp(dl_s/400) / exp(dl_ny/400);
1055:
1056: % ak we need interest-rate update equations:

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1057: % i_debt_fcy = r_debt_fcy_intrate_pers * i_debt_fcy{-1} + (1 -
1058: % r_debt_fcy_intrate_pers) * (ss_rstar_tnd + ss_dl_cpistar + ss_prem_debt_fcy)
1059: % i_debt_lcy = r_debt_lcy_intrate_pers * i_debt_lcy{-1} + (1 -
1060: % r_debt_lcy_intrate_pers) * (i_tnd + ss_prem_debt_lcy)
1061:
1062: intp_fcy_y = i_debt_fcy{-1}/400 * debt_fcy_y{-1} * exp(dl_s/400) / exp(dl_ny/400);
1063:
1064: intp_lcy_y = i_debt_lcy{-1}/400 * debt_lcy_y{-1} * (1) / exp(dl_ny/400);
1065:
1066: % External flow equilibrium (current account (RM=E-M) converted to mln $)
1067: % still need evaluate levels govt interest payments in usd for BOP
1068: DBG_usd = ( ( (def_fcy_y/100) * ny) / s ) * 1000; % for bor deficit RWFbln then to $mln
1069: NFG_usd = ( ( (grants_y/100) * ny) / s ) * 1000; % net foreign grants govt RWFbln then to $mln
1070:
1071: dNFA_usd = (nexp - nimp) / s * 1000 + NFG_usd + DBG_usd + dBP_usd;
1072:
1073: % ak 7/15/23 trace monetary flows, share of bank in domestic financing deficit
1074: def_lcy_bank_y = 0.5 * def_lcy_y; % nb still over Q GDP in %!
1075: NCG = NCG{-1} + (def_lcy_bank_y/100) * ny;
1076:
1077: NFA = NFA{-1} + dNFA_usd * s/1000; % ignores ER valuation changes
1078:
1079: % Monetary flow equilibrium in bln RWF, private credit residual
1080: dNCP = diff(md) - dNFA_usd * s / 1000 - diff(NCG);
1081:

```